

Surgical Treatment of Epilepsy

Last updated: February 2, 2026

DEFINITION OF PHARMACORESISTANCE	3
EPIDEMIOLOGY OF PHARMACORESISTANCE	4
Challenging the above figures	5
Surgery vs. Medications	7
MECHANISM OF PHARMACORESISTANCE	8
INDICATIONS	8
Referral to Epilepsy Center	9
Level of care and mortality	10
Underutilization of Epilepsy Surgery	10
Barriers for timely referrals	11
Timing of Surgery	14
Strategies to improve patient agreement	16
CONTRAINDICATIONS	16
DEFINITIONS OF SURGICAL ZONES	17
Node vs. Network disease	19
COMMON EPILEPTIC PATHOLOGIES.....	21
PRESURGICAL EVALUATION (NON-INVASIVE, PHASE I)	22
SCALP (SURFACE) EEG	22
High density EEG.....	23
NEUROIMAGING (ROUTINE).....	23
CT	23
MRI	23
ILAE protocol.....	24
HARNESS-MRI protocol.....	24
MRI-negative cases	24
NEUROIMAGING (ADVANCED).....	25
Special MRI postprocessing.....	25
Tractography (DTI)	25
Hippocampal Volumetry	26
Cortical Unfolding.....	26
¹ HMR (Proton) Spectroscopy.....	26
fMRI	27
Resting state fMRI.....	27
EEG-fMRI (event-related fMRI).....	27
nTMS (navigated transcranial magnetic stimulation)	29
Mapping for eloquent cortex	29
Epilepsy palliation.....	30
PET (positron-emission tomography)	30
SPECT (single-photon emission computerized tomography)	31
SISCOM (Subtraction Ictal Spect CO-registered to Mri)	31
Magnetoencephalography (MEG).....	33
MSI (Magnetic Source Imaging).....	35
MEG-based brain mapping	36
MEG-guided surgery	36
NEUROPSYCHOLOGICAL TESTING (NEUROPSYCHOMETRY)	37
GENETIC TESTING	37
MULTIDISCIPLINARY CONFERENCE.....	38

PRESURGICAL EVALUATION (INVASIVE, PHASE II)	38
CHEMICAL WADA (INTRACAROTID AMOBARBITAL TEST)	39
Indications	39
Procedure details	39
Side effects	40
Interpretation	40
Noninvasive alternatives to Wada	42
UAMS protocol	42
ELECTRICAL WADA	45
INTRACRANIAL EEG (icEEG)	45
ALGORITHM (TREATMENT)	46
FOCAL EPILEPSY	49
Temporal	49
Mesiotemporal	50
Nonmesiotemporal (neocortical temporal, extratemporal)	50
Extratemporal	51
Eloquent Cortex	51
Language localization / Lateralization	52
GENERALIZED EPILEPSY	52
Genetic (s. idiopathic, primary) generalized epilepsy	53
Lennox–Gastaut type (s. symptomatic, cryptogenic generalized epilepsy)	53
PREOPERATIVELY	53
ASM	53
TYPES OF SURGERY	54
Surgical Philosophy	54
1. RESECTION / ABLATION	54
Temporal Resections	55
Lesionectomy	55
Lesioning	55
Laser ablation (LITT)	55
RF-ablation	55
Stereotactic radiosurgery (SRS)	55
Focused ultrasound (FUS)	56
Tailored Neocortical Resection	56
USE (US Elastography)	58
Multilobar Resection	58
2. DISCONNECTIONS	58
Multiple Subpial Transections (MST)	58
Hemispherectomies (functional, anatomical)	59
Posterior Quadrant Disconnections	59
Corpus Callosotomy	59
Hippocampus Transections	59
3. NEUROMODULATION (“ELECTROCEUTICALS”)	59
Indications	59
Mechanism Of Action	60
Inward Interictal Inhibition	61
Comparison Of Neuromodulations	61
Seizure Outcomes	62
Stimulation Time	64
Recording Chronic Intracranial Data	64
Psychiatric Side Effects	65
So how to choose neuromodulation type?	65

Combination of neuromodulations	65
Deep Brain Stimulation (DBS)	66
Vagus Nerve Stimulation (VNS).....	66
Responsive Neurostimulation (RNS)	66
Transcranial Magnetic Stimulation (nTMS)	66
4. CELL THERAPY	67
POSTOPERATIVELY	68
ASM	68
Before discharge	69
SURGERY OUTCOMES - SEIZURES	69
CLASSIFICATION OF OUTCOMES	70
SEIZURE FREEDOM.....	71
Temporal Epilepsy	72
Resections.....	73
Comparison of different resection techniques	74
LITT	75
SRS.....	76
COST-EFFECTIVENESS	77
Temporal Epilepsy	77
SEIZURE WORSENING / DE NOVO SEIZURES.....	78
SURGICAL REFRACTORINESS / RECURRENT SEIZURES / SURGERY FAILURE	80
Temporal Epilepsy	81
SUDEP	82
SURGERY OUTCOMES - QUALITY OF LIFE.....	82
SURGERY OUTCOMES - PSYCHIATRIC	83
SURGERY OUTCOMES - COGNITIVE	84
Memory	84
Pediatric.....	84
Temporal Epilepsy	85
Role of structures.....	85
Open resections	86
LITT	86

DEFINITION OF PHARMACORESISTANCE

Drug-resistant $\neq$ Treatment-resistant

N.B. pharmacoresistance is diagnosed by history taking!

Medical intractability (pharmacoresistant epilepsy)

A. **Failed 2 medications**

2 trials of ASMs at optimal doses with appropriate medications are sufficient to consider referral of patients with focal seizure disorder for presurgical evaluation

- Commission on Therapeutic Strategies of the International League Against Epilepsy (ILAE) definition (officially adopted at the ILAE's 2009 meeting in Budapest, Hungary):
"DRUG-RESISTANT EPILEPSY - failure of adequate trials of **two** tolerated and appropriately chosen and used ASM schedules (whether as monotherapies or in

combination) to achieve **sustained seizure freedom**." – i.e. complete epilepsy control!

B. Failed 3 medications – obsolete!!!

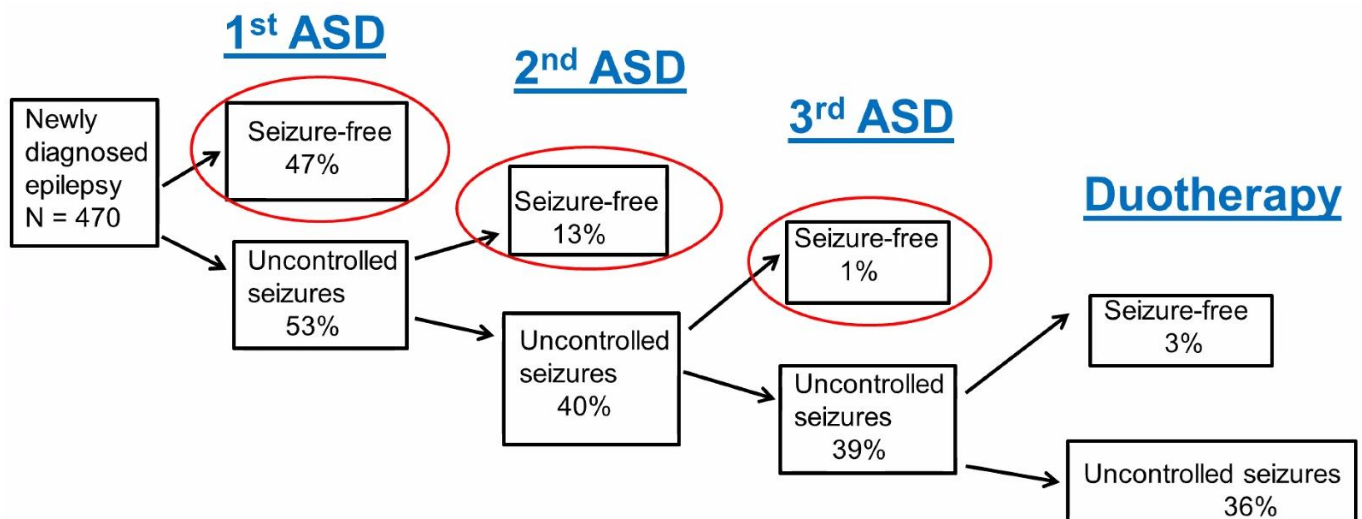
- some advocate at least 3 regimens, including *2 trials of high-dose monotherapy + 1 trial of 2-drug therapy*.
- these therapeutic trials can be accomplished within *6-12 months* - depending on frequency of seizures at baseline (i.e. how quickly can see treatment results).
- no rationale to try different medications with the same mechanism of action.

EPIDEMIOLOGY OF PHARMACORESISTANCE

20-30% patients are not controlled adequately with ASM!
($\approx \frac{1}{2}$ of them are potential surgery candidates)

Prevalence of pharmacoresistance: 17% (if count > 1 sz / month) or 26% if count > 1 sz / year)

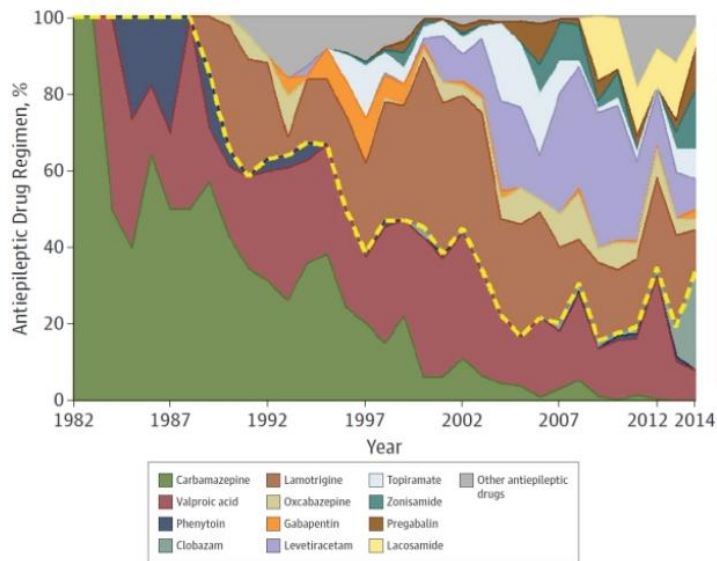
- among **all patients with epilepsy**, 60-70% are expected to become seizure-free with ASMs:
 - 47% of patients respond to first ASM
 - additional 14% respond to second ASM
 - additional 3% respond to third ASM
 - 36% of patients are refractory**
 - 20%* of patients with idiopathic generalized epilepsy are refractory (*35% of those with juvenile myoclonic epilepsy).



Kwan P, Brodie MJ. Early identification of refractory epilepsy. *N Engl J Med*. 2000;342(5):314–319

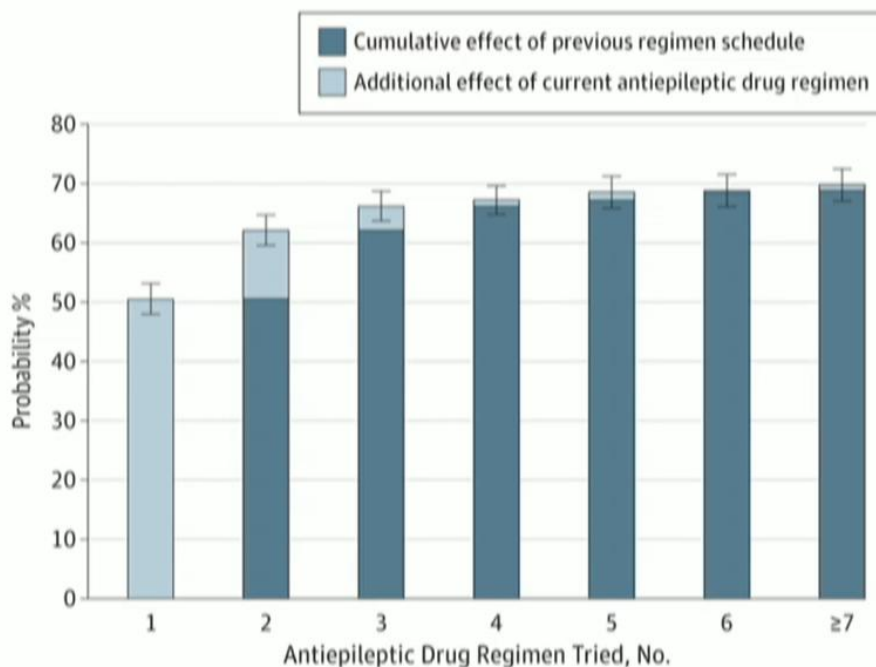
N.B. development of new medications did not change these figures! (same numbers in 2008 Kwan and Brodie study; 70% seizure freedom in **year 1881** data)

Response to first ASM is most important predictor of drug-resistant epilepsy!



	1982-1991	1992-2001	2002-2012
Seizure Free	89 63%	437 64%	591 61%
Continued Seizures	53 37%	247 36%	378 39%

Chen Z, et al. "Treatment Outcomes in Patients with Newly Diagnosed Epilepsy Treated with Established and New Antiepileptic Drugs: A 30-Year Longitudinal Cohort Study." *JAMA Neurol.* 2018



Chen et al., JAMA Neurology 2018

CHALLENGING THE ABOVE FIGURES

It is true that in Kwan and Brodie's seminal 2000 study, only 4% of the entire study cohort achieved seizure freedom after 2 failed medications. And in the 2018 follow-up by Chen et al, only 4.4% of the study cohort became seizure-free on a third medication regimen.

N.B. *most people in the 2000 and 2018 studies didn't try a third medication regimen* - so they cannot be included when calculating a success rate (i.e. seizure freedom rate for any

given medication regimen must include only the patients who tried that regimen, and exclude those who have not)

2018 Kwan and Brodie study: among people who tried a 3rd medication, 23.6% achieved seizure freedom. (15% - 4th, 14.1% - 5th, 14% - 6th medication):

Rates of 1-year seizure freedom with successive anti-epilepsy drug regimens

Regimen	Number of people attempting regimen	Of those who attempted, number achieving seizure freedom	Seizure freedom rate	Number eligible to try next regimen*
First	1,795	820	45.7%	975
Second	742	208	28.0%	534
Third	330	78	23.6%	252
Fourth	140	21	15.0%	119
Fifth	71	10	14.1%	61
Sixth	43	6	14.0%	37
Seventh	15	1	6.7%	14
Eighth	9	0	0%	9
Ninth	5	0	0%	5
Tenth	2	0	0%	2
Eleventh	1	0	0%	1

* This group includes all patients not achieving 1-year seizure freedom, including those who stopped taking medication due to adverse effects, pregnancy, or other concerns. Not all eligible patients go on to try subsequent regimens.

Data from Chen Z, Brodie MJ, Liew D, Kwan P. Treatment outcomes in patients with newly diagnosed epilepsy treated with established and new antiepileptic drugs: a 30-year longitudinal cohort study. *JAMA Neurol.* 2018;75(3):279-286. doi:10.1001/jamaneurol.2017.3949

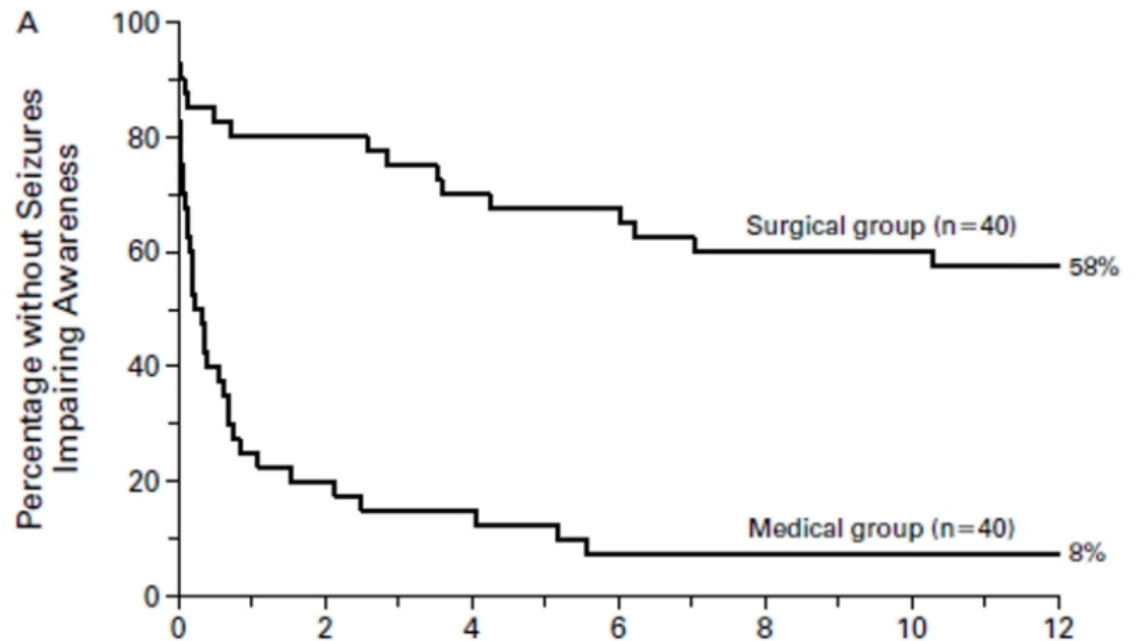
- another study showed slightly more optimistic results: 18.5% and 16.5% were made seizure free with a third and fourth ASM, respectively, and decreased to 0% after 5 or 6 previous ASM failures.

Schiller Y, Najjar Y. Quantifying the response to antiepileptic drugs: effect of past treatment history. *Neurology* 2008;70(1):54-65

- CRT (16 U.S. centers) of 38 patients with mesial temporal lobe epilepsy of < 2 years duration who failed 2 brand-name ASMs: 0 of 23 patients assigned to continued medical optimization achieved seizure freedom with drug adjustments alone during 2 years of follow up (vs. 11 of 15 in the surgical group became seizure free).

Engel J Jr, McDermott MP, Wiebe S, et al. Early surgical therapy for drug-resistant temporal lobe epilepsy: a randomized trial. *JAMA.* 2012 Mar 7;307(9):922-30.

- positive therapeutic expectations of ASMs vary according to epileptic syndrome:
 - 80-90% of patients with idiopathic generalized seizures are expected to become seizure-free;
 - only 50% of patients with focal seizure disorders are expected to become seizure-free (seizure-freedom may decrease to 30% in cases of temporal lobe epilepsy secondary to mesial temporal sclerosis or be as high as 95% in cases of benign rolandic epilepsy of childhood).

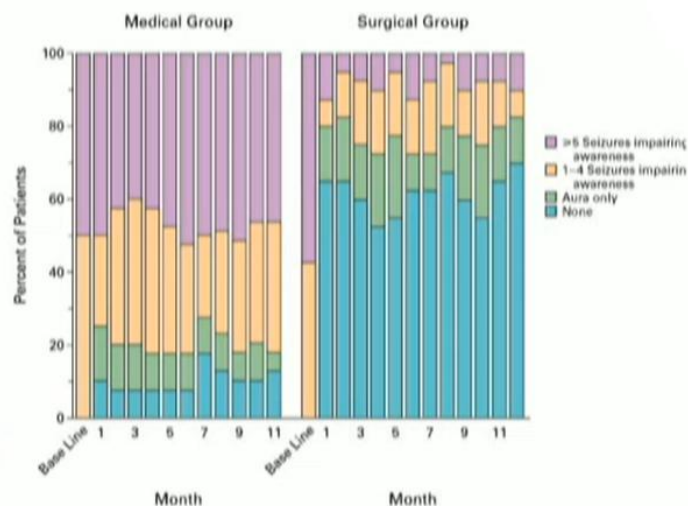


Wiebe S, et al. "A randomized, controlled trial of surgery for temporal-lobe epilepsy." *NEJM* 2001

SURGERY VS. MEDICATIONS

Surgery is more cost effective than continued meds in DRE

- 3 RCTs, **58-77% seizure freedom** w/ surgery vs. 7-8% w/ meds (Wiebe, *NEJM* 2001; ERSET *JAMA* 2012; Dwivedi, *NEJM* 2017)
- **NNT** for epilepsy surgery is **2** (1.5 – 2.7) (Engel 2003); NNT for Tylenol is 3.8 (Barden 2004)



MECHANISM OF PHARMACORESISTANCE

- remains incompletely understood.

Target hypothesis - changes in the ASM targets, such as ion channels, lead to decreased drug efficacy.

Transporter hypothesis - efflux pumps are thought to restrict ASM movement into cells and to be overexpressed in patients resistant to ASMs; P-glycoprotein (Pgp) is one such multidrug transporter that has been implicated in drug-resistant epilepsy.

INDICATIONS

Seizures **refractory***** to appropriate **medical management**

+ seizures seriously **limit** patient's **quality of life***

Ask - is patient impaired due to seizures?

+ **well-defined** epileptogenic **focus** not involving eloquent cortex – obsolete condition**.

*patients' **quality of life** must be included:

- high functioning pt maybe impaired with subclinical seizures.
- even as few as 2-3 seizures year may be disabling to individual whose occupation requires transportation with motor vehicle; vs. homebound patients who are not physically harmed by their seizures).
- short consciousness impairments in debilitated patients - no need to treat as treatment is more harmful;

**cortical resection must not intentionally produce significant neurologic deficit such as aphasia or hemiparesis; this is obsolete – RNS / DBS / VNS / cell therapy can treat those patients!

***there are more and more experts talking about **not waiting for refractoriness** (esp. if pathology is known to lead to refractory epilepsy) as medications have side effects.

FCD - if failed just one ASM, odds ratio that 2nd ASM will work is 1:346

Even pts on < 2 ASMs but lesion in noneloquent cortex need consider surgery

N.B. pharmacoresistance per se is not an indication for surgery!

DRUG-RESISTANT ≠ TREATMENT-RESISTANT

All patients are surgery candidates, just some are better than others

Make sure it is epilepsy and not something else (once a while a mistake is made when cardiac arrhythmias mimic epilepsy). “Not everything that shakes is epileptic seizure”

If insurance denies DBS (e.g. due to seizures not frequent enough), use SUDEP argument.

Complex partial seizures or **partial seizures with secondary generalization** are seizure types most amenable to surgical resection

Today, most “ideal” pathology for surgery is **right-sided temporal lobe epilepsy**.

N.B. temporal lobe epilepsy is **most medically refractory** but surgical results are the best!

- studies show that surgical results for frontal lobe epilepsy are also good.

Canadian Appropriateness of Epilepsy Surgery (CASES) tool - a highly sensitive guide for determining candidacy for epilepsy surgery

Jette N, Quan H, Tellez-Zenteno JF, et al. Development of an online tool to determine appropriateness for an epilepsy surgery evaluation. Neurology. 2012;79:1084–93

Online: www.epilepsycases.com

REFERRAL TO EPILEPSY CENTER

- **individuals who have continued seizures after treatment with ≥ 2 ASMs has failed** are very unlikely to achieve seizure freedom with medical treatment alone → guidelines recommend that these patients be **referred to a comprehensive epilepsy center**.

Engel J Jr, Wiebe S, French J, et al. Practice parameter: temporal lobe and localized neocortical resections for epilepsy: report of the Quality Standards Subcommittee of the American Academy of Neurology, in association with the American Epilepsy Society and the American Association of Neurological Surgeons. Neurology. 2003;60(4):538–547.

Labiner DM, Bagic AI, Herman ST, et al. Essential services, personnel, and facilities in specialized epilepsy centers-revised 2010 guidelines. Epilepsia. 2010;51(11):2322–2333.

Cross JH, Jayakar P, Nordli D, et al. Proposed criteria for referral and evaluation of children for epilepsy surgery: recommendations of the Subcommittee for Pediatric Epilepsy Surgery. Epilepsia. 2006;47(6):952–959.

Patients in whom ≥ 2 ASMs (appropriately chosen and tolerated) have failed to completely control epilepsy (sustained seizure freedom) should be considered drug resistant and referred to a comprehensive epilepsy center for surgical evaluation

National Association of Epilepsy Centers >>>

- if you see a **family doctor**, and you are continuing to have seizures, you should ask for **another opinion after 3 months**.
- if you are seeing a **neurologist**, and the seizures have not been brought under control **after 9 to 12 months**, then you should ask for a referral to a **specialized epilepsy center**.

N.B. only one-third of cases general neurologists aligned with epileptologists' opinions.

Erba G, Moja L, Beghi E, et al. Barriers toward epilepsy surgery. A survey among practicing neurologists. Epilepsia. 2012;53:35–43.

- NAEC recommends that patients whose seizures are not fully controlled after treatment for **one year** be referred to a level 3 or 4 specialized epilepsy center.

N.B. patients benefit from the advanced care provided by epilepsy centers if they continue to have seizures despite treatment with **two medications**, experience **unacceptable side effects**, are / want to become **pregnant**.

Epilepsy surgery results in significantly:

- 1) reduced mortality

Sperling MR, Barshow S, Nei M, Asadi-Pooya AA. A reappraisal of mortality after epilepsy surgery. Neurology. 2016;86:1938–44

- 2) increased quality of life

Choi H, Sell RL, Lenert L, et al. Epilepsy surgery for pharmacoresistant temporal lobe epilepsy: a decision analysis. JAMA. 2008;300:2497–505

LEVEL OF CARE AND MORTALITY

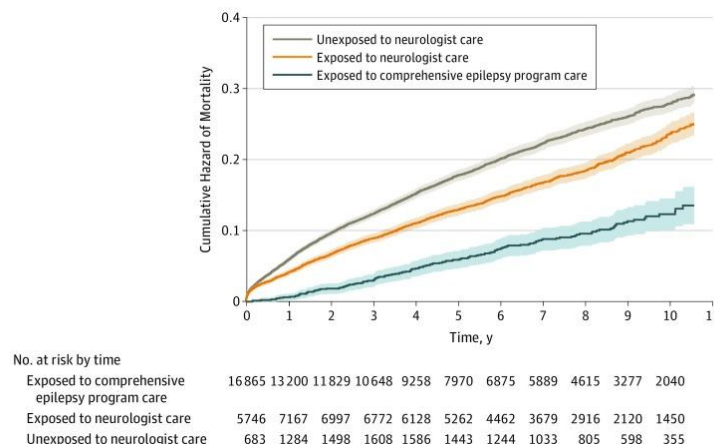
- patients with epilepsy are at an elevated risk of premature mortality.

Canadian study – all-cause premature mortality in epilepsy vs level of care

Lowerison MW, Josephson CB, Jetté N, et al. Association of Levels of Specialized Care With Risk of Premature Mortality in Patients With Epilepsy. JAMA Neurol. 2019;76:1352-1358.

- retrospective study of 23 653 adults (2002-2016).
- carefully control for age, sex, sociodemographic factors, disease severity, and comorbidities.
- 60% were not exposed to specialist neurological care, 40% received care by a neurologist, and 9% received care in the comprehensive epilepsy program.
- standardized mortality rate was 7.2% for the entire cohort; by level of care:
 - 9.4% - for those receiving nonspecialist care;
 - 5.6% - for those seen by a neurologist (HR 0.85; 95% CI 0.77-0.93);
 - 2.8% - for those seen in the comprehensive epilepsy program (HR 0.49; 95% CI 0.38-0.62).
- exposure to specialist care** (non-neurologist vs. neurologist vs. comprehensive epilepsy program) is associated with an incremental reduction in the risk of premature mortality from all-causes. Those referred to a comprehensive epilepsy program received the greatest benefit.
 - N.B. importance of early referral!
- possible causes of the observed benefit:
 - expediting an accurate initial diagnosis
 - selecting the right antiepileptic drug at the first opportunity
 - experience in using rational polytherapy
 - greater chances of receiving a new drug with fewer adverse effects
 - access to epilepsy surgery

Mortality Over Time



UNDERUTILIZATION OF EPILEPSY SURGERY

Underutilization of epilepsy surgery is a public health crisis!

Only 15% of epilepsy surgeons survive only doing epilepsy surgery!

- 30-40% of patients with epilepsy have DRE and therefore are surgical candidates.

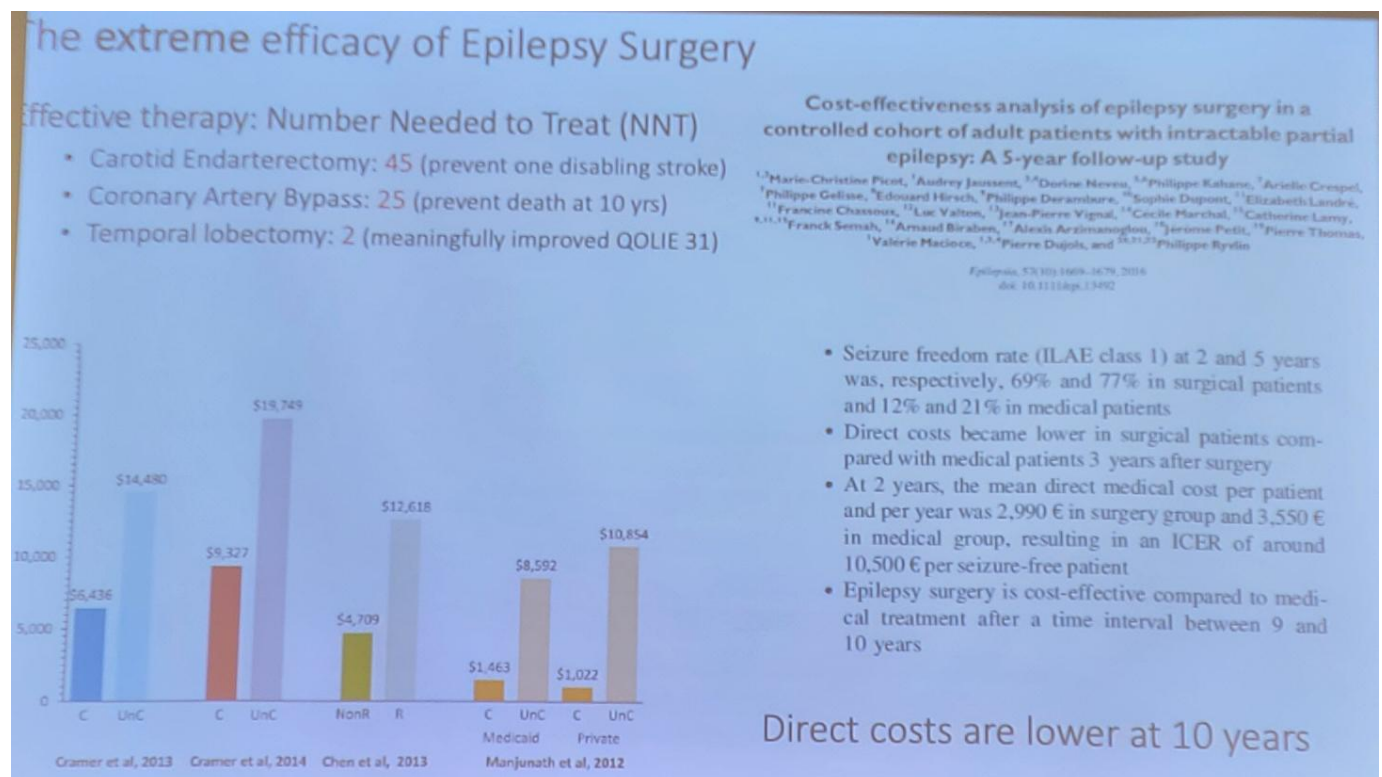
- 75% of patients with DRE are not referred to an epilepsy specialist.
 - nonreferral by **neurologists** is greater than by PCPs.
- < 3% of surgical candidates receive surgery (4000 epilepsy surgeries performed per year out of 150,000 potential surgical) and number of epilepsy surgeries is decreasing.
 - 1.2 M people in US are surgery candidates but only 6.5K get operated annually - **Public Health Crisis!**
- candidates - 2003 joint position paper from the American Academy of Neurology, American Association of Neurological Surgeons, and the American Epilepsy Society); subsequent studies have demonstrated no change in epilepsy surgery utilization.
- average time from diagnosis to surgery for a medically refractory patient is reported to be around **18 years**.

Now that we know how great surgery is, what do you think is the average time from diagnosis of refractory epilepsy to referral for surgery?

"Surgical treatment for epilepsy is arguably the most underutilized of all proven effective therapeutic interventions in the field of medicine." Engel 2003

20 Years

Ryvlin et al., 2008; de Tisi et al. 2011



BARRIERS FOR TIMELY REFERRALS

- referral rates from physicians and approval rates by patients for presurgical assessment remain constantly low.

- neurologists from several countries defined medically refractory epilepsy as a failure of 3-5 ASM monotherapy trials = delaying referral to epilepsy surgery
Hakimi AS, Spanaki MV, Schuh LA, Smith BJ, Schultz L. A survey of neurologists' views on epilepsy surgery and medically refractory epilepsy. Epilepsy Behav. 2008;13:96–101.
Erba G, Moja L, Beghi E, et al. Barriers toward epilepsy surgery. A survey among practicing neurologists. Epilepsia. 2012;53:35–43.
- in two recent surveys from Canada and Italy, > 50% of affected patients stated that they considered epilepsy surgery to be very dangerous and > 60% saw it as a last resort treatment only.
Erba G, Messina P, Pupillo E, Beghi E; OPTEFF Group. Acceptance of epilepsy surgery among adults with epilepsy—what do patients think? Epilepsy Behav. 2012;24:352–8
Hrazdil C, Roberts JJ, Wiebe S, et al. Patient perceptions and barriers to epilepsy surgery: evaluation in a large health region. Epilepsy Behav. 2013;28:52–65.

Study of 185 German patients in tertiary epilepsy care center (Charité University Hospital) - how many patients with intractable focal epilepsy were recommended by their epileptologists to undergo presurgical evaluation with noninvasive v EEG and how many patients followed this recommendation.

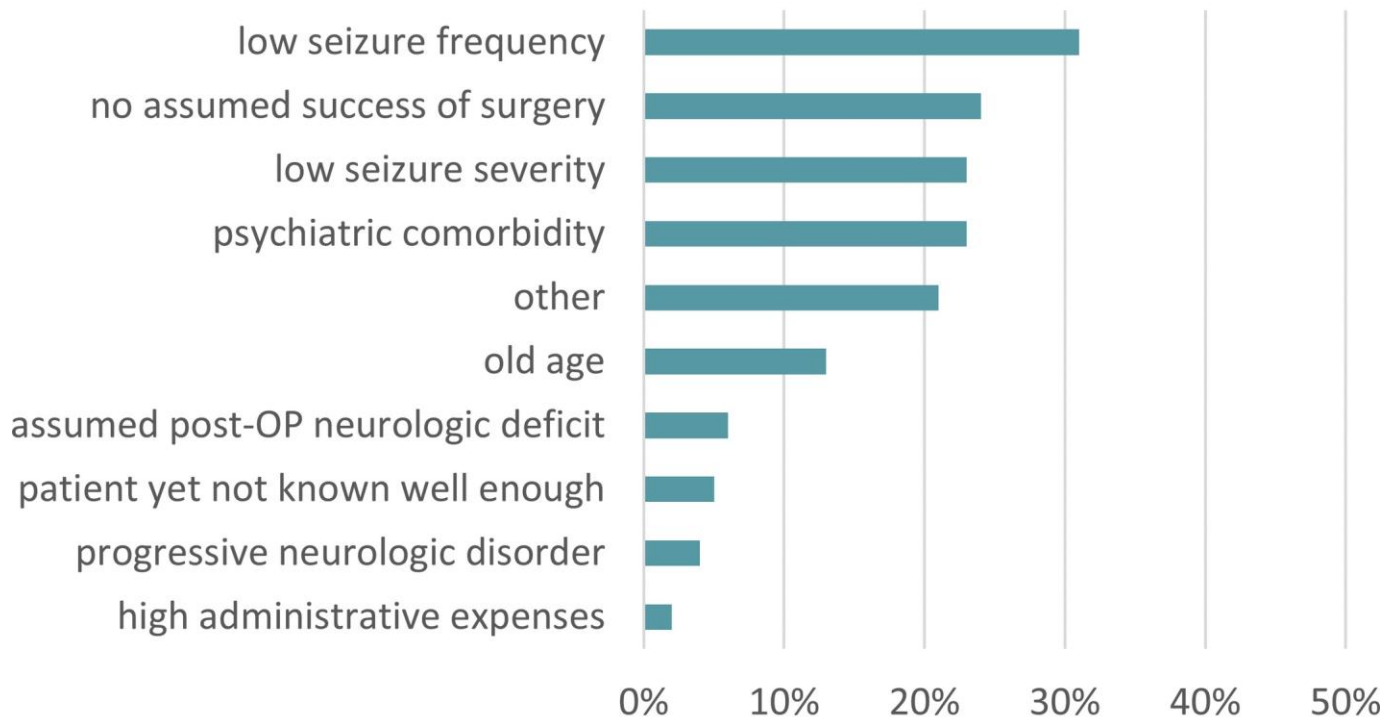
Mirja Steinbrenner et al. Referral to evaluation for epilepsy surgery: Reluctance by epileptologists and patients. Epilepsia. 17 January 2019

- high decision rates against presurgical assessment:
 - 43% were recommended presurgical evaluation by their epileptologists, and only 30% of these patients consented (79% of these had vEEG).
 - 10% of all patients actually underwent noninvasive presurgical assessment, and 5% of these eventually proceeded to resection.

Epileptologists

- the most frequent reason for nonreferral by epileptologists - low seizure frequency (31%):

Epileptologists' reasons for non-referral



Comments:

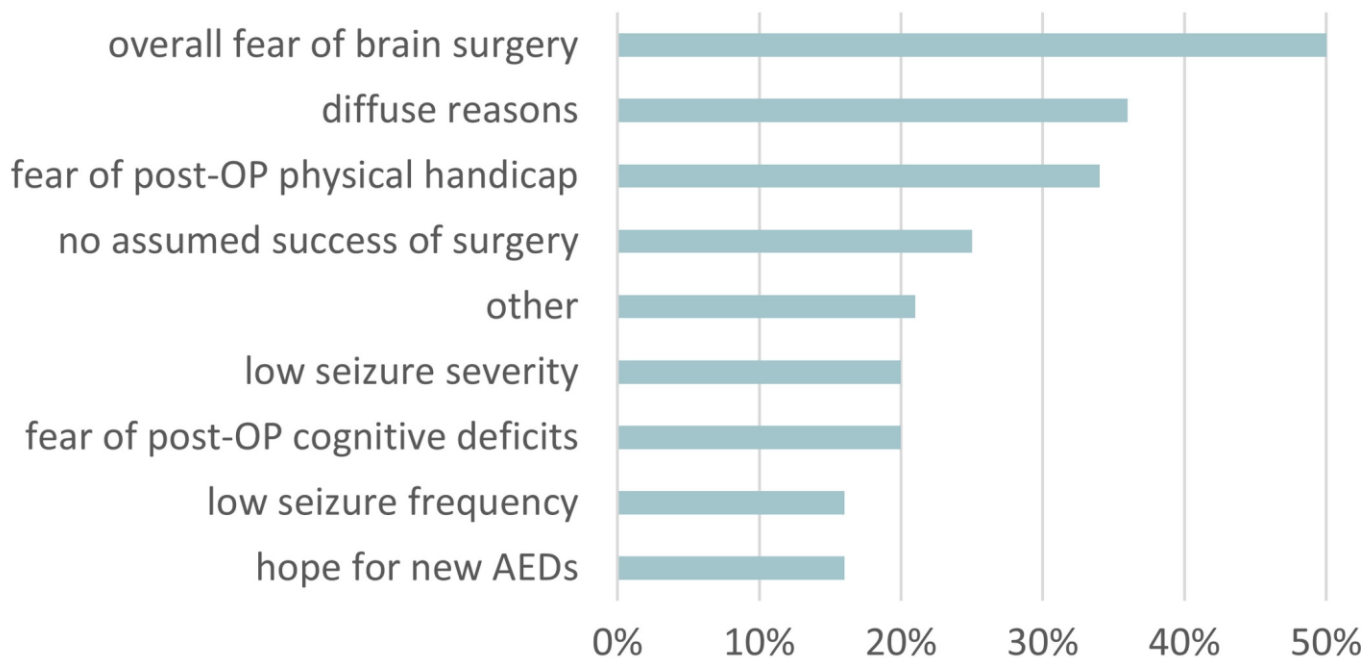
- low seizure frequency and severity may be reasonable in some cases (e.g., patients who have one or two focal seizures with impaired awareness per year or who have focal aware seizures exclusively) ;
- severe intellectual disability - even patients with an IQ of 50-69 have been reported to become seizure-free in 37% of cases.

- age > 60 years is not associated with a more unfavorable outcome after seizure focus resection, and patients of this age group should also be offered surgery if eligible
- variables independently associated with nonreferral by epileptologists:
 - 1) older age of patients (odds ratio [OR] = 1.03)
 - 2) no previous evaluation for epilepsy surgery (OR = 4.04)
 - 3) presence of legal guardianship (OR = 4.29)
 - 4) ≥ 11 years of professional experience (OR = 4.62) - reflects higher vigilance of younger generation of epileptologists due to an update in training on the benefits of epilepsy surgery.

Patients

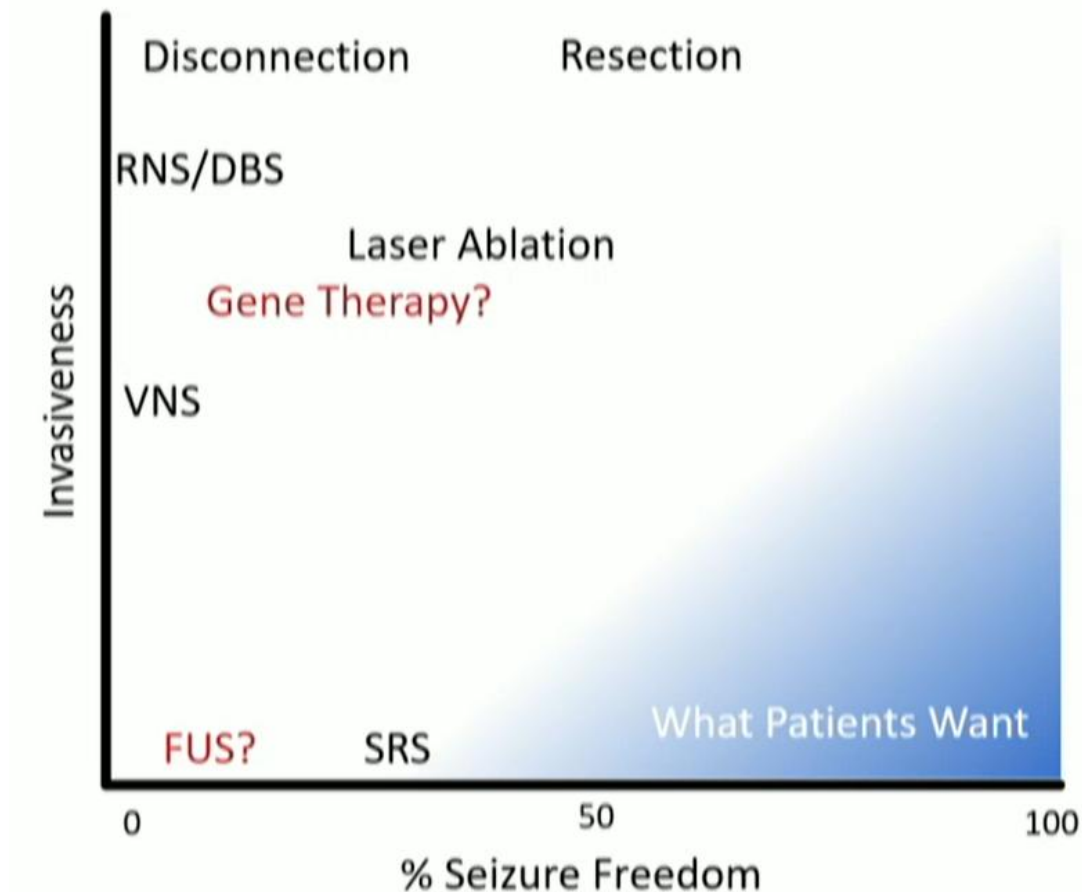
- patients declined most often due to overall fear of brain surgery (50%):

Patients' reasons for rejecting referral



Other reasons:

- wanting to wait for the evaluation until their children were older in case something “bad” happened to them.
- being content with the current seizure situation and not wanting any change (e.g. seizure freedom may result in reduction or loss of disability pension).
- independent predictors for patients' rejection of presurgical evaluation:
 - 1) older age (OR = 1.08)
 - 2) lifetime number of antiepileptic drugs ≥ 5 (OR = 4.47)
 - 3) focal aware seizures (OR = 4.37)
 - 4) absence of focal seizures with impaired awareness (OR = 11.24).



TIMING OF SURGERY

Pediatric aspects on cognitive outcomes – see below >>

Surgery should be performed as early as possible!!! (surgery is no longer therapy of last resort)

Epilepsy is progressive disease! (kindling, new foci, network damage)

- intractable seizures, the longer they persist, portend poorer prognosis for seizure remission and psychosocial outcome (e.g. "mirror" foci can become established as independent foci).
- strategy of *trying all combinations of drugs* is **not acceptable** in syndromes known to have excellent chances of benefiting from surgery.

Kindling - in popular language: teaching to seize independently.

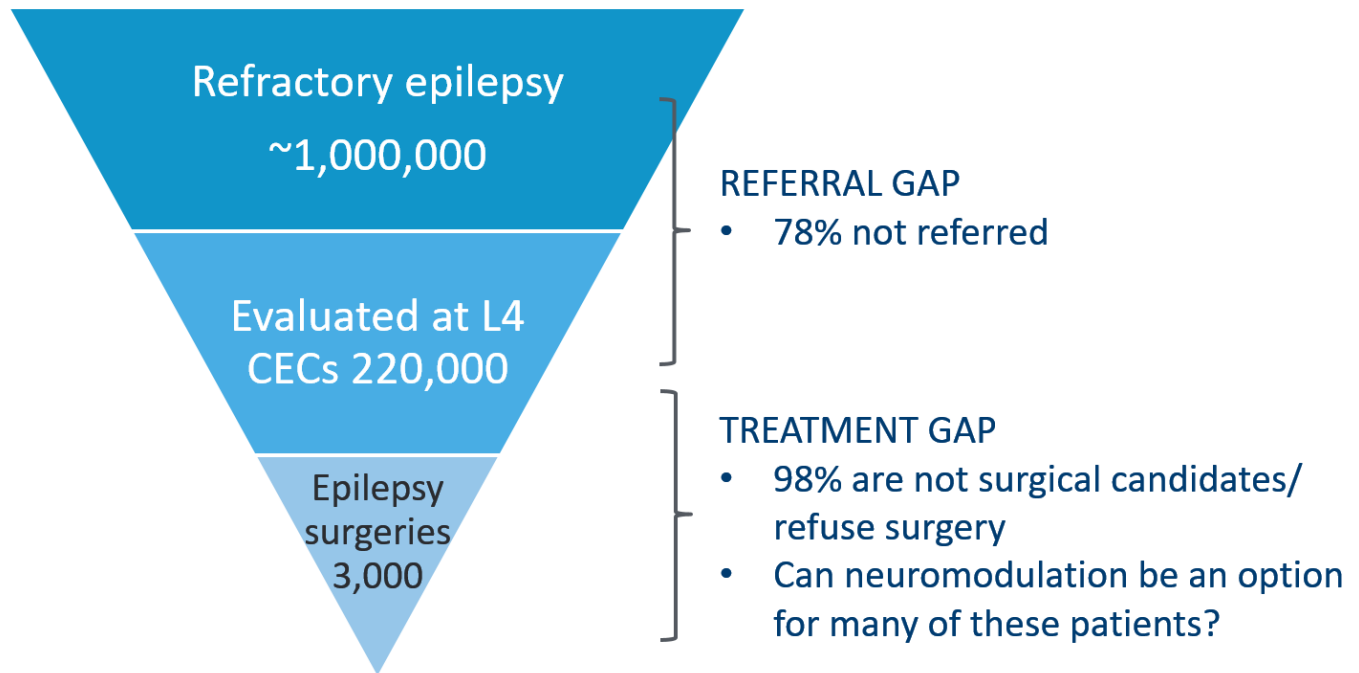
Despite reported success, surgery for pharmacoresistant seizures is often seen as a last resort. Patients are typically referred for surgery after 20 years of seizures, often too late to avoid significant disability and premature death.

- neurosurgery in patients' minds is notorious for poor outcomes and mortality in severe TBI, malignant tumors, vascular pathology.
- patients don't know that neurosurgery for epilepsy is very safe (made possible by technological advances and multidisciplinary team work).

Tell patients about the risk of SUDEP in uncontrolled epilepsy! It is not to scare your patient; it is your obligation ("I am obligated to tell you about SUDEP, the same as I tell my patients about diagnosis of cancer and life expectancy")

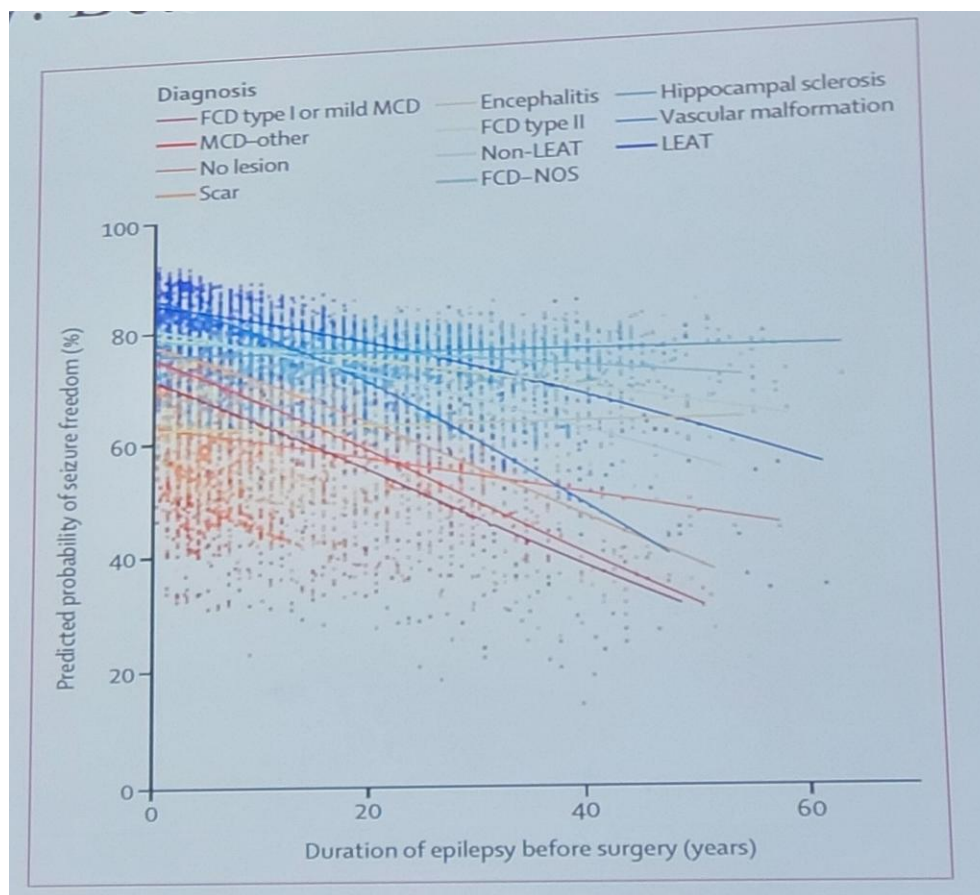
Given results of Brodie (2000) and Schiller (2008) studies, timing of surgery should be individualized:

- for **MTS** in which the seizure-free rate is high and the risk low after both ablative and resective surgery, surgery may be considered *earlier, that is, after 2 drug failures*.
- when the chance of seizure freedom after surgery is possibly lower, such as in the **absence of MTS**, and risk is correspondingly higher in the setting of preserved neurocognitive functions, ablative surgical treatment might be delayed until *after a more exhaustive regimen of ASMs or neuromodulation have been attempted*.



Do not delay neuromodulation – involved networks may get damaged as time goes.

Earlier surgery = Better outcome



Lamberink et al. Lancet Neurology 2020

STRATEGIES TO IMPROVE PATIENT AGREEMENT

- we work in strong **multidisciplinary team**.
- reassure - we **do not remove normal brain tissue**.
- when counseling patient, **talk about complications at the last step** (cf. car sales people – do not start with “with the car you can get into accident and die”).
- "do not refer for surgery; refer for comprehensive multidisciplinary evaluation" - evaluation for surgery should be done earlier; surgery itself can happen later.
- **PCPs** are afraid of seizures so they refer; need to target **general neurologists** - they are comfortable with treating seizures with medicines.

CONTRAINDICATIONS

- 1) benign partial childhood epilepsy
- 2) significant noncompliance
- 3) progressive neurologic diseases
- 4) severe concurrent medical illness

Relative contraindications:

- 1) generalized epilepsy (indication for corpus callosotomy or VNS or DBS)
- 2) mental retardation / low IQ (< 70)

- 3) psychiatric disease (psychosis or other serious psychiatric disorder)
- 4) multiple seizure types arising from different brain regions (unless one seizure type is most frequent and disabling)
- 5) coexistence of epileptic and nonepileptic seizures.

Not contraindications:

- 1) epileptogenic focus in *dominant hemisphere*
- 2) epileptogenic focus in *eloquent cortex*
- 3) *bilateral* or *multifocal* epileptogenic foci
- 4) *neurological deficits* on examination

DEFINITIONS OF SURGICAL ZONES

Epileptogenic lesion s. Seizure Onset Zone (SOZ) – lesion / area able to *produce seizures*; needs to be included in resection.

Epileptogenic zone (EZ) (ictal onset zone + ictal forming zone) for topography see p. E1 >>
– cortical area that *needs to be resected / disconnected / thermo-coagulated / thermo-ablated / desynchronized by multiple transections* in order to make patient seizure free.

- it *cannot be determined directly* (unless surgery is done and patient became seizure free) but it is deduced by outlining related cortical areas, including the irritative zone, the seizure onset zone, the epileptogenic lesion, the symptomatogenic zone, and the functional deficit zone.
- may be *larger than lesion* visible on MRI (includes lesion and surrounding margin, sometimes extends beyond brain lobe boundaries; e.g. cavernoma itself is not epileptogenic but adjacent hemosiderin-lade cortex is) – role of **intracranial EEG**.

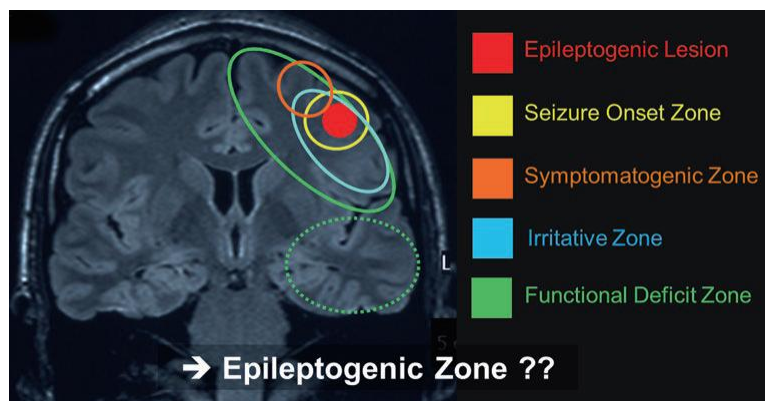
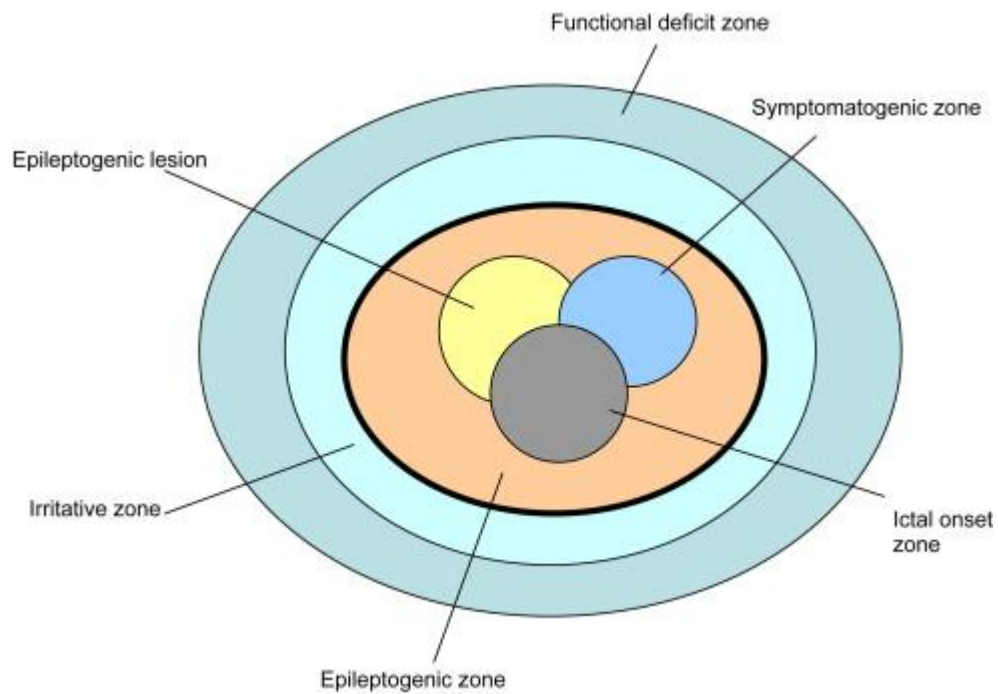
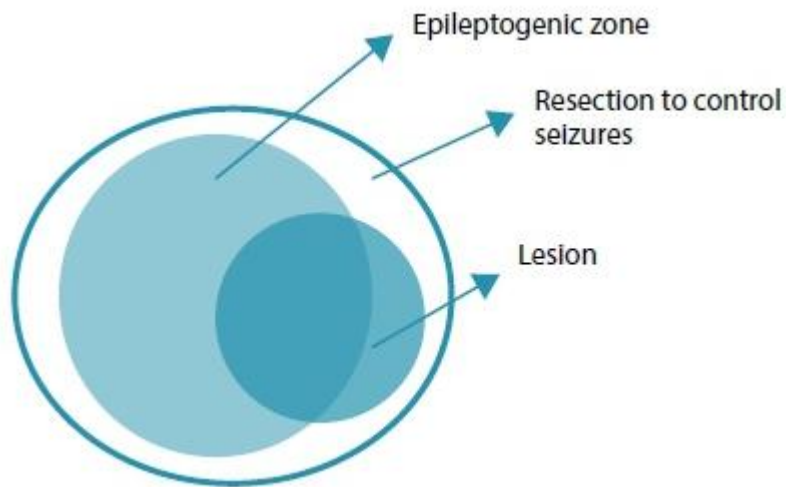
N.B. it is essential to resect EZ and not just SOZ

Irritative zone – cortical area generating epileptiform discharges but whose *resection is not necessary**; usually *larger than epileptogenic zone*.

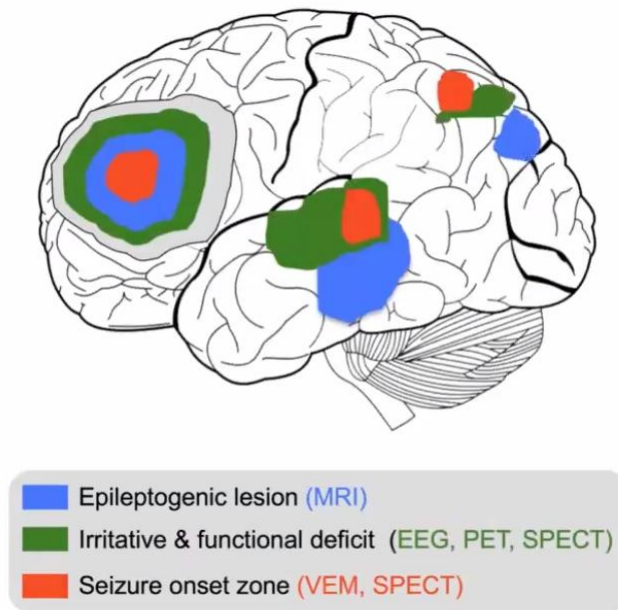
*irritative spikes cease after surgical resection of epileptogenic zone!

Symptomatogenic zone – cortical area that *produces clinical symptoms* but whose removal is not necessary (e.g. seizures may begin silently in frontal lobe and produce typical temporal lobe complex partial seizure when discharge has spread there); cortical stimulation studies have shown that often region producing auras is *much larger than epileptogenic zone*.

Functional deficit zone – cortical area showing *hypometabolism* on **FDG-PET**; *much larger than epileptogenic zone* (e.g. in medial TLE, hypometabolism involves both medial and lateral temporal lobe cortex ± subtle regions of hypometabolism in frontal lobe).



Zones may or may not overlap:



NODE VS. NETWORK DISEASE

There were/are proponent experts in each theory:

In the 1950s, Penfield and Jasper established **surface electrocorticography** as the mainstay for defining an “**epileptogenic focus**,” and Bailey and Gibbs wrote that “surgical eradication of focal seizure activity was comparable to eradicating a tumor.”

vs.

SEEG philosophy (originated by Talairach and Bancaud) focuses on determining the regions of cortex generating clinical manifestation of the seizure whereby the chronologic occurrence of ictal clinical signs (semiology) is crucial for elucidating “**anatomo-electro-clinical**” **organization of seizures**.

Paradigm shift in 2020s: EPILEPSY SURGERY AS NETWORK SURGERY!

(Epileptogenic zone → Epileptogenic network)

old school - if a resection is performed and the patient is not seizure free, the interpretation is that one “didn’t get enough” or did not find “the right focus”;

new school - **take the seizure network offline**, that is, to prevent emergence of seizures by **disrupting the critical nodes of seizure organization** (emphasis is shifted from **resection & seizure freedom** to **modulation & improved quality of life**, with the ultimate goal of arresting seizures indefinitely).

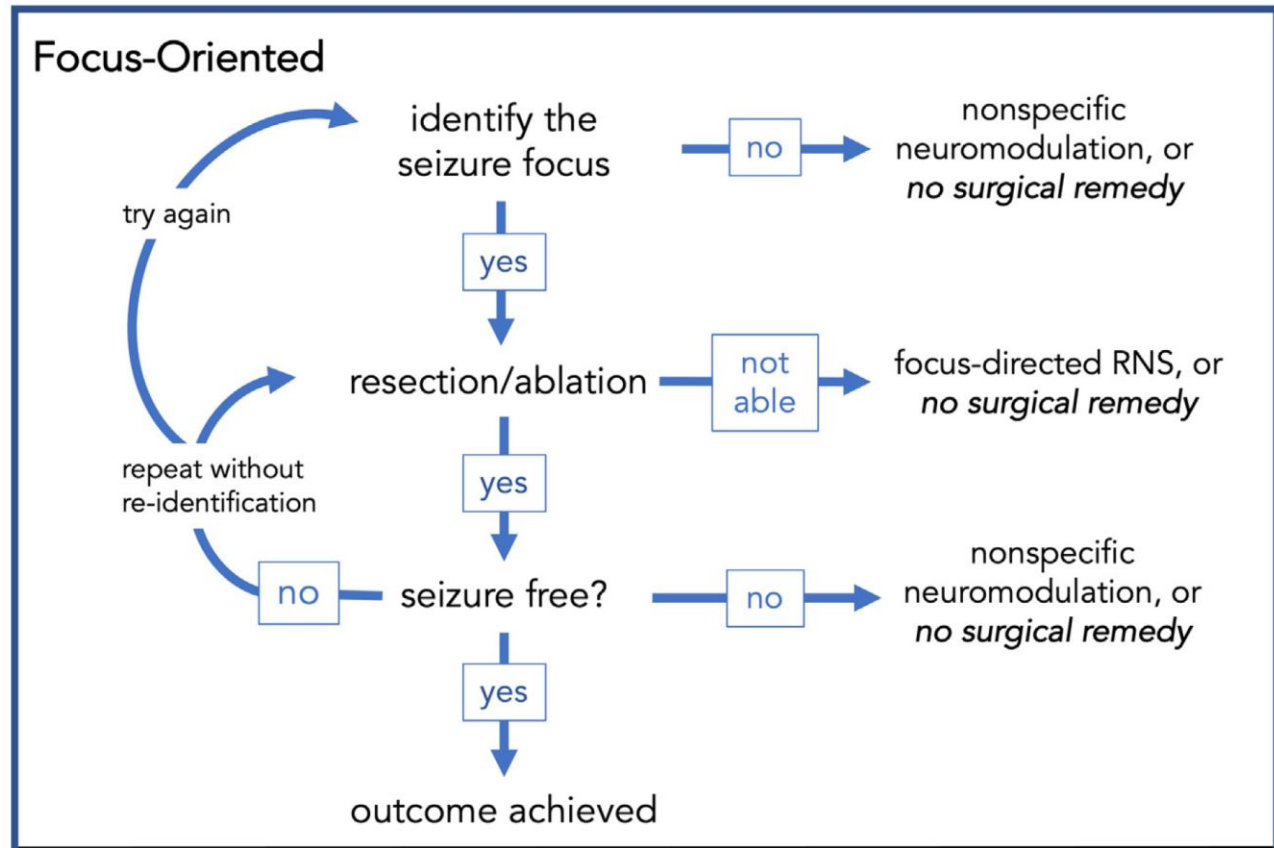
New questions for intracranial monitoring:

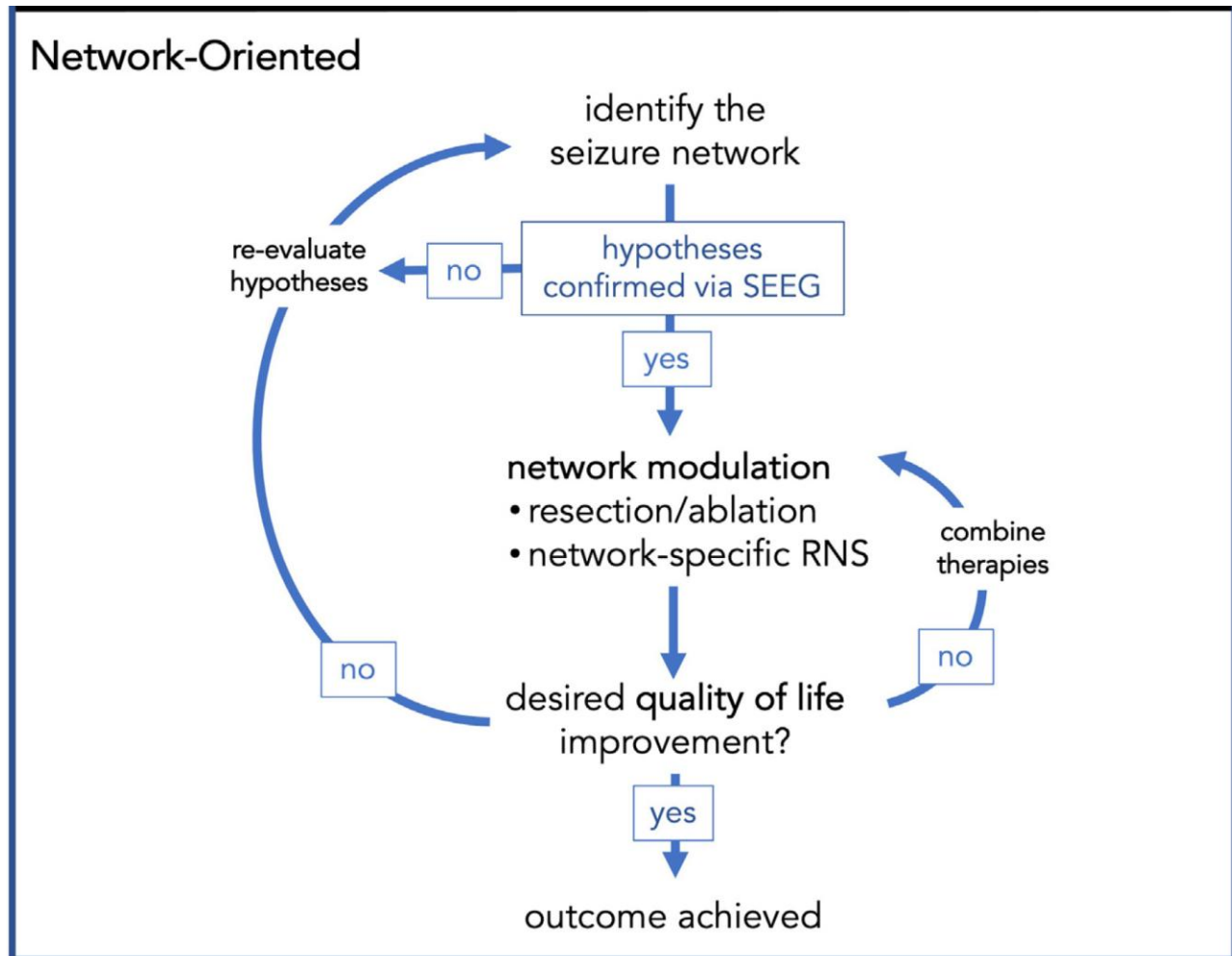
1. Whether a resection can be accomplished
2. How to take the network offline

- **generalized epilepsy** – EZ (node) is entire brain (**meganode**)
 - even generalized epilepsies likely have a focal trigger.
- why some patients become **seizure free after callosotomy** if it is node disease - two nodes across corpus callosum **can no longer synchronize** oscillations.
- **focal epilepsy** – small local network (**mininetwork**).

Definition for focal epilepsy: seizure onset at ≤ 4 contiguous contacts
--

- focal epilepsy - originates in network within ONE hemisphere
- treatment goal – resect / ablate **primary source** and neuromodulate **network nodes** (“take seizure network offline” – role of SEEG exploration).
In 2020s emphasis is shifted from *resection and seizure freedom* to *modulation and improved quality of life*, with the ultimate goal of arresting seizures indefinitely.





Richardson 2022

- downfall of approaching epilepsy surgery solely from the perspective of focus hunting: if a resection is performed and the patient is not seizure free, the interpretation is that one “didn’t get enough” or did not find “the right focus.”

Common Epileptic Pathologies

- **hippocampal sclerosis** is #1 among adults (44.5% surgical specimens; only 15% in children)
- **malformations of cortical development** are #1 among children (39.3% surgical specimens with focal cortical dysplasia occupying 70.6%).
- **tumors** (mainly ganglioglioma) are #2 in all age groups.
- **vascular malformations** (mainly cavernomas) - found in 6.1% of specimens.
- TBI, intracranial infections, neurosurgical procedures may leave **glial scars (gliosis)** - found in 4.9% of specimens - hyperintense on T2 and often associated with cortical atrophy.
- **infections** - Rasmussen's encephalitis, herpes simplex encephalitis, neurocysticercosis (most common cause of infection-related epilepsy worldwide).

Presurgical Evaluation (Non-Invasive, Phase I)

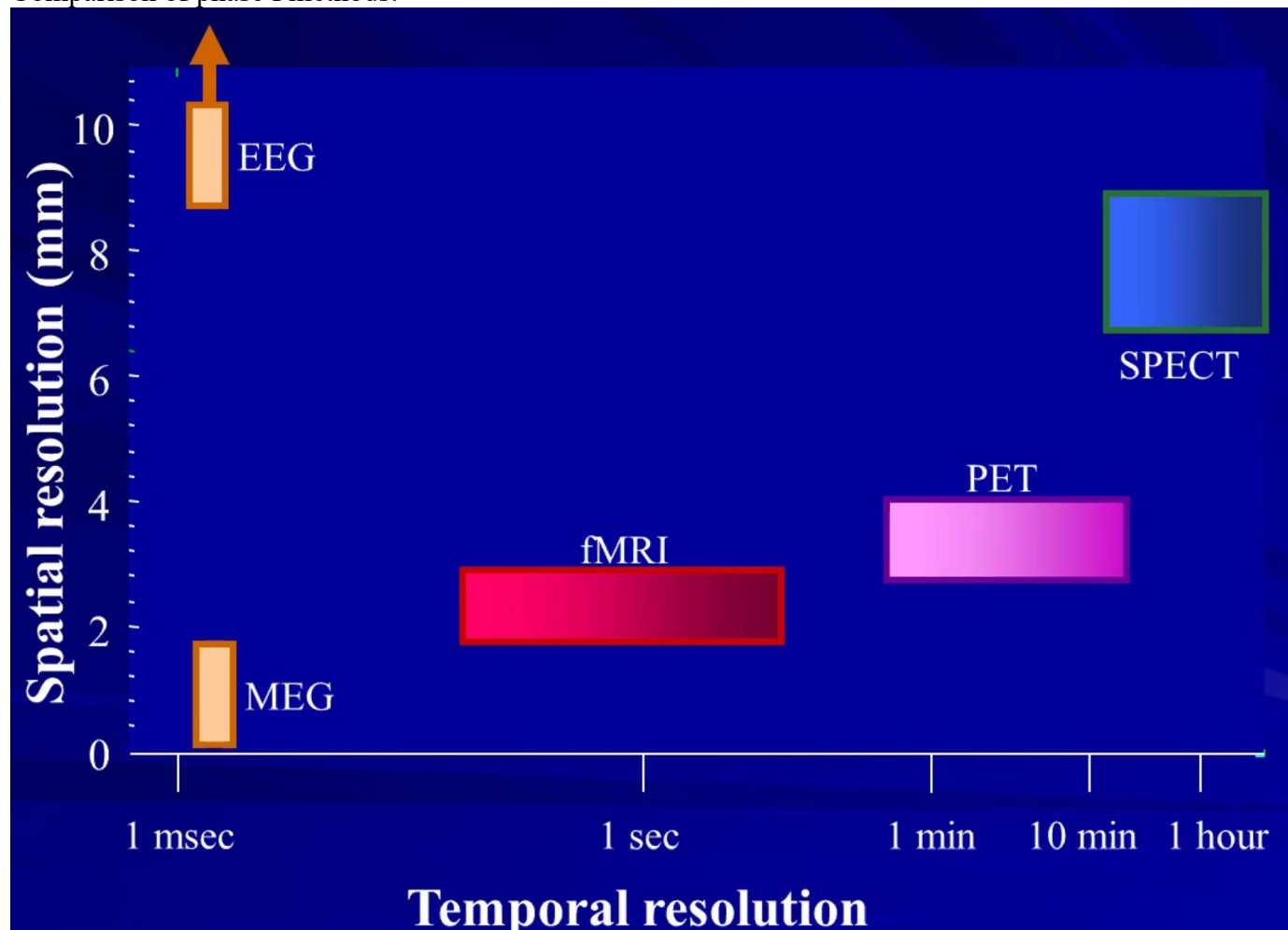
- goal is to define **EPILEPTOGENIC ZONE**.

Surgical treatment is presently limited by our *inability to localize epileptogenic focus*.

Four pillars:

1. Electrophysiology
2. Imaging
3. Neuropsychology
4. Genetic testing

Comparison of phase I methods:



- all methods can be done for patients of all age groups.

SCALP (SURFACE) EEG

All surgical candidates should begin presurgical evaluation with **EEG-audio-video monitoring** to record actual seizures (*weaning from anticonvulsants* may be necessary) – so called PHASE I MONITORING.

N.B. *seizures* may begin in areas distant from (or even contralateral to) location of *interictal* epileptiform activity - *ictal discharges* are most reliable means of localization!

- interictal EEG is normal in 30% cases; remaining of cases often have nonspecific findings.
- since it is impossible to record from all cortical & subcortical structures from which seizures may arise, exact onset may not involve recording electrodes until spread of discharge has occurred (i.e. early ictal changes are often not identified in scalp recordings).
- foci in *mesial* or *basal* cortical areas are particularly apt to escape detection; *extratemporal seizures* are more difficult to localize (e.g. epilepsy of occipital or frontal lobe origin may have interictal activity at temporal region).
- **modified electrode placements, dense array, and semi-invasive techniques** increase yield.

HIGH DENSITY EEG

- the same as MEG for temporal seizures; for extra-temporal - MEG is better.

NEUROIMAGING (ROUTINE)

see also p. E1 >>, p. E9 >> (temporal lobe epilepsy, diagnosis)

CT

- it is now replaced by MRI (only need for high-resolution CT is for intraoperative neuronavigation planning)
- seizure focus may enhance with IV contrast shortly following a seizure.
- small cortical lesions, particularly in orbitofrontal and medial temporal regions, are not detectable on CT.

MRI

All patients with **focal** epilepsy should undergo MRI without and with IV contrast.

MRI is usually not indicated for genetic and self-limited epilepsy syndromes (such as benign childhood epilepsy with centrotemporal spikes).

Lesional MRI = less chances for medication success, more chances for surgical success

- desirable ≥ 3 Tesla
N.B. higher field strength – more geometrical distortion, motion and susceptibility artifacts (caused by air-filled cavities such as nasal cavities)
- 7T detects lesions in 22-29% of 3T MRI-negative cases (MAP postprocessing increases it to 43%)
- there is software (e.g. FreeSurfer) for automated cortical thickness analysis – esp. useful for focal cortical dysplasia (FCD) detection.

N.B. MRI and ictal EEG can be discordant or negative in up to 40% of potentially preoperative cases!

ILAE members can take hands-on practice sessions to review major neuroanatomical landmarks:

<https://www.epilepsy-imaging.org/#/>

ILAE PROTOCOL**HARNESS-MRI PROTOCOL**

- mandatory and optional sequences for epilepsy.

Bernasconi et al., 2019

Mandatory sequences

1. **3D magnetization-prepared rapid gradient-echo** (MPRAGE, Siemens), Spoiled gradient-echo (SPGR, GE), Turbo field echo (TFE, Phillips)
2. **3D fluid attenuation inversion recovery** (FLAIR) - very useful for evaluation of epileptogenic lesions; sagittal FLAIR provides a complete antero-posterior view of signal distribution along the length of the hippocampus and parahippocampus.
N.B. FLAIR images are not sensitive to epilepsy-associated pathology in **neonates and infants < 2 yo**, as the myelination process is not yet complete.
3. **2D coronal turbo spin echo (TSE)** with acquisition plane perpendicular to the hippocampus long axis) – T2 high in-plane resolution for optimal visualization of hippocampal volume, shape, and internal structure on coronal cuts.

Optional sequences

1. **3D gadolinium-enhanced T1** – to rule out tumor-like lesions, vascular malformations, or infectious processes.
2. **3D susceptibility weighted imaging** – T2 sensitive to iron deposits, blood products and calcifications.

Surgery planning sequences

1. **3D gadolinium-enhanced T1** – to plan trajectories avoiding vasculature.
2. **3D FGATIR** – to plan ANT DBS.
3. **M2PRAGE** – to plan CM DBS / RNS.

MRI-NEGATIVE CASES

- even experienced observers may miss small and subtle lesions - preferably, an **observer with expertise in epilepsy imaging** should review the scans.
- it is often necessary to **repeatedly review the MRI**, in the context of new information such as PET, EEG, MEG.
- most commonly missed etiology – FCD (esp. bottom of sulcus); then temporal pole encephaloceles, hypothalamic hamartomas and subtle double cortex.
- if truly MRI-negative, proceed with **advanced outpatient imaging: MEG, PET** (if nonrevealing → **inpatient imaging ictal SPECT**).
Positive result from advanced imaging often requires **verification by depth electrode SEEG recordings**.

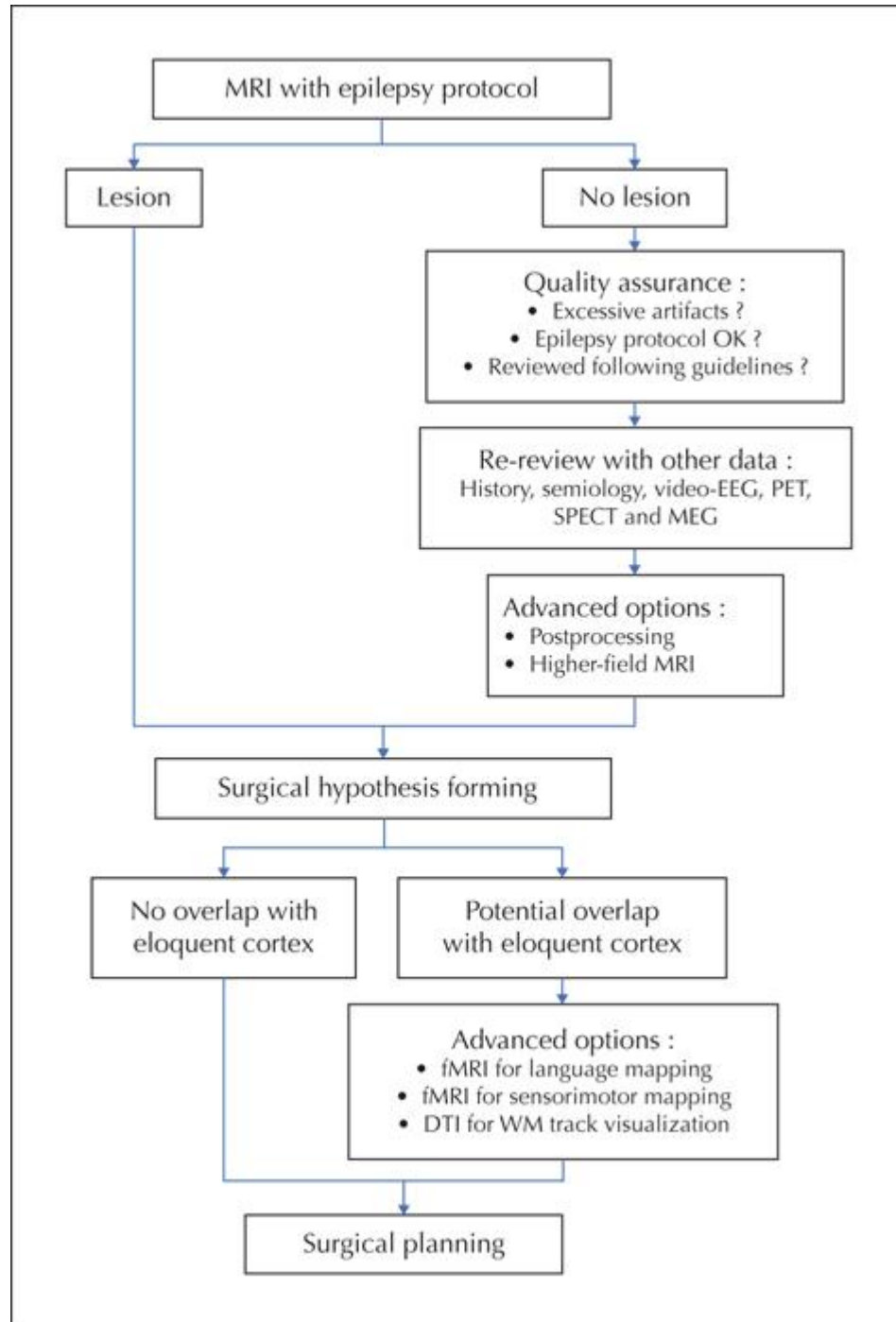
Methods to increase MRI sensitivity

7 Tesla scanners

Quantitative MRI

Voxel-based morphometry - MRI postprocessing to direct radiologist to look closer at certain areas

NEUROIMAGING (ADVANCED)



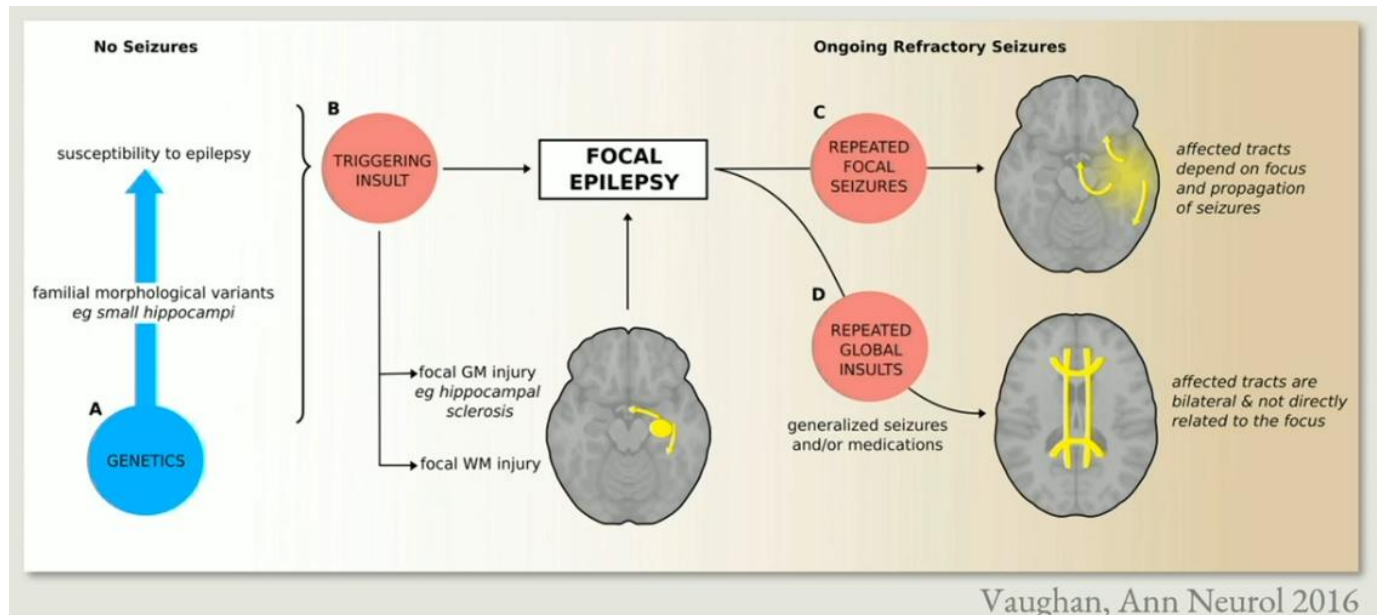
Special MRI postprocessing

TRACTOGRAPHY (DTI)

- displays white matter (WM) tracts.

Is WM microstructure abnormality important predisposing factor for epilepsy? - not known yet.

- **WM integrity correlates with cognitive abilities**, i.e. cognition depends on network efficiency.
 - *abnormally low FA correlates with lower cognitive scores* (cognitive phenotypes can be derived from white matter characteristics).
- WM microstructure is very heritable.
- WM microstructure changes with aging!
- TLE - white matter is affected in entire brain (not just temporal lobe); even siblings (of TLE patients) have reduced WM volume

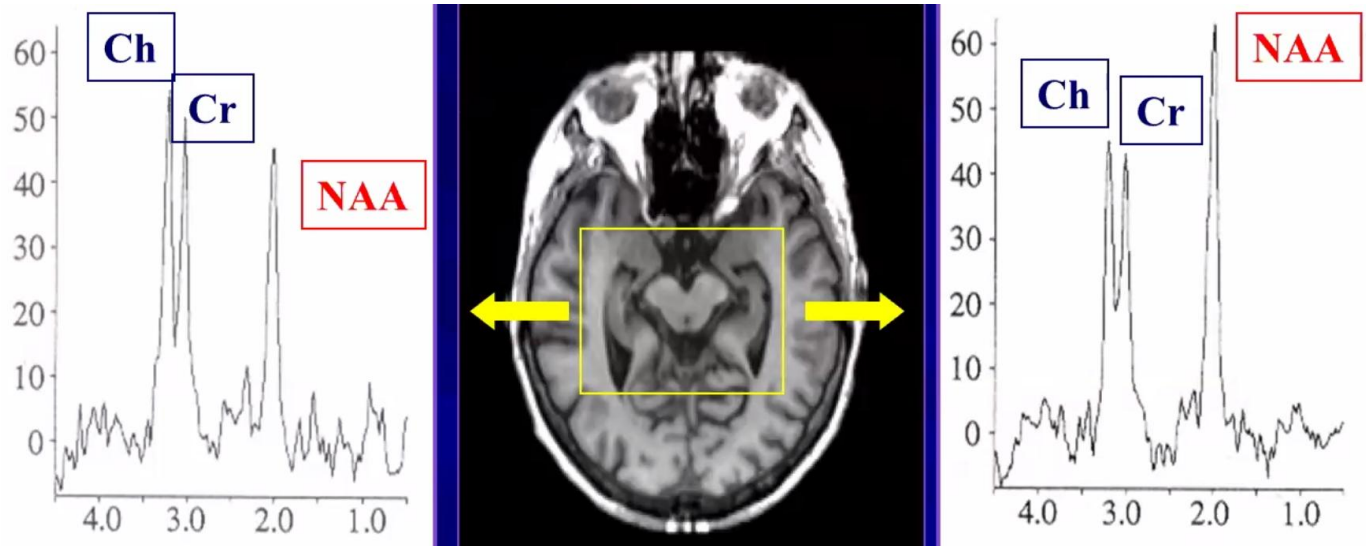


HIPPOCAMPAL VOLUMETRY

CORTICAL UNFOLDING

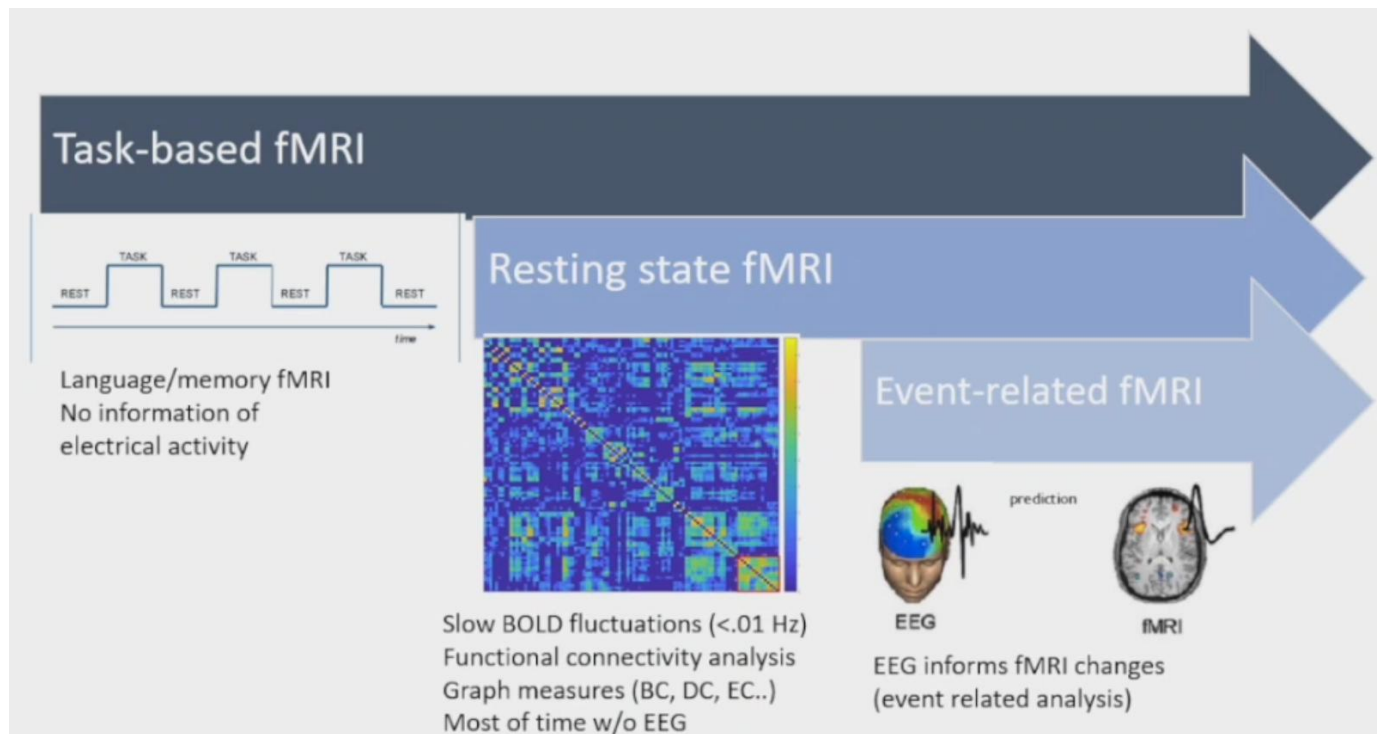
¹HMR (Proton) Spectroscopy

Helps with pathology lateralization:



fMRI

- used to determine language dominant side; less commonly – relation of eloquent cortex to EZ

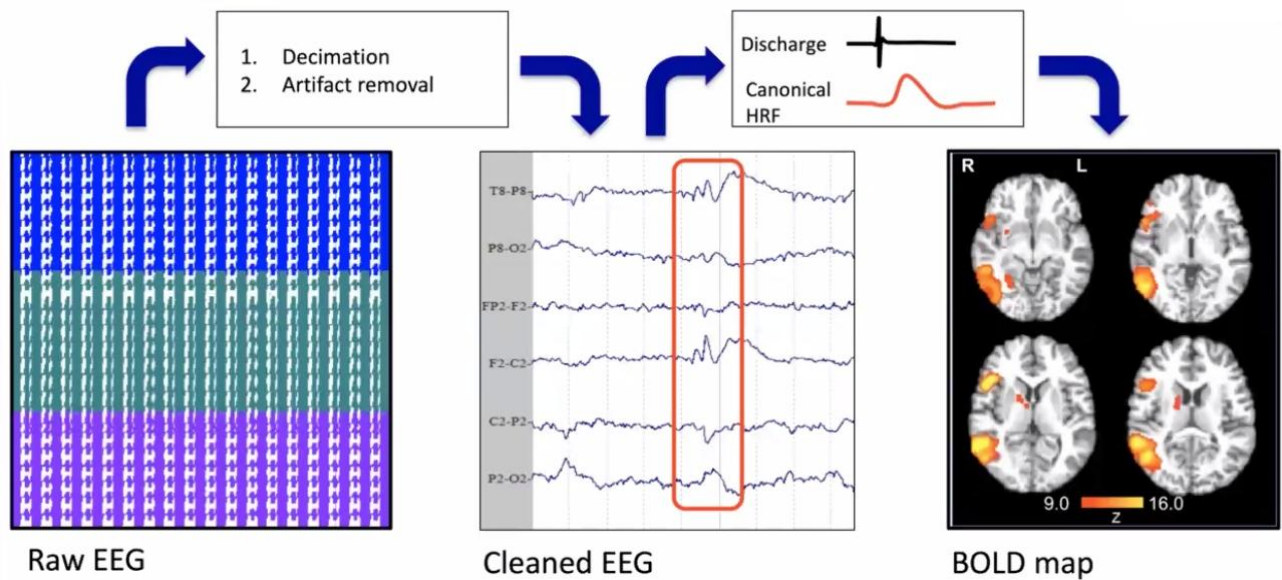


RESTING STATE FMRI

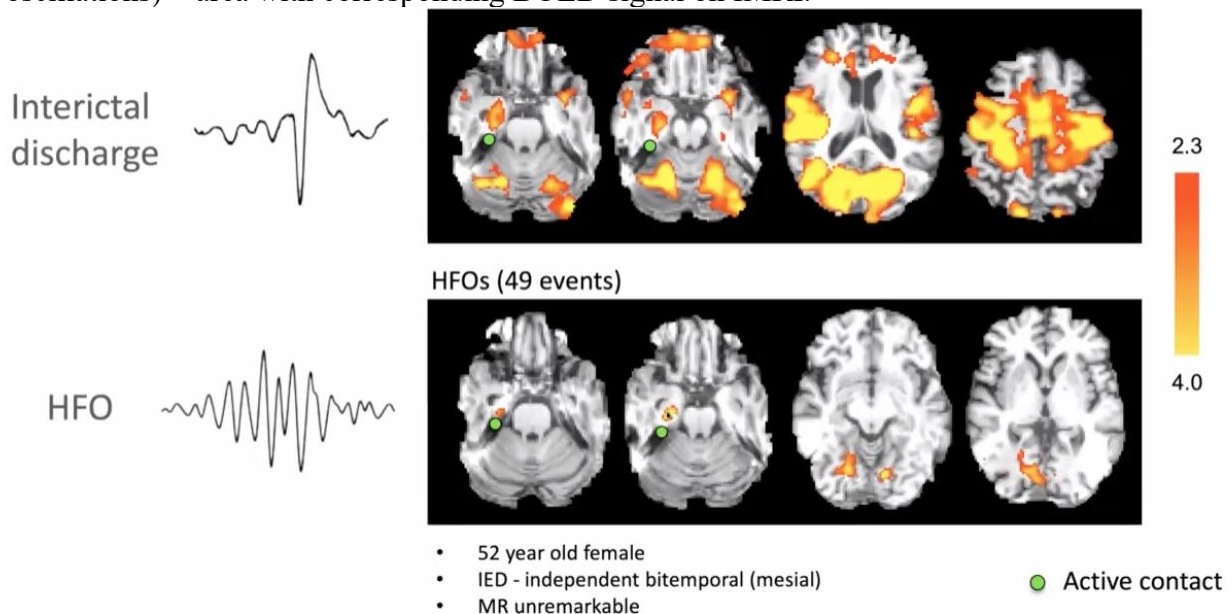
- network analysis; becoming popular for developing surgical strategies and predicting surgery outcomes.
- TLE patients have different subcortical connectivity seen on rs-fMRI (“recovers” after successful surgery).

EEG-FMRI (EVENT-RELATED FMRI)

- combines scalp EEG (good **temporal** resolution) and fMRI (good **spatial** resolution) - especially useful for investigating **deep-seated foci**.



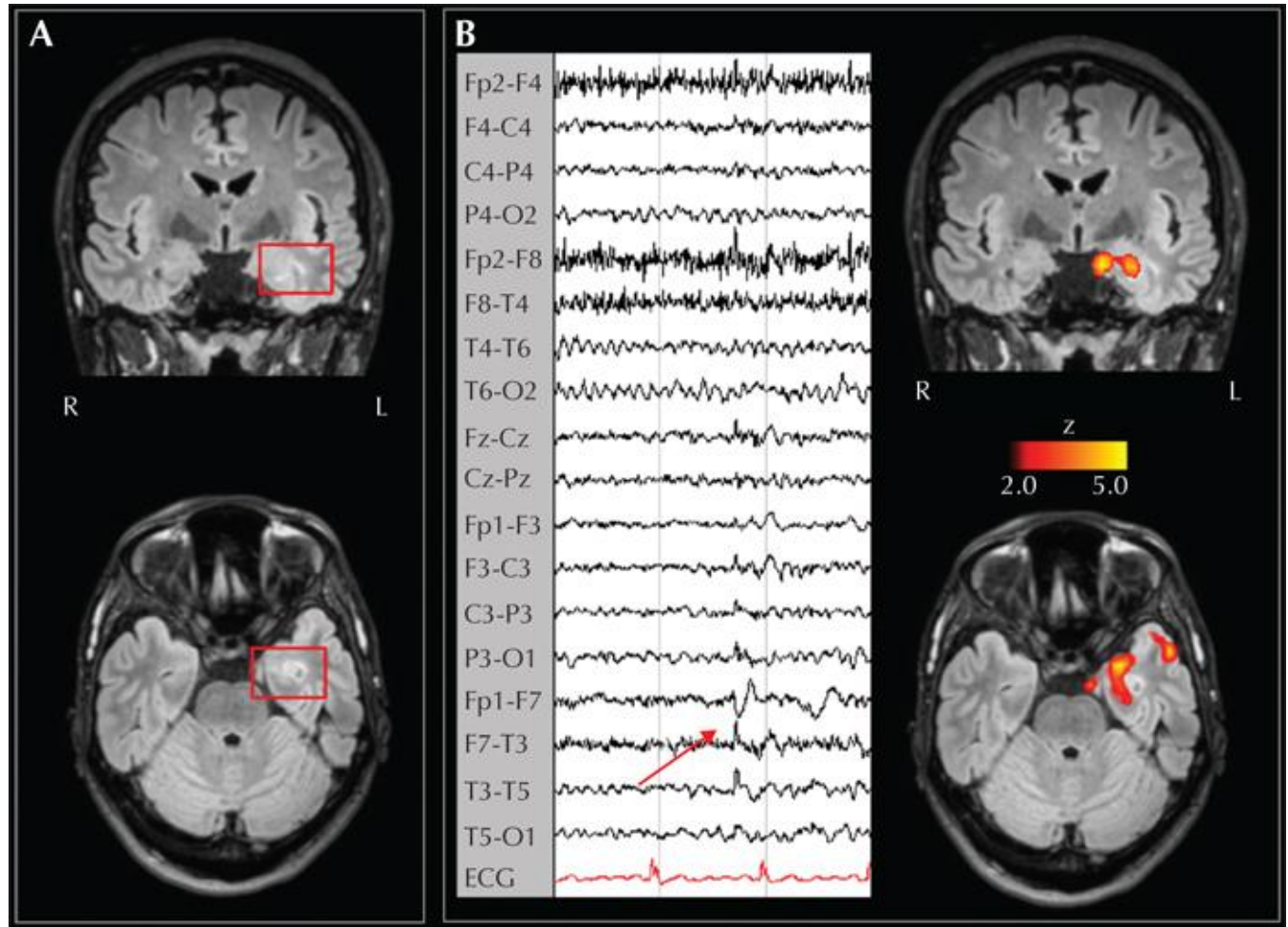
- developed in 1993 by Ives et al.
- indicated for patients that have ≥ 1 epileptiform discharges / min.
- An et al (2013) demonstrated 87.5% sensitivity, 76.9% specificity, 70% PPV, 90.9% NPV – for ILAE class I & II surgery outcomes if resection included BOLD signal areas.
- can be combined with SEEG – resect area around electrode contacts with HFO (high-frequency oscillations) + area with corresponding BOLD signal on fMRI.



Patient with refractory seizures arising from a brain tumor (possibly DNET) in the left mesial temporal region associated with blurring of the grey-white matter interface in the ipsilateral temporal pole:

(A) FLAIR displaying the lesion (red box);

(B) Scalp EEG-fMRI analysis of left frontotemporal discharges (F7-T3) shows a maximum BOLD response on the left amygdala and hippocampus + additional smaller BOLD cluster at the temporal pole (superior temporal gyrus).



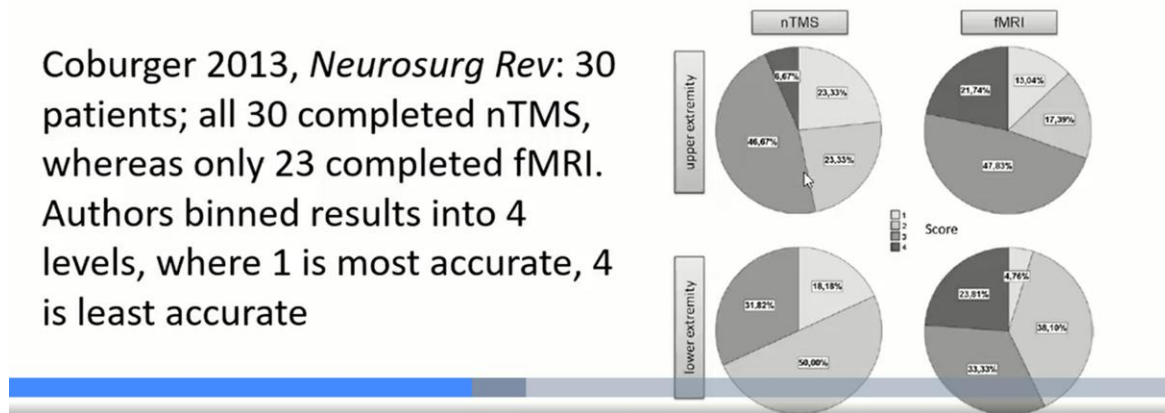
nTMS (navigated transcranial magnetic stimulation)

MAPPING FOR ELOQUENT CORTEX

- FDA approved for “presurgical motor and language mapping”.
- possible for **noncooperative patients** (vs. MEG, fMRI)
- cannot reach **deep parts of brain** (solution – TDCS – transcranial direct current stimulation).
- studies show it may be **more accurate than fMRI** (as compared to gold standard – Direct Cortical Stimulation):

- Several studies have evaluated accuracy of motor mapping with nTMS vs fMRI (with DCS as gold standard)
 - Forster 2011, *Neurosurgery*: 10 pts, mean distance to DCS hotspot 10.5 +/- 5.7 mm for nTMS vs 15.0 +/- 7.6 mm for fMRI
 - Mangraviti 2013, *Neurol Sci*: 7 patients, mean distance to DCS hotspot 8.5 +/- 4.6 mm for nTMS vs 12.9 +/- 5.7 mm for fMRI

Coburger 2013, *Neurosurg Rev*: 30 patients; all 30 completed nTMS, whereas only 23 completed fMRI. Authors binned results into 4 levels, where 1 is most accurate, 4 is least accurate



EPILEPSY PALLIATION

- see below >>

PET (positron-emission tomography)

(FDG-PET, ^{11}C flumazenil-PET)

- determines **functional deficit zone** in focal epilepsy.

- shows **hypometabolism** in:
 - Epileptogenic zone**
 - Diffuse **cerebellar** – due to ongoing antiepileptic therapy.
 - Mild symmetric **thalamic** - metabolic alteration/corticothalamic diaschisis secondary to temporal lobe epilepsy.
 - Temporal lobe** (in normal conditions):
 - N.B. temporal lobe is relatively hypometabolic compared to other lobes.
 - N.B. PET often shows hypometabolism in temporal lobes of pediatric patients – ignore it (esp. if symmetric).
- FDG is made in an expensive medical cyclotron and "hot-lab" (automated chemistry lab for radiopharmaceutical manufacture), and then delivered immediately to scanning sites because of the natural short (110-minute) half-life of Fluorine-18.
- after injection need to scan the patient within 30 minutes (vs. SPECT – have 4 hours).
- PET:
 - has highest accuracy for temporal lobe foci, which are diagnosed most easily with MRI and EEG.
 - is least reliable for extratemporal nonlesional foci, which also are most difficult to define with MRI and standard EEG.
 - PET is superior to interictal SPECT.

- indications:
 - a) discordant MRI and EEG findings
 - b) normal MRI findings.
- cannot do PET if given **insulin** recently (glucose goes into muscle).
- PET can be combined with **simultaneous EEG** – helps to **avoid false lateralizations from hypermetabolism** due to unrecognized ictal / postictal events as well as frequent interictal epileptiform discharges.
- **quantitative PET** – patient's PET subtracted from population-normalized template PET.

SPECT (single-photon emission computerized tomography)

- gamma-emitting tracer used is 99mTc-HMPAO (hexamethylpropylene amine oxime) - emits gamma rays that can be detected by a gamma camera.

- HMPAO allows 99mTc to be taken up by brain tissue in a manner proportional to brain blood flow (i.e. SPECT is assessment of regional cerebral blood flow in 3D).
- SPECT is more widely available (than PET), because the radioisotope is longer-lasting and far less expensive.

Interictal SPECT - shows **hypoperfusion**; resolution is inferior to that of PET; useful only for SISCOM.

Ictal SPECT - **hyperperfusion** during seizure - high localizing value* (only if ictal injection occurs within 20 seconds of ictal onset - then may scan within next 4 hours**).

*70-100% in temporal epilepsy vs. 33-63% in extra-temporal epilepsy

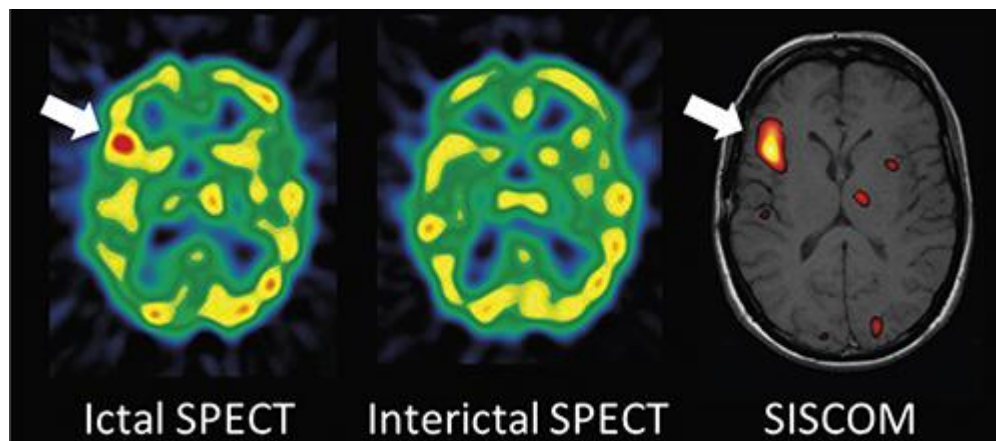
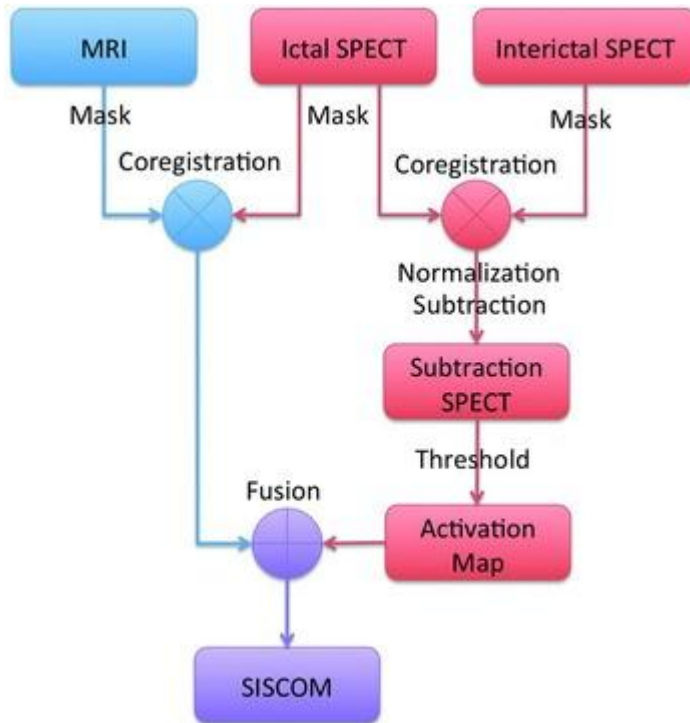
**vs. PET – within 30 minutes

- may also see hyperperfusion in *ipsilateral basal ganglia, contralateral cerebellum*.
- for ictal SPECT to be meaningful, seizure duration has to be > 40 seconds.

SISCOM (Subtraction Ictal Spect CO-registered to Mri)

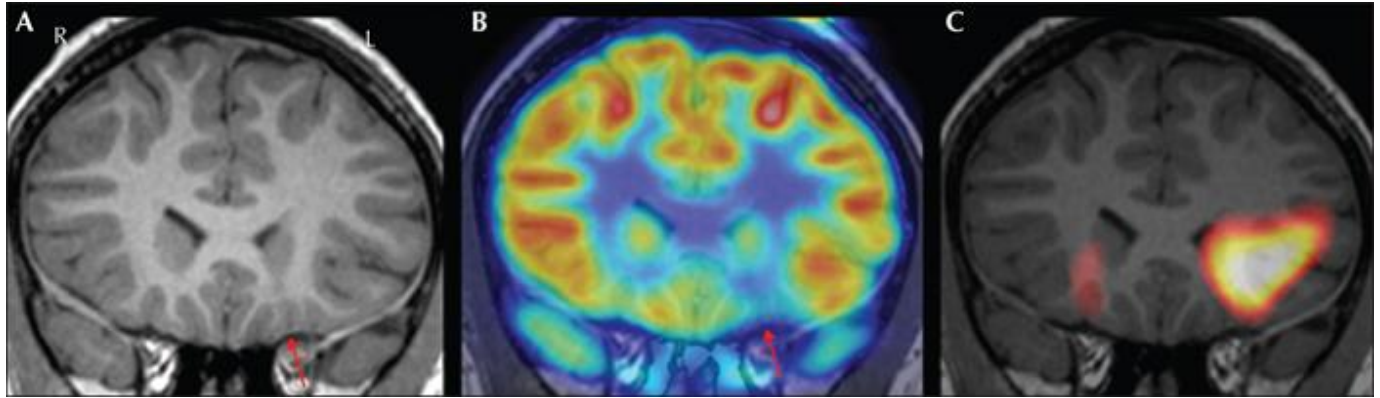
- much higher accuracy than either ictal or interictal SPECT! (may provide alternative to depth electrode studies)

- requires two SPECT scans (separated by ≥ 48 h to accommodate radionuclide washout) - during *interictal* period and *within seconds of seizure onset*.
- using computer software, these scans are subtracted from each other.
- subtracted scan then can be co-registered onto MRI to provide support for focus location.



N.B. in *MRI-negative* cases, reliability of functional neuroimaging is much reduced, and positive result from *functional imaging* generally requires verification by *depth electrode recordings*!

Multimodal imaging with interictal FDG-PET (middle) and ictal SPECT (right) which guided focused re-review of MRI (left, T1-weighted MPRAGE sequence) - hypometabolism on interictal PET (arrow) and hyperperfusion on ictal SPECT both pointed to a region in the left orbitofrontal cortex → re-review of the MRI revealed a cortical malformation characterized by gray-white blurring and thickened cortex (arrow). Resection of this region led to postoperative seizure freedom and surgical pathology-confirmed FCD:



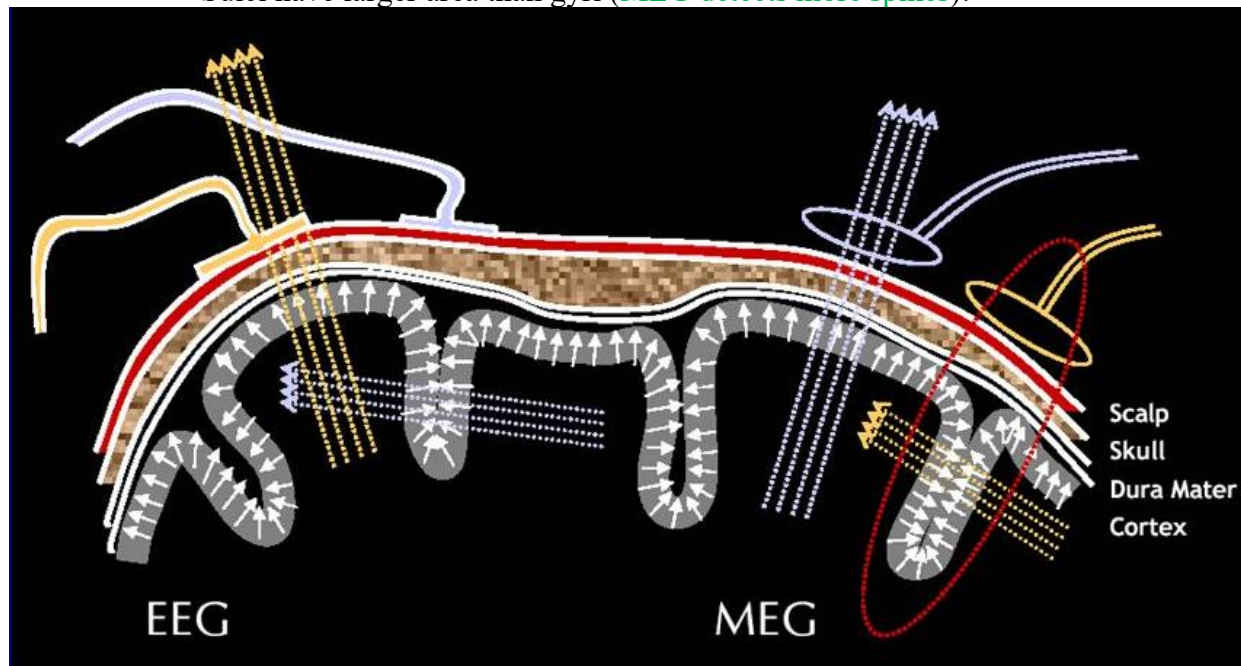
MAGNETOENCEPHALOGRAPHY (MEG)

- method for 3D detection of deep epileptic foci; records magnetic fields produced by brain electrical activity

- can localize the source of magnetic field generated mainly by **neurons tangentially oriented to the skull**.

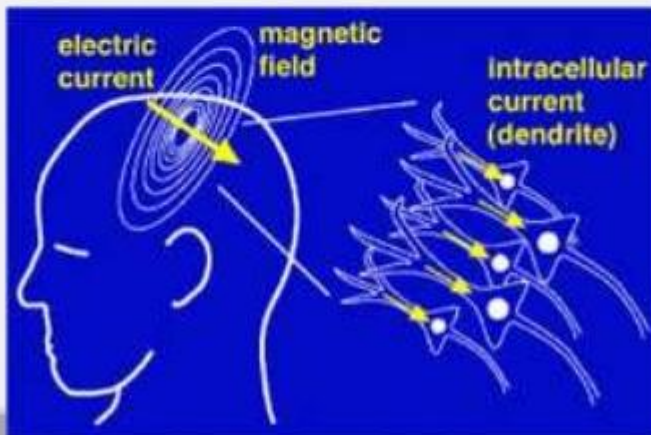
EEG detects gyri (radial currents) vs. MEG detects sulci (tangential currents) – not all MEG dipoles have a counterpart spikes on EEG.

Sulci have larger area than gyri (**MEG detects more spikes**)!



50,000 neurons need to fire to generate a Readable signal.

Neurons near the outside of the brain generate the strongest signals.



- MEG is closer to pathophysiology (electrical discharge) than PET, SPECT.
- MEG provides 30% nonredundant information (vs. PET is useful in only 10% of cases).
- attempt ictal MEG – occasional MEG scans capture seizures.

Procedure details

- requires sophisticated expensive suite (well isolated, far from highways).



- stimulators have to be turned off;
 - no need to turn off ahead off time (to reduce residual magnetism), MEG suite uses demagnetizers.
 - brain leads have their own artefacts.
- if need sedation, may use **CLONIDINE** or **MELATONIN**.

MSI (MAGNETIC SOURCE IMAGING)

- routine way to export MEG results onto MRI.

- dipole analysis has 80% concordance between MEG source and MRI lesion within 10 mm radius.
- UK no longer does spike analysis but does averaging.

Cluster is ≥ 5 -6 dipoles with ≤ 1 cm between sources.

MEG resolution:

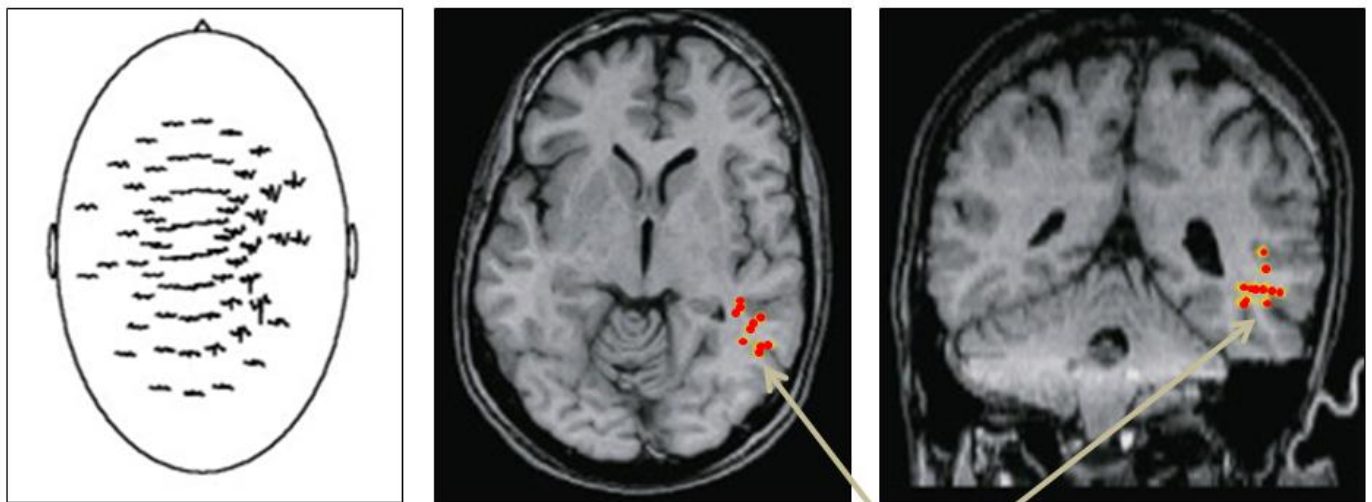
Spatial resolution:

Superficial cortex - 1 mm.

Deeper structures - 1 cm.

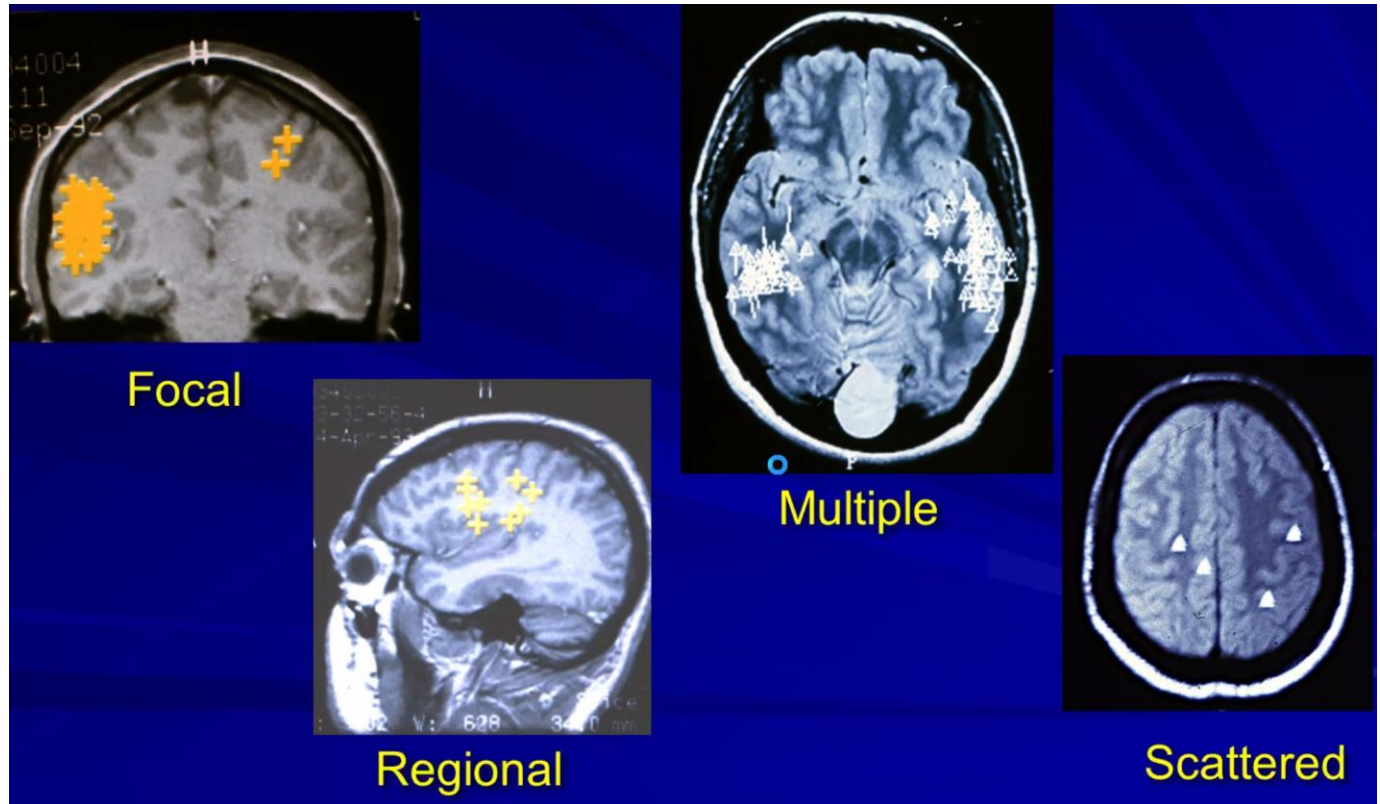
Temporal resolution - 1000 Hz and more.

Identifying Epileptic Spikes (Merrifield et al., 2007)



Right Temporal Lobe Epilepsy

MSI patterns:



- diagnostic accuracy 57-70% in temporal cases (44% - extratemporal cases).

MEG-BASED BRAIN MAPPING

- **language** – MEG sensitivity is poor (fMRI is much better).
- **visual cortex** – good quality, not a part of routine protocol (so need to specifically request for it).

MEG-GUIDED SURGERY

Case series

Gregory W. Albert et al. Magnetoencephalography-guided resection of epileptogenic foci in children.
J Neurosurg Pediatrics 14:532–537, 2014

- 16 pediatric resections based on MEG data and no intracranial recordings.
- all patients had **localized spike clusters*** on MEG (results of other noninvasive evaluations are concordant with the MEG data) and 93.7% had abnormal findings on MRI (1 patient had normal MRI).
 - * ≥ 6 dipoles with ≤ 1 cm between sources
 - most had either presumed cortical dysplasia or postoperative changes from previous surgery (most of these previous surgeries were prior attempts at surgical control of epilepsy, 1 patient had undergone previous resection of a DNET).
 - pathology results: FCD, heterotopia, gliosis, postoperative changes, normal / nondiagnostic.
- outcome:
 - 62.5% of patients - seizure free.
 - 20% of patients - improvement in seizures (Engel II-III).
 - 2 cases – Engel IV → additional surgery was performed subsequently with intracranial monitoring in attempts to obtain seizure control.

NEUROPSYCHOLOGICAL TESTING (NEUROPSYCHOMETRY)

- routinely, *all surgical candidates* undergo extensive neuropsychological testing (possible exception – VNS candidates).
- testing is not standardized between centers.
- test battery contains:
 - 1) **personality** inventory (e.g. Minnesota Multiphasic Personality Inventory)
 - 2) tests of **memory** and **language**
 - Temporal mesial sclerosis in dominant hemisphere – memory↓
 - Verbal memory is always on the left!
 - 3) tests for **interhemispheric transfer** (before callosotomy) - cross-retrieval and naming of objects, cross-replication of hand postures, cross-localization of fingertips.
 - 4) other tests, depending upon interests of neuropsychologist.
- goals of testing:
 - 1) to help *localize epileptogenic focus* (subtle deficits in cognitive functioning might provide additional localization that neurologic examination misses); it is not reliable because very few tests reliably measure frontal and temporal lobe function.
 - 2) identification of *significant cognitive problems* (e.g. memory problems - might not be a candidate for temporal lobectomy).
 - 3) predicting *postoperative cognitive deficits*.
 - 4) formulating *postoperative vocational goals*.
- testing results are valid for max **18 months**.
- patients are instructed to **continue their medications as usual** but **abstain from marijuana for ≥ 72 hours**
 - *ASMs are known to affect processing speed*; many tests are timed so ASMs may have effect on results; there are no tools to differentiate ASM side effect from true encephalopathy, thus, neuropsychologist's expertise and insight are needed.
 - **TOPIRAMATE** has well-researched negative side effect on **verbal** processing and memory.
 - **LEVETIRACETAM** has **mood** effects.

GENETIC TESTING

Also see p. E1 >>

The National Society of Genetic Counselors (NSGC) published 2022 evidence-based practice guidelines recommending comprehensive genetic testing, such as whole exome, whole genome, or large multigene panel, in any-aged individual with unexplained epilepsy.¹

Additionally, as of 2022, the Genetics Commission of the International League Against Epilepsy (ILAE) recommends consideration of genetic testing in presurgical evaluation of non-acquired medically refractory, lesional or non-lesional, focal epilepsy.²

Practically indicated for **every surgical patient*** (co-existing genetic causes predict outcomes – genetic testing may exclude unsuitable surgical candidates), but especially:

*including acquired structural MRIs – lesion may be incidental (9%) in addition to genetic cause

- 1) structural lesions with genetic causes (tuberous sclerosis, FCDs, MCDs, cavernomas)
- 2) focal MRI-negative epilepsies (e.g. all suspected FCD/MCD cases - 37-58% of operated FCD cases)
- 3) patients with intellectual disabilities, family history, dysmorphic features, autistic spectrum

Material:

- A. Germline mutations – blood
- B. Somatic mutations – brain tissue (resected, SEEG electrode residuals)

MULTIDISCIPLINARY CONFERENCE

- final results of noninvasive testing (phase I) are presented to **multidisciplinary epilepsy surgery conference** (neurologist specializing in epilepsy, epilepsy neurosurgeon, neuropsychologist, epilepsy nurse, speech pathologist, neuroradiologist, and psychiatrist).

Conference decides on:

- 1) need for WADA test (phase II)
 - 2) need for intracranial EEG (phase III)
 - 3) surgical plan
- in most scenarios, there are several reasonable options, which are not mutually exclusive, and some differences among centers, and even among epileptologists, are expected - this is why all comprehensive epilepsy centers make decisions after discussions at multidisciplinary conferences.

Presurgical Evaluation (Invasive, Phase II)

Two goals:

1. Further define **epileptogenic zone** – intracranial EEG (icEEG) >>
2. **Map** brain function:
 - 4) fMRI >>
 - 5) MEG >>
 - 6) nTMS >>
 - 7) WADA: chemical (>>), electrical (>>)

CHEMICAL WADA (INTRACAROTID AMOBARBITAL TEST)

- so called PHASE II TESTING
- named after **Japanese epileptologist Juhn Atsushi Wada**, of the University of British Columbia - he developed the test while he was a medical resident in Japan just after World War II, when he was receiving training in neurosurgery. Wada developed the technique of transient hemispheric anesthetization through carotid amytal injection to decrease the cognitive side effects associated with bilateral electroconvulsive therapy.

Wada J (1949). "A new method for the determination of the side of cerebral speech dominance. A preliminary report of the intra-carotid injection of sodium amytal in man". Igaku to Seibutsugaki, Tokyo. 14: 221–222.
- injection of 100 mg SODIUM AMOBARBITAL (AMYTAL®)* / PROPOFOL into carotid artery - to temporarily anesthetize (inactivate) hemisphere in ipsilateral carotid artery distribution (includes amygdala and anterior hippocampus) - allows independent **testing function of contralateral hemisphere**.

*barbiturate derivative that activates GABA_A receptors
- does WADA need to be **bilateral** even if **pathology unilateral**? Wada is the difference score between the two hemispheres as they relate to each other and to baseline (if the second hemisphere couldn't happen due to a seizure, decision can be made after unilateral WADA, it is just not ideal)

INDICATIONS

- preparation for ablative surgery (once epileptogenic focus has been identified) - injection ipsilateral to epileptogenic zone:

1. Which hemisphere contains **LANGUAGE function** (dominant hemisphere inactivation → aphasia; nondominant → dysarthria); fMRI is much less useful here.
2. Functional adequacy of contralateral hippocampus to **sustain MEMORY** (before anterior temporal lobectomy to avoid permanent amnesia).

N.B. failure of memory function is contraindication to resection of hippocampus and parahippocampal structures on injected side!

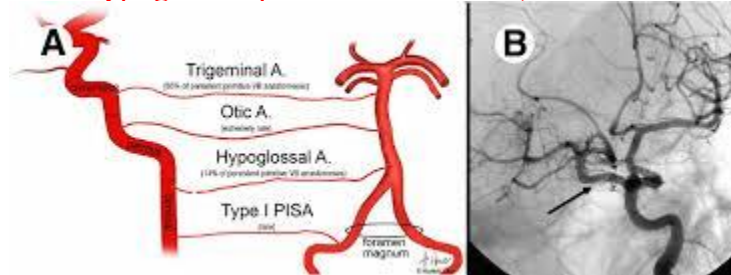
Some experts believe that WADA is more important for temporal neocortical resections (and not so much for SAH).

3. Prognosing **seizure-free OUTCOME**.

- never has been standardized - there is no such thing as *the* Wada Test - major efforts to establish procedural uniformity across epilepsy centers have been made, but many different *versions* of the test still remain.
- some centers insist on Wada tests even when clearly right-handed patient is diagnosed with right temporal focus

PROCEDURE DETAILS

- instruct patient as to what is expected.
- catheter is passed into ICA (as for standard carotid arteriograms).
- **4-vessel arteriography** must verify that blood flows to corresponding hemisphere and there is:
 - 1) no **cross flow**
N.B. **significant cross-flow** is relative contraindication to anesthetizing side of dominant supply (patient goes to sleep).
 - 2) no **persistent embryonal communication between anterior and posterior circulation** (e.g. **trigeminal, otic, hypoglossal, pro-atlantal arteries**) – will cause brainstem failure!



N.B. **fetal PComA** is not contraindication but will cause cortical blindness.

- 3) no significant **vascular abnormalities** - Wada test may be grossly inaccurate with high flow AV fistulas, such as AVM
- start on side of lesion (Wada testing is almost always performed with bilateral carotid amobarbital injections).
 - have patient hold both arms in air and count loudly - inject Amytal rapidly; effect starts almost instantaneously, begins to subside after ≈ 8 minutes.
 - **contralateral hemiparesis** + **ipsilateral EEG slowing** ($> 50\%$) must appear (confirmed adequacy of injection); if not, add 25 mg more of drug.
 - patient is monitored to make certain that **recirculation** has not affected both sides simultaneously during testing.
 - **assess language** by showing patient pictures of objects and ask them to name each one out loud and remember each one.
 - **assess memory** by asking patient to name as many of pictures as they can 15 minutes after test: if they have difficulty, ask them to pick out pictures from a group that contains additional ones not shown to patient.
 - may repeat procedure on other side (use lower Amytal doses with each subsequent injection).

Caveats:

- **portions of hippocampus** may be supplied by **posterior circulation** (not anesthetized by ICA injection).
- despite aphasia, sometimes **ability to sing** is preserved - music and singing uses a different part of the brain than speech and language.

SIDE EFFECTS

1. Drastic **personality changes** are rarely noted, but disinhibition is common.
2. **Shivering** are often seen.
3. Recovery of speech is **dysphasic** (contains errors in speech or comprehension) after a language dominant hemisphere injection.

1% risk of **vessel dissection and stroke!**

INTERPRETATION

Memory scores of > 50-67% have been deemed passing scores.

- operational definition of failure = **corrected score < 5**

ICA WADA mimics ATL

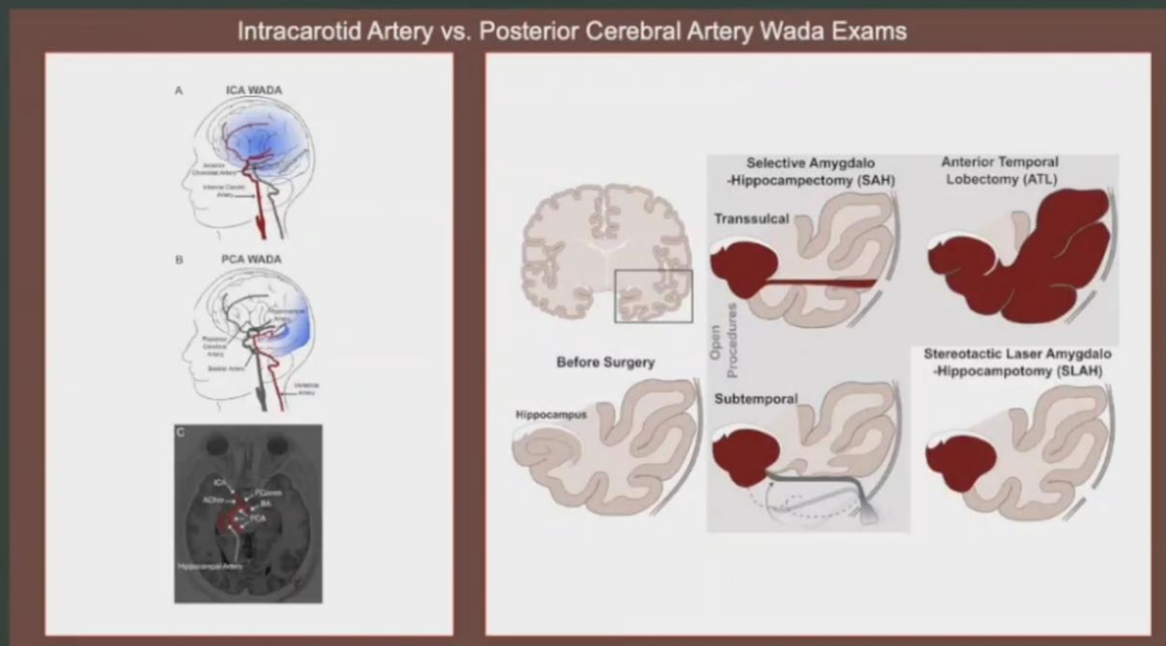
PCA WADA mimics LITT or SAH

- if ICA WADA failed but ablation is still highly recommended:
 - proceed to **selective PCA WADA** - if that is OK, then may ablate MTL (but avoid ATL!!!)

N.B. WADA test is still associated with false positives and negatives!

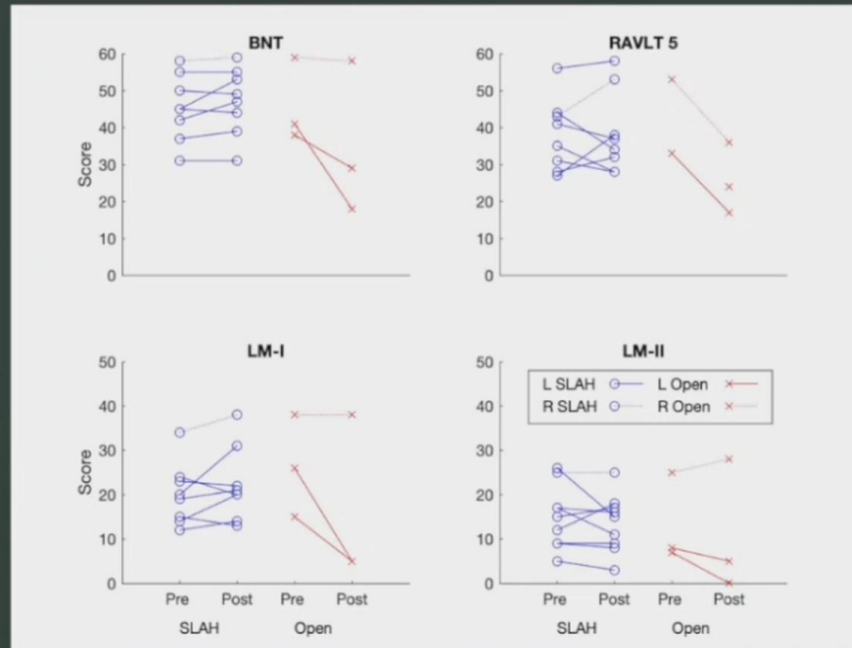
N.B. PCA WADA may miss head of hippocampus!

We examined cognitive and functional outcome in patients (N=13) who failed the ICA- but passed the PCA-Wada (Considered high risk for memory decline)



Drane, D. L.,* Acerbo, E., Rogers, A., Pedersen, N. P., Williamson, A., Stern, M. A., Dickey, A. S., Howard, B. M., Bearden, D. J., Okada, N., Staikova, E., Gutekunst, C. A., Alwaki, A., Gershon, T., Jirsa, V., Gross, R. E., Loring, D. W., Kheder, A., & Willie, J. T.* (preprint). Posterior cerebral artery (PCA) Selective posterior cerebral artery Wada better predicts good memory and naming outcomes following selective stereotactic thermal ablation for medial temporal Lobe Epilepsy Than Internal Carotid Artery Wada. MedRxiv, 2024 Mar 26:2024.03.24.24304488.

Comparison of SLAH vs. ATL/SAH following a failed ICA Wada but a passed PCA Wada (high risk patients)



Drane, D. L.,* Acerbo, E., Rogers, A., Pedersen, N. P., Williamson, A., Stern, M. A., Dickey, A. S., Howard, B. M., Bearden, D. J., Okada, N., Staikova, E., Gutekunst, C. A., Alwaki, A., Gershon, T., Jirsa, V., Gross, R. E., Loring, D. W., Kheder, A., & Willie, J. T.* (preprint). Posterior cerebral artery (PCA) Selective posterior cerebral artery Wada better predicts good memory and naming outcomes following selective stereotactic thermal ablation for medial temporal Lobe Epilepsy Than Internal Carotid Artery Wada. MedRxiv, 2024 Mar 26:2024.03.24.24304488.

NONINVASIVE ALTERNATIVES TO WADA

Wada test has angiography risks – so check for alternatives:

Electrical WADA

fMRI - can lateralize **language**

nTMS

MEG - can lateralize **language**

Near-infrared spectroscopy

H₂¹⁵O-PET

UAMS PROTOCOL

Prerequisites:

Providers-Neurology, Neuropsychology and Interventional Neuroradiology

Location: IR suite

Ancillary: EEG tech with vEEG setup (full montage)

Medication: Amytal (amobarbital sodium preparation) (at least 1 vial for each side)

Team member specific considerations

EEG tech

1. Must be vEEG (live) with patient's face and hand in clear unobstructed view
2. Audio recording should be ON with appropriately placed microphone
3. EEG tech at the machine side annotating the record
 - a. Injection times

- b. Delta slowing
 - c. 50% delta-theta reduction
 - d. Strength assessment by MD
 - e. EEG return to baseline
4. EEG screen should also be clearly visible to the Epilepsy MD
5. Perform a baseline at least 15 min recording before the start of the procedure (during baseline assessment by Neuropsychology, prepping etc)

Nursing

Amytal (amobarbital sodium preparation)

1. Needed 20 cc of sterile water and 500 mg of amobarbital for EACH side
2. Reconstitute *MAXIMUM* 30 min before use
3. Add 5 cc of sterile water to amobarbital vial.
4. **Do not shake.** Only rotate vial.
5. The dose should also be mixed and allowed to **dissolve for > 5minutes.**
6. Withdraw solution into a 25 cc syringe
7. Wash amobarbital vial with additional 5 cc of sterile water. Withdraw into 25 cc syringe containing first 5 cc of amobarbital solution
8. Add another 10 cc sterile water to 25 cc syringe containing amobarbital solution. Mix
9. **Final concentration: 25 mg/cc**
10. Carefully note and label the syringe with the time the medication was mixed.
11. Discard if not clear in 5 minutes and discard if a precipitate forms.
12. Attach **0.22 micron filter (yellow disc)** to Amytal syringe. **Give 6 ml (150 mg) Amytal to IR MD by pushing** it through the 0.22 micron filter (not pulling up through filter) into 12 ml sterile syringe MD is holding.
13. Amytal once mixed it must be used within 30 minutes. Be prepared to mix a second dose.
 - a. The time gap between two sides injection must be minimum 30 min and preferably 40 min so the previously reconstituted via CANNOT be used for the other side.
14. Usual injection Amount
 - a. First injection (either side); 3 ml (75 mg)
 - b. Repeat injections in 1 ml (25 mg) bolus up to 3 ml (75 mg)
15. Patient's both hands should be out of the sterile field and freely accessible to Epilepsy MD for testing

IR MD

1. Baseline assessment by Neuropsychology before catheter placement
2. Patient's both hands should be out of the sterile field and freely accessible to Epilepsy MD for testing
3. Side planned as the surgical side to be injected first.
4. Pre (amytal)-injection contrast run-through
 - a. Cross-filing?: Yes/ No
 - b. Fetal PCOM?: Yes/ No
5. Pre-injection assessment by Neuropsychology (please see below)
6. Re-explanation of strength testing to the patient by Neurology MD
7. Patient to raise both hands in air and count from 1 to 20 at a paced rate
8. Inject at cue of Epilepsy MD around Count ~ #5-6
16. Inject 3 cc (75 mg) over 3-4seconds
 - a. If flaccid contralateral hemiparesis does not occur after 5 secs, inject additional 1 cc (25 mg – total 100 mg)
 - b. If flaccid contralateral hemiparesis still does not occur after 5 secs, inject additional 1 ccs (25 mg – total 125 mg).

- c. If flaccid contralateral hemiparesis still does not occur after 5 secs, inject remaining 1 ccs (25 mg – total 150 mgs).
 - d. If flaccid contralateral hemiparesis still does not occur, stop procedure and check catheter position.
 - i. If the lack of hemiparesis was due to inappropriate catheter placement, wait 45 minutes from initiation of injection, then repeat the original injection sequence.
 - ii. If catheter placement is appropriate, wait 45 minutes from initiation of injection.
 1. Draw up 175 mg (7 cc) in 10 ccs sterile water. Hand inject 6 cc (150 mg) over 4 seconds. If flaccid contralateral hemiparesis still does not occur after 5 secs, inject the final 1 cc (25 mg).
- Usual range of Amytal dose 75-125 mg
 - Preferred maximum dose: 150 mg
 - Absolute maximal dose: 175 mg
 - *Time difference between two side injections, minimum 30 min (or complete recovery, whichever is later) preferably 40 min.*

Neurology MD

1. Baseline strength or arm, sensation, vision AND grip
 2. Contralateral hemiparesis achieved? Yes/ No ___/5
 3. Check grip strength
 4. Speech arrest? Yes/ No
 5. Amytal effect on EEG
 - a. Delta/theta slowing (ipsilateral)?
 - b. 50% reduction of EEG delta/theta ipsilateral to injection (or primarily theta background):
for memory item presentation by Neuropsychology
 6. Regularly check strength (may need repeat dose if strength improving fast)
 7. No items presented after strength had recovered to three-fifths in the hand contralateral to injection.
 8. Repeat Neuro exam between side injections
- *Time difference between two side injections, minimum 30 min (or complete recovery, whichever is later) preferably 40 min.*

Neuropsychology

1. Once the patient is arrived and prior to beginning of procedure (catheter placement) approximately 5-10 minutes of baseline testing
2. Rule out cross-filing and fetal PCOM before Amytal injection
3. Two minutes prior to injection the first test stimulus is shown
4. One minute prior to injection the second test stimulus is shown
5. Re-explanation of strength testing to the patient by Neurology MD
6. Patient to raise both hands in air and count from 1 to 20 at a paced rate
7. Inject Amytal at cue to Epilepsy MD ~ at Count #5-6
 - a. Neurology MD to assess and confirm
 - i. Hemiparesis
 - ii. EEG ipsilateral delta slowing
8. Neuropsychology and neurology run Wada protocol (Neurology monitors strength and EEG; Neuropsychology does language/memory testing per Emory Clinic protocol).
 - 50% reduction of EEG delta/theta ipsilateral to injection (or primarily theta background),
 - Per Neurology MD: Memory item presentation
9. No items presented after strength had recovered to three-fifths in the hand contralateral to injection.

10. Assessment of both spontaneous and recognition memory:

- a. After full motor recovery and return of a 10-s epoch of the patient's usual baseline EEG frequencies.
 - b. At 12 minutes post injection recognition memory testing occurs
- *Time difference between two side injections, minimum 30 min (or complete recovery, whichever is later) preferably 40 min.*

ELECTRICAL WADA

- allows high-precision testing in preparation for highly-selective ablations (esp. when chemical Wada was nonpermissive).
- using depth electrodes.
- simultaneously stimulating across all 8 electrode contacts while patient completes consecutive Wada based memory tests.
- protocol (Dr. Gross): different amplitudes (5, 7.5, 10 mA), pulse frequency 100 Hz, and 500 msec pulse duration; more details – see “Emory case report”

Emory case report

– 51 yo F patient failed chemical Wada (0/8) but electrical Wada from longitudinally along hippo axis implanted 8-contacts was permissive (8/8) → patient had RFA and then SLAH → 12 months postop **no evidence of verbal or visuospatial memory decline!**

Kaestner, E. ... Gross, R.E. and Drane, D.L. (2022), Electrical Wada for pre-surgical memory testing: a case report. Epileptic Disorders, 24: 411-416.

Proposed paradigm

Step 1: Fosphenytoin load provided for anti-seizure prophylaxis during cortical stimulation

Step 2: High frequency stimulation at 50Hz, pulse width 1msec, duration 5 sec, stimulus intensity increase by 0.5mA till after discharges or clinical symptoms noted up to maximum of 8mA

Electrodes stimulated

Step 1: Low frequency stimulation at 1Hz, pulse width 1msec, duration 60sec, stimulus intensity 80% of maximum stimulus intensity at 50Hz which either produced after discharges or clinical symptoms

- during this, patient is presented with 8 pictures and asked to recognize and then memorize during 60sec stimulation. Once stimulation is completed patient is asked to repeat the images seen.

- a 2nd 60 sec stimulation period is applied where images are shown again. Following stimulation, patient is asked to repeat the images seen.

- 10-15min later, the patient is asked to repeat the images shown (Free recall).

- Recognition task: In addition to the 8 images, 16 other images are added and patient is asked to identify if they were part of the original stack or not. Errors are counted.

INTRACRANIAL EEG (icEEG)

(subdural strips and grids, depth electrodes) – so called PHASE III MONITORING – only when noninvasive studies fail to adequately delineate epileptogenic zone. see p. E13 >>

- it may include Stimulation Cortical Mapping - determines **areas of eloquent cortex** that should not be encroached upon at time of operation. see p. E13 >>

ALGORITHM (TREATMENT)

Seizure **semiology** is extremely important in diagnosing ictal zone and seizure propagation network!

Medically refractory disabling focal seizures

↓
Focus can be localized → NO → **VNS** or **DBS** or **thalamic RNS**

↓ YES
How many foci → 2 → **RNS**; more than 2 → **VNS** or **DBS** or **thalamic RNS**

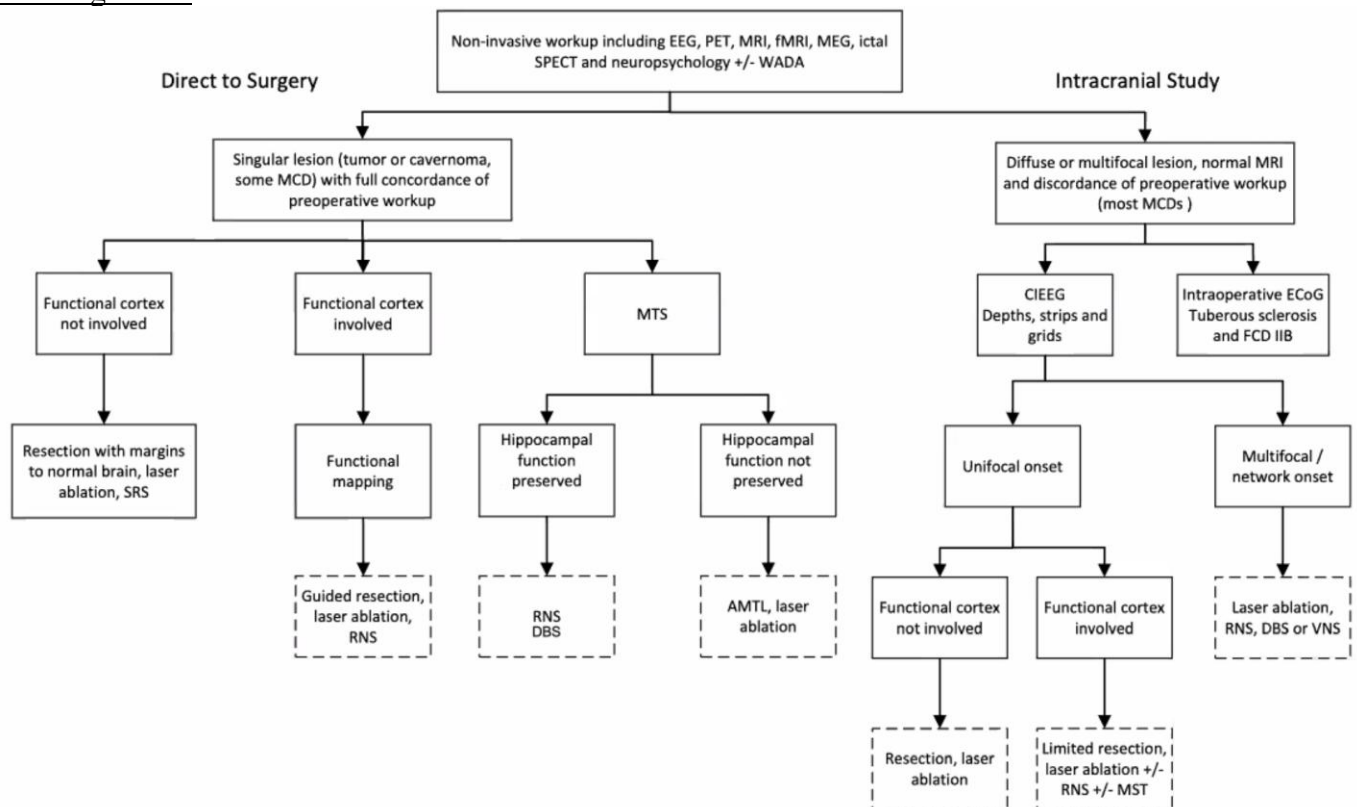
↓ 1
Safe to resect and cognitive risk is low → NO → **RNS** (or **multiple subpial transections**?)

↓ YES
Will it be curative → NO → **RNS**

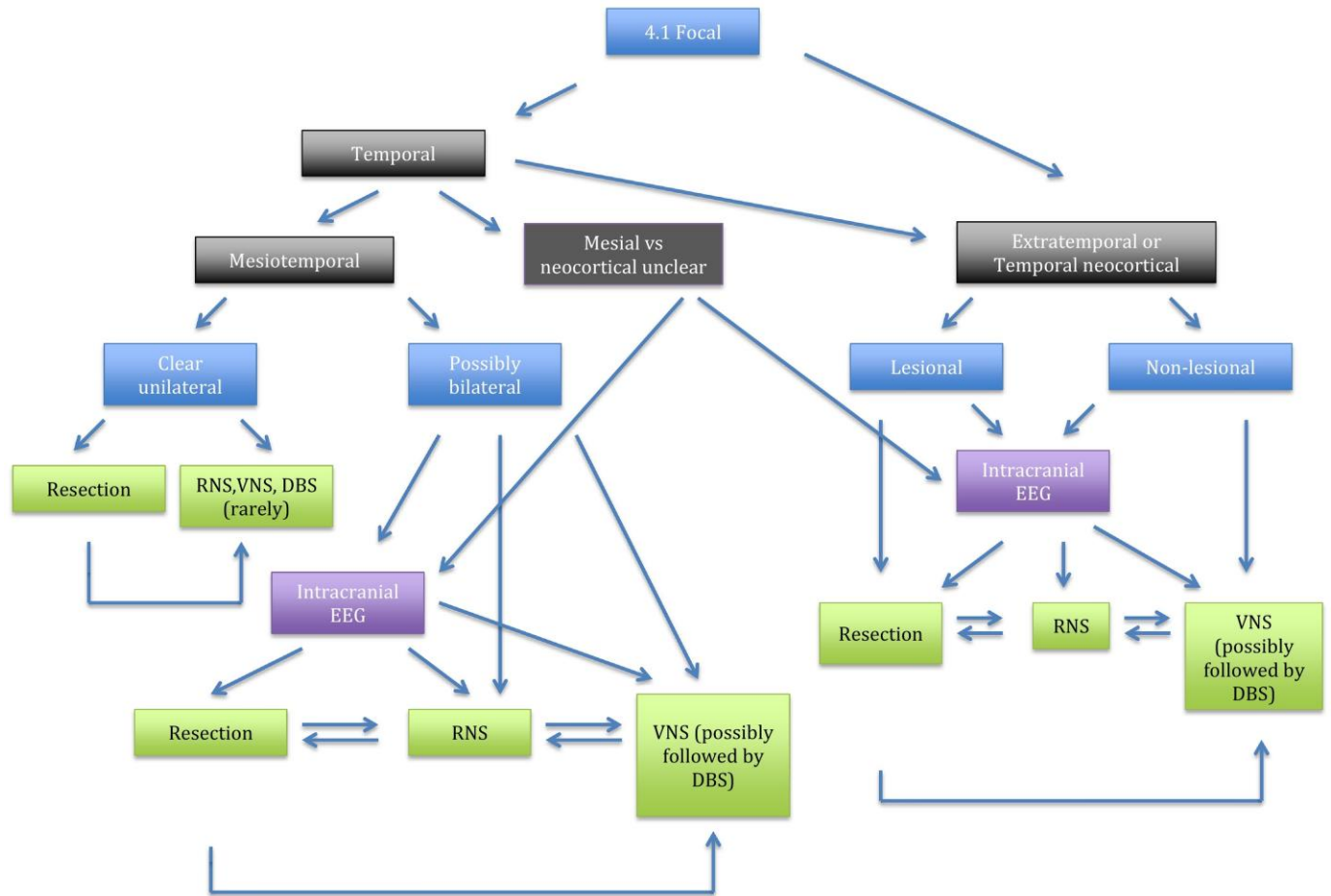
↓ YES
Ablation (resection, laser, SRS, RF, FUS)

N.B. RNS data (in bitemporal epilepsy) may support later resection / laser ablation

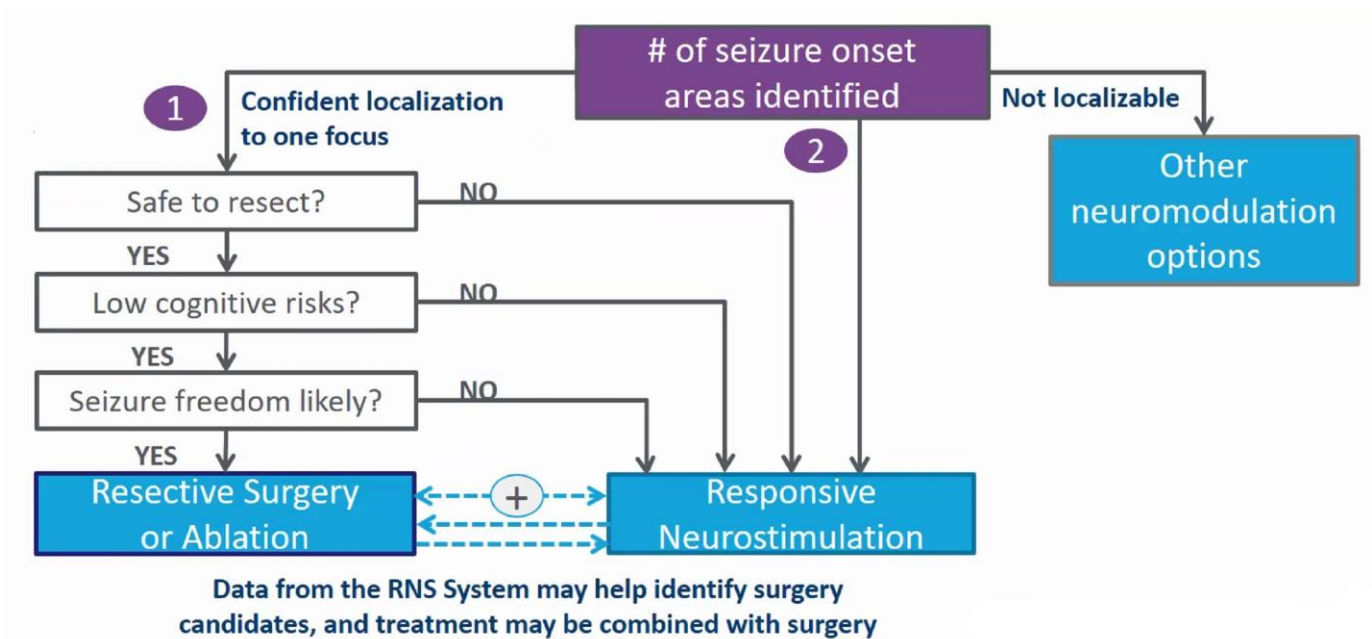
Yale algorithm



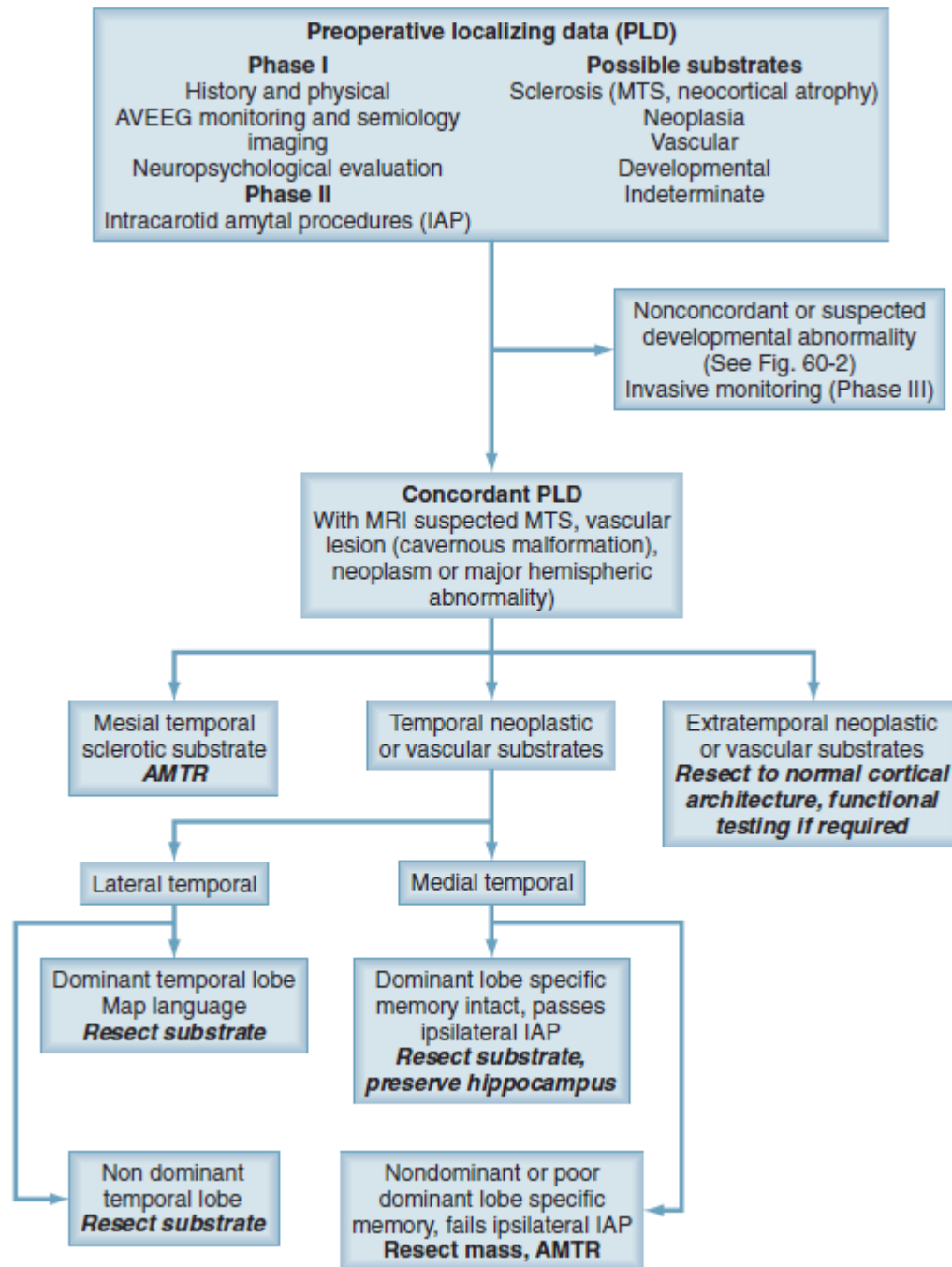
Gerard J et al.



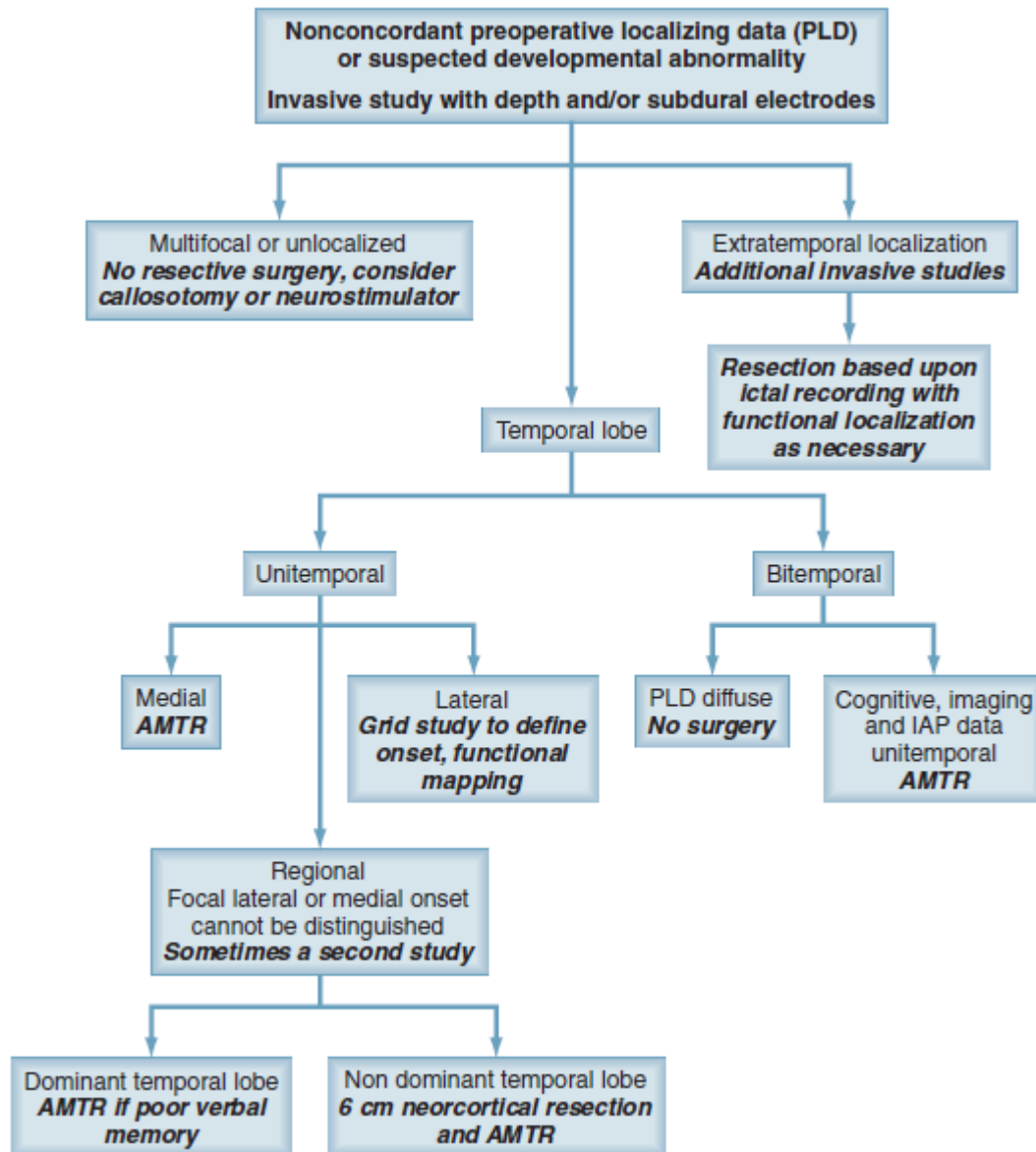
Benbadis et al. 2018



Alternative algorithm



Alternative algorithm (fig. 60-2)



FOCAL EPILEPSY

- key dichotomies: temporal vs. extratemporal



mesial vs. neocortical

- distinguishing mesial vs neocortical temporal may at times require intracranial EEG, but once that is done, the neocortical temporal group generally follows the same course as extratemporal.

TEMPORAL

Outcomes in bilateral or non-resectable temporal lobe

1. VNS – 60-62.5% response
2. ANT DBS – SANTE trial - 44.2% response
3. RNS - 74 % response in hippocampal epilepsy

4. **Hippocampal DBS** – 25-100% response in hippocampal epilepsy
5. CM DBS – not as effective on focal epilepsy
6. STN DBS – too limited data
7. TMS – probably ineffective for mesial TLE

MESIOTEMPORAL

Unilateral

- resective surgery or ablation is by far the treatment of choice.
- traditionally, **en bloc resection of anterior lateral and mesial temporal lobes structures** has been performed. However, the potential for neuropsychological decline (especially in dominant temporal lobe surgery) has led to the development of **selective amygdalohippocampectomy (LITT of the mesial structures** is an effective option).
- very rarely, **neurostimulation** may be preferable if memory is superior on that side, but this should be exceptional because a minor verbal memory decline is usually offset by seizure freedom.

Bilateral

- the issue is “how bilateral is it?” (continuum from minimal or weak evidence on one side, to definitely and clearly bilateral):
 - a) high concern for bilateral disease (independent seizures on surface EEG, bilateral mesial temporal sclerosis (MTS) on MRI, conflicting lateralizing findings)
 - b) low concern for bilateral disease (very rare interictal spikes on the opposite site)
 - c) most often - anywhere in between high and low concern.
- options:
 - a) straight **unilateral resection**
 - b) bitemporal **intracranial EEG**
 - c) high concern for bilateral disease* → **RNS** (data may allow an eventual resection should there be a strong unilateral predominance).
 - *RNS is preferred over short-term intracranial EEG in order to ascertain lateralization based on a larger sample of seizures (hundreds), rather than on a limited number of seizures obtained in EMU (and usually on reduced doses of ASMs).

NONMESIOTEMPORAL (NEOCORTICAL TEMPORAL, EXTRATEMPORAL)

Lesional

Here, much of the approach depends on the nature of the lesion and, specifically, its suspected nature (tumor, vascular malformation, cortical dysplasia, etc.), size, location (with respect to eloquent cortex), and the degree of certainty (based on semiology and EEG) that the lesion is indeed causing the seizures. If the lesion is a suspected neoplasm, then its potential for evolution into higher grades becomes part of the equation, favoring resection for tumor reasons rather than epilepsy reasons. If the lesion itself is static and not concerning for a neoplasm, then lesionectomy has the highest probability of seizure freedom and will usually be preferred, with neurostimulation being a distant second choice, and probably only appropriate if the resection would involve critical eloquent cortex. Here, a partial resection can be combined with neurostimulation, though RNS can interfere with serial MRI surveillance of lesions. Another complicating factor in lesional focal epilepsy is the potential for dual pathology. Resection can be done with or without intracranial EEG or ECoG. Hemispherectomy represents an extreme example of lesional resection (e.g., hemimegalencephaly, Rasmussen's Syndrome) and also belongs in this category.

Nonlesional

This scenario (neocortical + nonlesional) is the most challenging and a very common scenario at level-4 epilepsy centers.

- A. Strongly lateralized (hemisphere known) Strong lateralization can be based on interictal discharges, ictal EEG, or seizure semiology (e.g., consistent head version prior to secondarily generalized convulsions). This scenario, most commonly frontal, will require intracranial EEG to cover the area in question (frontal, frontotemporal, parietal, occipital, etc.). Since the area is usually extensive, sEEG is increasingly preferred to subdural grids and strips, but both can be used.
- B. Not confidently lateralized (hemisphere unclear) Such cases have often been referred to as “fishing expeditions.” In the past, they could only be pursued with abundant amounts of hardware (bilateral strips and grids), resulting in a higher rate of complications and a relatively poor outcome. For that reason, many centers have tended to not pursue these cases for possible resective surgery. The advent of frameless robot-assisted sEEG, allows better surveying both in space (numerous depth electrodes on both sides) and time (electrodes can remain in place for weeks with little morbidity). Therefore these cases can now be reasonably pursued, “displacing” palliative neurostimulation treatments. One approach here is initially bilateral frontotemporal sEEG, which after the first few seizures can be focused by adding depth electrodes on the confirmed side of seizure onset once clarified [24]. Subdural electrodes and sEEG can also be combined. In this scenario, neurostimulation is also a reasonable option, since it carries less morbidity and the chances of seizure freedom with resective surgery are lower than in temporal or lesional cases.

EXTRATEMPORAL

1. VNS – efficacy in focal epilepsy does not appear to be limited to temporal lobe
 2. RNS – 37% response rate in neocortical epilepsy
 3. CM DBS – not as effective on focal epilepsy
 4. ANT DBS – Sante trial did not demonstrate efficacy in this subgroup
 5. STN DBS – was efficacious in small #'s tested and FLE was main target
 6. TMS – may be effective for MCD
 7. Hippocampal DBS - N/A
- posterior quadrant epilepsies – counsel for hemianopsia risk as often big resections (or disconnection) is needed.

ELOQUENT CORTEX

A. Resections

- need **mapping** - grids are ideal (but preceding SEEG may guide grid implantation)
- resections in eloquent cortex are feasible and safe (as long as preservation of functional areas is observed)
- outcomes are excellent in **lesional** epilepsies.
- in *non-lesional* epilepsies, if intracranial recordings provide convergent data, seizure freedom has high probability
- role of LITT is limited – cannot provide mapping

B. Neuromodulation / Cell therapy

- reserved for nonresectable areas (as determined by direct cortical mapping).

- can be combined with resection (limited topectomy) and mapping (precisely guides RNS electrode placement).

LANGUAGE LOCALIZATION / LATERALIZATION

Chemical Wada

Electrical Wada

fMRI - can lateralize **language**

Practice Guideline of American Academy of Neurology:

Szaflarski J.P. et al. Practice guideline summary: use of fMRI in the presurgical evaluation of patients with epilepsy: report of the Guideline Development Dissemination and Implementation Subcommittee of the American Academy of Neurology. Neurology. 2017;88:395-402.

- fMRI using **verbal memory or language encoding** **should be** considered for predicting postoperative **verbal memory** outcome (Level B recommendation).
- fMRI using **non-verbal memory encoding** **may be** considered for predicting postoperative **visuospatial memory** outcomes (Level C recommendation).

nTMS

MEG - can lateralize **language**

High-density EEG

Near-infrared spectroscopy

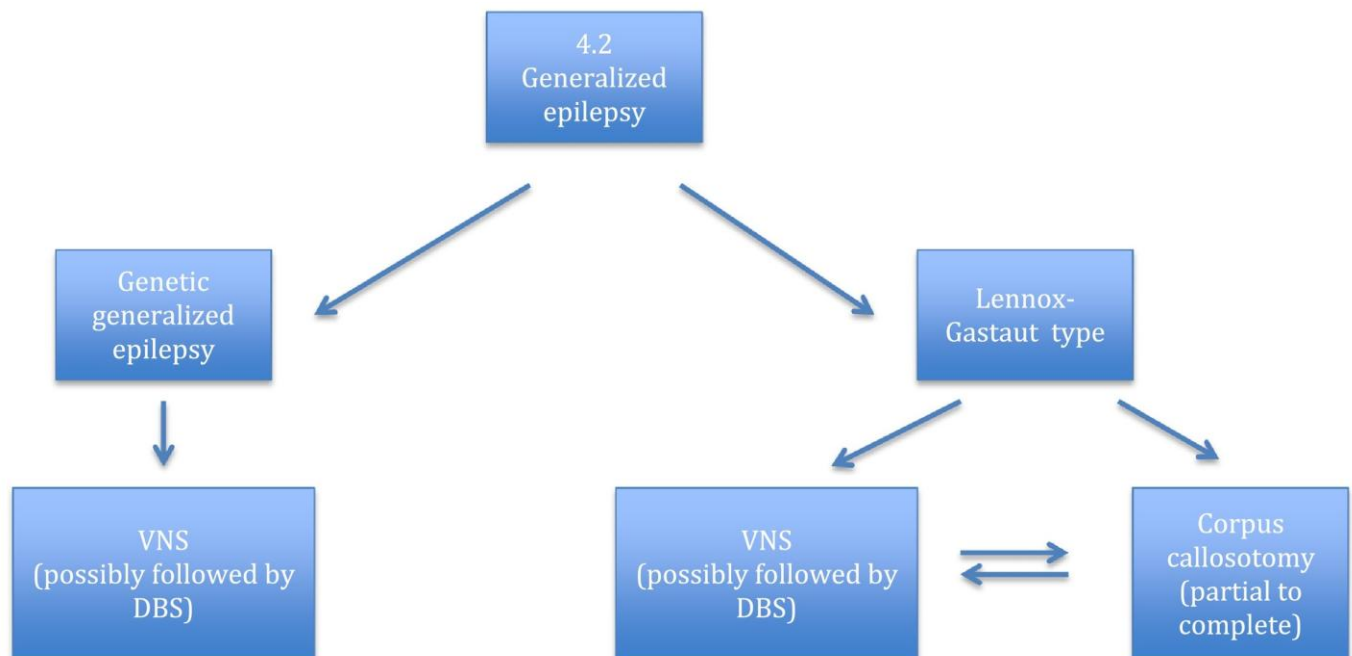
H₂¹⁵O-PET

Cortical stimulation (intraop vs implanted electrodes)

Epilepsy onset < 6 yo - high chances of language dominance on R side.

Language on right - in 3.4 % general population (but 25% of pediatric epilepsy population)

GENERALIZED EPILEPSY



GENETIC (S. IDIOPATHIC, PRIMARY) GENERALIZED EPILEPSY

- this group is usually not medically intractable, but it can be.
- occasionally, when bifrontal epileptiform discharges mimic generalized spike-wave complexes (“secondary bilateral synchrony”), distinguishing a GGE from a frontal lobe epilepsy can be challenging, and it is generally safer to err on the side of GGE and only pursue surgically if there is strong evidence for a focal onset.

LENNOX-GASTAUT TYPE (S. SYMPTOMATIC, CRYPTOGENIC GENERALIZED EPILEPSY)

- usually with cognitive impairment (developmental intellectual disability), neurologic abnormalities such as cerebral palsy, and characteristic EEG abnormalities.
- cause may be genetic, structural, or metabolic but is often unknown.
- this group can also be considered multifocal rather than “generalized” (epilepsy with ≥ 3 “foci” is equivalent to “generalized”).

Resective surgery and RNS are never options for the generalized epilepsies, but corpus callosotomy, VNS, and DBS can be. Technically, VNS for generalized epilepsy is off label in the US but is nonetheless one of the most common uses of VNS (endorsed by AAN guidelines). Corpus callosotomy is almost never performed in patients with GGE since they are usually mentally and neurologically healthy, but is a viable option for LGS type and while it is most often used in children, it can be performed in adults. Compared with neurostimulation, corpus callosotomy is more invasive, more ambitious, and may have more immediate effects. Neurostimulation is less invasive than callosotomy (VNS less than DBS), and the therapeutic effect is delayed, taking weeks to months. Preferences to use callosotomy or VNS first vary, and both are acceptable strategies that are not mutually exclusive. Additionally, since both are palliative procedures, some patients will benefit from both sequentially. Very rarely, patients with LGS-type of generalized epilepsy with a lesion can benefit from a focal (lesional) resection.

Lastly, patients with true multifocal epilepsy, i.e., related to truly multifocal discrete lesions such as tubers or cavernomas, can be considered for resection if it is confidently proven that one lesion is the source of a majority of the seizures.

Outcomes

1. VNS – 44-67% response
2. ANT DBS – Sante trial limited to CPE
3. CM DBS – 0-100% response
4. RNS – by definition excluded
5. Hippocampal DBS - by definition excluded
6. STN DBS- N/A
7. TMS – N/A

PREOPERATIVELY

ASM

- VALPROATE can cause **bleeding disorders** - routinely check coags and bleeding time - if values are abnormal, decrease or discontinue* valproate and recheck values before surgery!
*at least 3 weeks prior to surgery (replace with another medication).

- on surgery morning, patient receives usual medication dosage with few sips of water.
- some centers taper anticonvulsants, and completely D/C 1 day before surgery (??).

TYPES OF SURGERY

ABLATIONS S. RESECTIONS (curative) - resection of seizure focus (up to entire hemisphere):

- 1) lesionectomy
- 2) neocortical resections

DISCONNECTIONS (palliative*) - used when eloquent brain is involved to disconnect seizure focus from other functional parts of brain:

- 1) callosotomy
- 2) hemispherotomy
- 3) multiple subpial transections

STIMULATION S. NEUROMODULATION (palliative*)

- 1) RNS
- 2) DBS
- 3) VNS

CELL THERAPY

*term **palliative** (alleviating symptoms but not treating the underlying disease) is incorrect and unhelpful because intracranial neuromodulation has a neuroplastic effect on brain circuitry!

Only 10-30% of pharmacoresistant patients are candidates for **resective** surgery (rest of patients – **neuromodulation**)

SURGICAL PHILOSOPHY

Nodes vs. Networks – biggest debate about *pathophysiology of epilepsy* and may affect **surgical philosophy**.

Goals of surgery – **node removal / disruption** (i.e. lesion resection or stimulation with RNS / DBS) vs. **network disconnection** (lesioning) or **neuromodulation** (DBS / RNS / VNS).

Analogy with oncology – hope to remove *critical portion of abnormal neurons*, so then local inhibitory mechanisms will take care of the reminder.

More brain removed / ablated → better seizure control (“if remove entire brain → 100% seizure freedom”).

1. RESECTION / ABLATION

Posterior insula and **dorsal amygdala** - most dangerous areas for epilepsy surgeon!

Temporal Resections

see p. E15 >>

Lesionectomy

– for **structural lesions** delineated by **MRI ± electrocorticography**.

- most often in frontal lobe.
- all larger arteries or veins adjacent to or crossing gyri should be preserved.
- avoid resection of deep fiber pathways in white matter, i.e. stay in cortex (white matter is not epileptogenic).
- most common mistake is to remove only gross tumor and not immediate surrounding tissue (leaving epileptogenic tissue and clinical seizures).
e.g. small vascular abnormalities surrounded by hemosiderin can be extremely epileptogenic.

N.B. lesion in **eloquent cortex** – need **ECoG** to guide resection (else high risk of *postop status epilepticus*)
– if planning awake mapping, tell the patient “if you become noncooperative, we will implant grid and will send you to EMU for extra-operative mapping”.

Lesioning

- a) for **structural lesions** delineated by **MRI ± electrocorticography**.
- b) for **disconnecting** (e.g. callosotomy)

LASER ABLATION (LITT)

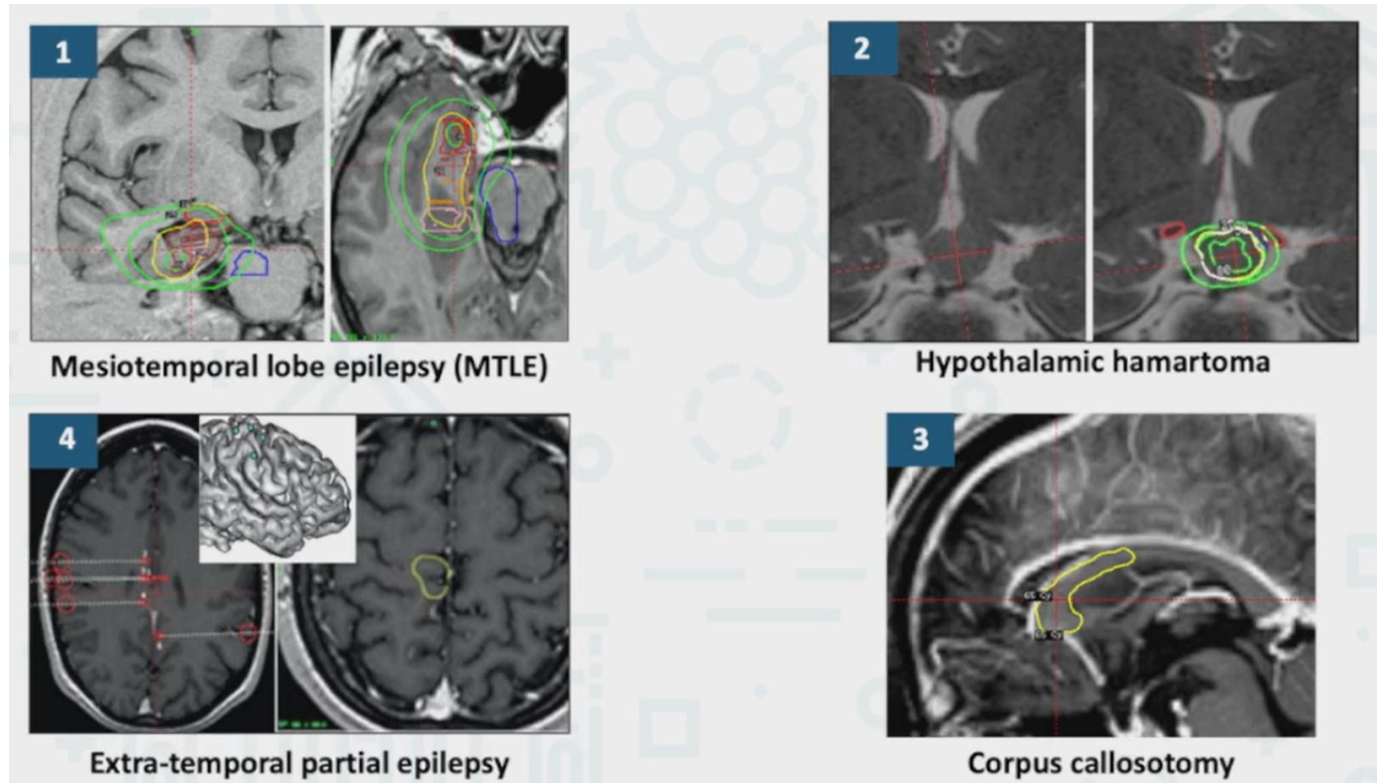
Temporal lobe epilepsy - see p. E15 >>

RF-ABLATION

(e.g. guided by and delivered through SEEG)

see p. E13 >>

STEREOTACTIC RADIOSURGERY (SRS)



FOCUSED ULTRASOUND (FUS)

see p. Rx15 >>

- best for hypothalamic hamartomas.
- FUS of ANT trial is under way.
- may have role in terminating status epilepticus via neuromodulation.

Tailored Neocortical Resection

– for (nonlesional) extratemporal epilepsy

Tailored - no two operations are identical!

- *eloquent cortical regions* are spared.

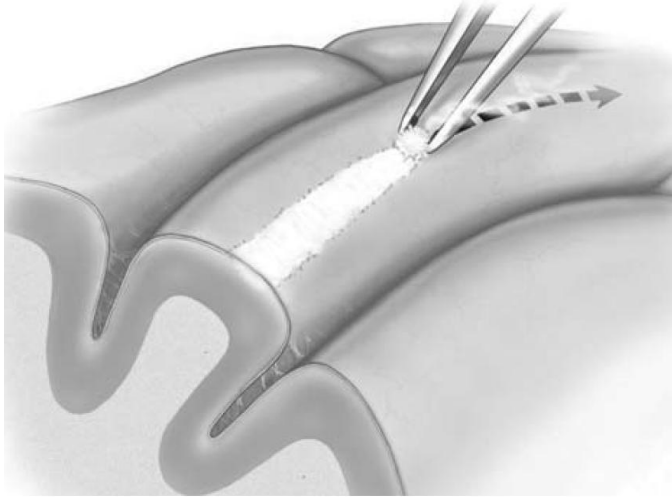
Role of SEEG and/or intraoperative electrocorticography (iECoG)

- resection guidance!

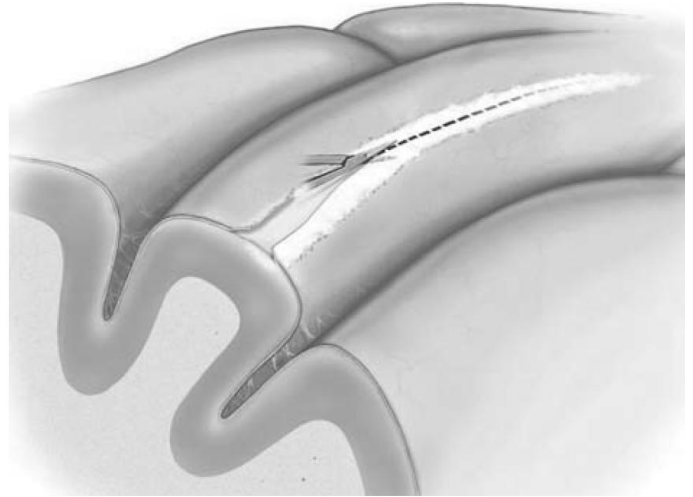
- epileptiform discharges recorded acutely during surgery define boundaries of cortical resection.
N.B. be prepared for larger craniotomy as epileptiform discharges may encompass larger area than expected!
- Brazilians call intraop ECoG as an alternative to SEEG in poor countries: ECoG uncovers (interictal spikes) and guides resection of “dormant” EZs that would emerge later if only “active” EZ was removed.
- iECoG significantly improves seizure-free rates in glioma patients operated for tumor-related epilepsy.
see p. E9 >>

Technique for topectomy:

(a) The pia over the center line of the gyrus is coagulated with bipolar electrocautery and (b) divided with microscissors. (c) The pial flap is then elevated and the gray matter evacuated with gentle suction. Bipolar electrocautery is used for small vessel hemostasis, and occasionally to disrupt tissue that is not easily suctioned. (d) The resection is carried down to the bottom of the gray matter of the sulcus. White matter in the gyrus may be resected to this level, but not further.



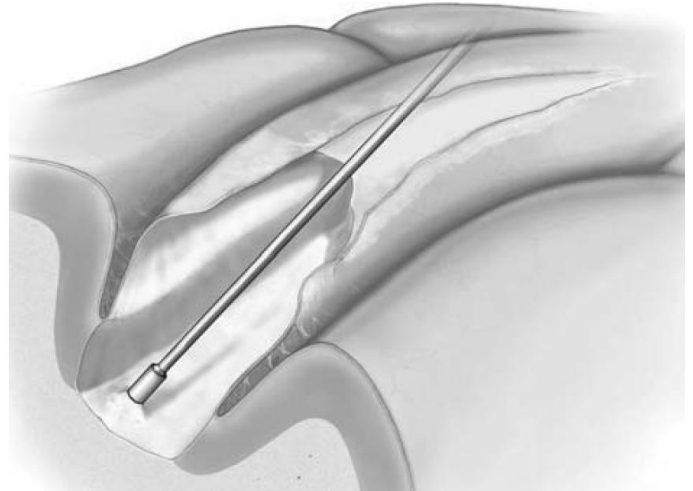
(a)



(b)



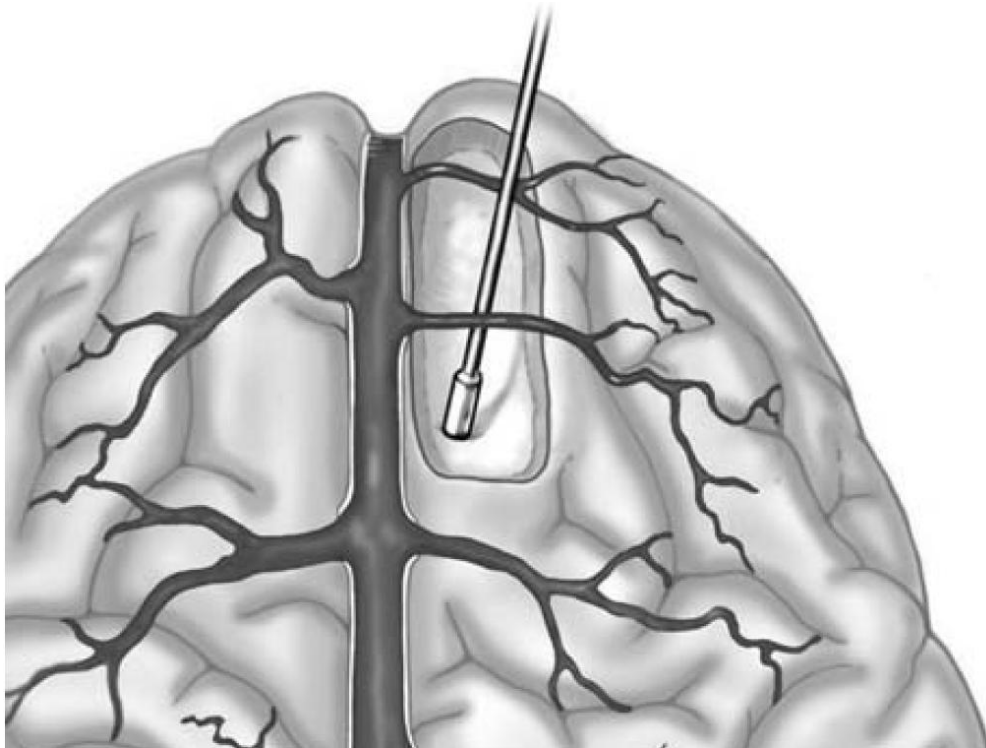
(c)



(d)

Source of picture: Lüders "Textbook Of Epilepsy Surgery" 1st ed. (2008); Publisher: CRC Press; ISBN-10: 1841845760; ISBN-13: 978-1841845760 >>

Preservation of vessels in topectomy. Respect for the vasculature is of the utmost importance, as sacrifice of en passage arteries or bridging veins may result in infarction of adjacent, functional cortex, leading to loss of function or a new epileptogenic focus:



Source of picture: Lüders "Textbook Of Epilepsy Surgery" 1st ed. (2008); Publisher: CRC Press; ISBN-10: 1841845760; ISBN-13: 978-1841845760 >>

- resection near eloquent areas where cannot expect complete sz freedom - leave RNS leads (no device) during the same resection procedure (if needed, can add RNS IPG later).

USE (US ELASTOGRAPHY)

- helps detect FCD and LGG intraop.
- significant learning curve, artefacts - might be confusing (advisable to have radiologist helping).
- it is OK to use US gel on cortex (just rinse off at the end).
- use US first before opening dura.

Multilobar Resection

- a) corticectomy (resection of grey matter)
- b) lobe excision (resection of grey and white matter)
- c) lobe disconnection
- d) combination.
- usually involves *frontoparietal*, *parieto-occipito-temporal*, or *parieto-occipital lobes*.
- indications as for functional hemispherectomy.

2. DISCONNECTIONS

Multiple Subpial Transections (MST)

see p. E17 >>

Hemispherectomies (functional, anatomical)

see p. E19 >>

Posterior Quadrant Disconnections

see p. E16 >>

Corpus Callosotomy

see p. E21 >>

Hippocampus Transections

see p. E15 >>

3. NEUROMODULATION (“ELECTROCEUTICALS”)

- electrical stimulation was first used therapeutically for epilepsy in the 1970s, when cerebellum became the first therapeutic target for electrical stimulation in human epilepsy, with mixed results.

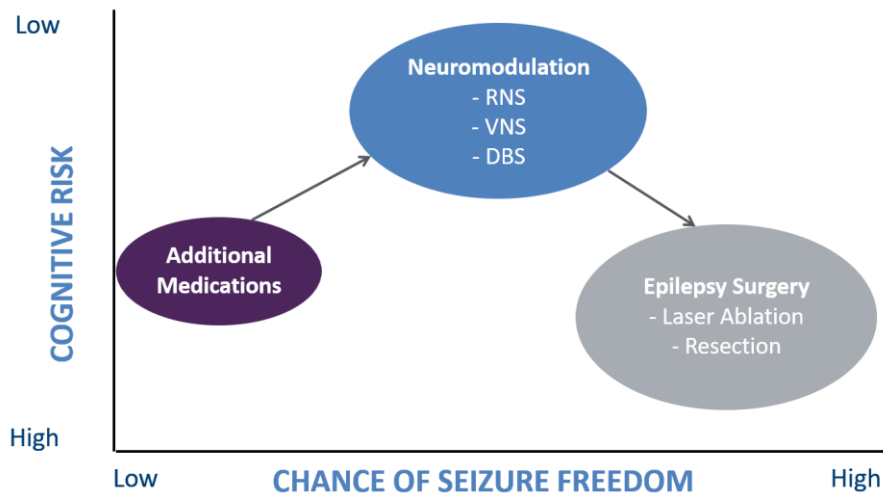
“The future of neurosurgery is not what we take out, it’s what we put in” – Richard D Penn, MD

Neuromodulation is not once and done. Programming is lengthy, lifelong process. Physician should be involved in programming!

Indications

1. **Unable to localize** seizure focus
2. **Unable to resect / ablate** seizure focus:
 - a) eloquent cortex
 - b) multiple foci
3. Need to collect **ambulatory ECoG data**

Neuromodulation may help bridge the treatment gap for many refractory patients:



"Patients tend to accept less effective but also less invasive procedures"

Mechanism Of Action

Two theories (mutually nonexclusive and maybe overlapping):

1. **Modulation** of Network / Node.
 2. **Disruption** of Network / Node abnormal resonance / synchrony.
- most modern approach – **thalamic neuromodulation** (for regional, multifocal, or generalized onsets) - influencing cortical seizure onset zones via diffuse cortical projections from the thalamus.
 - stimulating tracts is more important than regions (role of connectomics).
 - *stimulated networks may not be pathological*, they may be normal compensatory tracts!
 - connectivity in standardized connectome atlas (1000 normal subjects) may not be the same as in particular patient.

Two stimulation concepts:

- A. **Closed loop** (on-demand stimulation) – RNS philosophy: acute **abortive** effect of stimulation during a seizure might act through **acute desynchronization** to arrest a seizure (like electroconvulsive therapy in depression)
 - Penfield and Jasper noted: inhibitory polarization caused by transmembrane currents of applied current → flattened local electrocorticography pattern.
- B. **Open loop** (continuous stimulation) – DBS philosophy: chronic **neuromodulatory** (disease-modifying) effect.
 - synapses are saturated after the seizure event (impairing long-term network plasticity after a seizure) vs. stimulation during periods of relative ictal quiescence leads to **long-lasting functional network changes**.

Effect determinants

1. **Target** matters!
2. Stimulation **parameters**:
 - a) *don't matter* (e.g. lesion and DBS in the same target cause same effects).
 - b) *matter a lot* - maybe a reason for some failures (need lots of fine tuning to find the right parameters); most important is **stim frequency**!

INWARD INTERICTAL INHIBITION

- precise mechanism leading to long-term seizure reduction is still debated.
- evidence is emerging of inward interictal inhibition from brain regions not involved in ictogenesis onto the seizure onset zones during rest and single-pulse stimulation.

Johnson GW, Doss DJ, Morgan VL, et al. The Interictal Suppression Hypothesis in focal epilepsy: network-level supporting evidence. Brain. 2023;146(7):2828-2845

Comparison Of Neuromodulations

Feature	VNS	DBS	RNS
Closed-loop	Partially (newer models may detect tachycardia)	No (newer models record LFPs)	Yes (detects electrographic activity)
Tonic (continuous) stimulation	Yes (by design)	Yes (by design)	No (by design, but can be turned on)
Recording capability*	No	Low resolution sensing: 6 data points/hr, 5 Hz range	High resolution sensing: 250 data points/sec, 1-125 Hz range
Biomarker (for epilepsy) collection	Very limited	Limited (LFPs)	Yes
Newest model	Sentiva 1000 (2017), Sentiva DUO (2023)	Percept PC (2020)	320 (2019)
Generalized or multifocal epilepsy	Yes	Probably (thalamic CM nucleus)	Probably (thalamic CM nucleus)
Invasiveness (intracranial)	No	Yes	Yes
Ease of surgical explantation	Usually, coils are left on nerve	Easiest	Leaves skull defect (needs cranioplasty)
Recording capacity	No	LFPs	Yes (1 MB)
Indication for depression	Yes	No	No
Positive effects on mood & cognition	Yes	Unknown	Probably
Children	Yes (age ≥ 4 years)	Off-label	Off-label (predicted in 2024)
Patient compliance needed	Not needed	Not needed	Yes
Feedback to the patient	None / swipe magnet to feel stimulation	Yes	Yes
Patient control	None	Yes	None
MRI	Yes, exclude C7-T8 region (up to 3 T)	Yes, full body (up to 3 T with Percept®)	Yes, full body (up to 1.5 T)
MRI exclusion zones	Yes	No	No
Side effects during stimulation	Yes	No (memory?, depression?)	No
IPG site	Infraclavicular	Infraclavicular	Cranial
Regulatory approval (2018 December)	The whole world (FDA – 1997; Europe - 1994)	USA (FDA - 2018), Europe (CE mark - 2010), Canada,	USA only (FDA - 2013)

		Australia; not available in UK	
Outcomes			
Biomarker predictors of therapeutic response**	Non-lesional epilepsy, generalized seizures	Temporal theta-band desynchronization, hippocampal-evoked potentials, increased functional connectivity between ANT and the default mode network	None
Median seizure reduction at 1 year	45% (EO5)	41% (SANTE)	67% (Post-approval)
Best median seizure reduction***	76% at 10 yrs	75% at 7 yrs 56% at 5 yrs (MORE)	75% at 9 yrs (trial data), 82% at 3 yrs (post-trial data)
Responder rate (> 50% seizure reduction)		41% at median 3.5 years (MORE)	
Seizure remission****	Long term follow up: 8% were seizure free	At 7 yrs: 18% of patients experienced at least one 6-month seizure-free period, 7% were seizure-free for the preceding 2 years. At 3.5 yrs: 4% (MORE)	At 7 yrs: at least one seizure-free period lasting: ≥ 3 months - 39% of patients ≥ 6 months - 29% of patients ≥ 1 year - 16% of patients

*value of recorded data is evident in at least 6 domains: informing additional surgical procedures, seizure forecasting, medication management, lifestyle management, spell characterization, and biomarker detection.

- 60% of individuals exhibit multidien seizure cycles, and 89% - circadian cycle.
- RNS may predict whether a new antiseizure drug would be efficacious within 1-2 weeks.

**reliable predictors of therapeutic response are lacking across neuromodulation options! (also failure of one neuromodulation option does not predict response to another neuromodulation option)

***follow up studies dismissed patients who stopped using devices or were lost for follow up – if to count those as treatment failures, responder rate (50% seizure burden reduction) remains stable over time at ≈ 40% across different devices.

******medical therapy alone** in drug-resistant epilepsy gives 12-month seizure-remission in 33.4% of patients at 7 years!

Handforth, A et al. Vagus nerve stimulation therapy for partial-onset seizures: a randomized active-control trial. Neurology. 1998;51(1):48-55.

SEIZURE OUTCOMES

With neuromodulation **there is no endpoint** as **programming options are infinite!**

Therapy optimization in neuromodulation is slow but **never plateaus!**

VNS, RNS, and DBS all have RCTs (Rolston et al., 2012)

TABLE 1: Randomized controlled trials of VNS compared with other stimulation-based therapies*

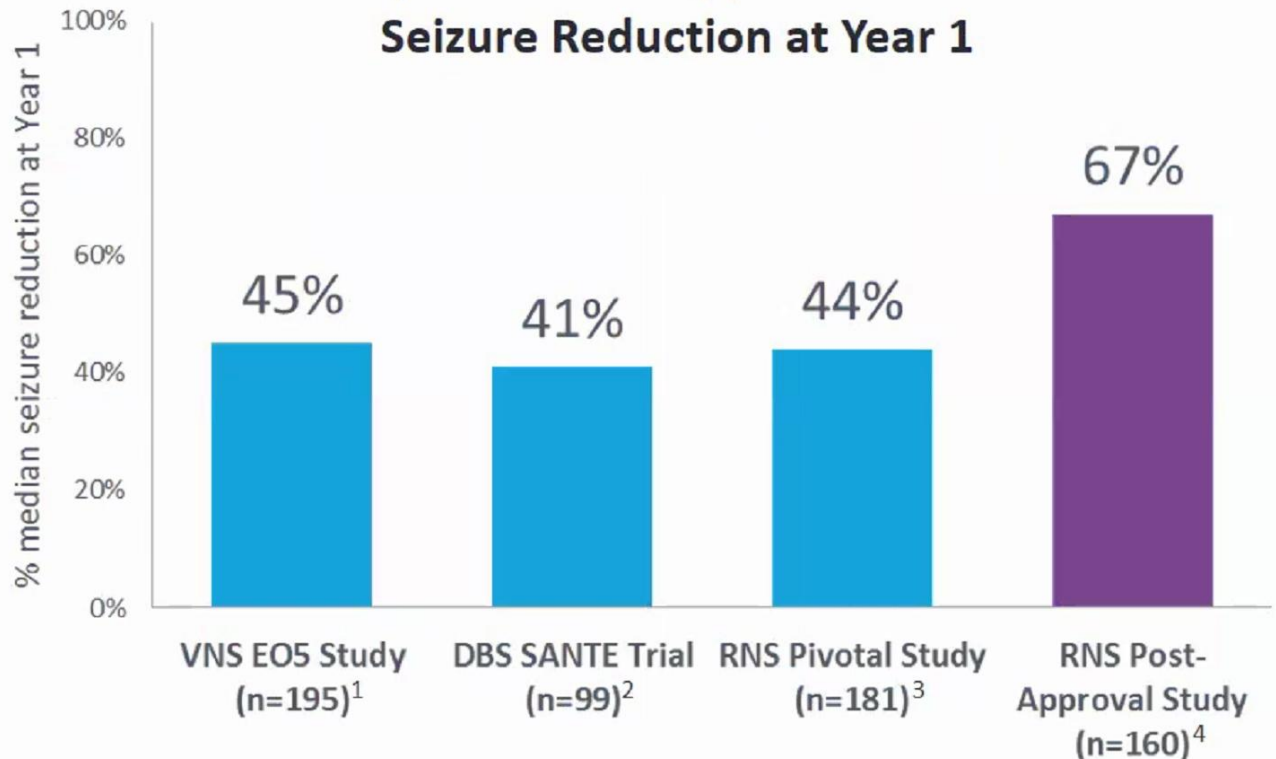
Study	No. Patients (no. in active group)	% w/ Seizure Reduction, Blinded (95% CI)	% w/ Seizure Reduction, 1 Yr	% Responder Rate, Blinded	% Responder Rate, 1 Yr	Regulatory Approval	
						FDA	CE Mark
VNS						yes	yes
EO3	114 (54)	24.5 (14.1–34.9)	43	31			
EO5	196 (94)	27.9 (21.0–34.8)	45	23.4	35		
thalamic DBS—SANTÉ	109 (54)	40.4 (NR)	41	NR	43	†	yes
cortical stimulation—RNS	191 (97)	37.9 (27.7–46.7)	NR	29	43	†	†

* Seizure reduction is defined as change in actively treated patients compared with their baseline seizure frequency. Responder rate is defined as a $\geq 50\%$ reduction in seizure frequency experienced in actively treated patients. The $\geq 50\%$ responder rate is not reported in the SANTÉ trial, although it was not significantly different from the untreated group. NR = not reported.

† Pending review.

	VNS (LilaNova)	RNS (NeuroPace)	DBS (Medtronic)
Median Seizure Reduction Versus Baseline (%):	Year 1	• 45% (N=195, EO5) • 33% (N=102, EO3)	• 44% (N=181)
	Year 2	• 41% (N=51, EO3)	• 53% (N=174)
	Year 3	• 40% (N=49, EO3)	• 60% (N=214)
	Year 4		• 63% (N=204)
	Year 5		• 65% (N=172)
	Year 6		• 66% (N=115)
	Year 7		• 72% (N=185)
	Year 8		• 73% (N=163)
	Year 9		• 75% (N=147)
Responder Rate > 50% Seizure Reduction:	Year 1	• 35% (N=195, EO5)	• 44% (N=181)
	Year 2	• 39% (N=51, EO3)	• 55% (N=174)
	Year 3	• 39% (N=49, EO3)	• 58% (N=214)
	Year 4		• 61% (N=204)
	Year 5		• 61% (N=172)
	Year 6		• 59% (N=115)
	Year 7		• 66% (N=185)
Seizure Freedom Rate:	≥ 3 Months		• 39%
	≥ 6 Months		• 28%
	≥ 1 Year		• 16% (5 years)
	≥ 2 Years		• 5.4% (5 years)

Prospective FDA-Approved Studies Seizure Reduction at Year 1



Note: Therapies were studied in different clinical trials. Caution must be exercised when comparing results.

1) DeGiorgio CM, *Epilepsia*, 2000 2) Salanova et al., *Neurology*, 2018 3) Heck et al., *Epilepsia*, 2014 4) Szaflarski et al., Poster AES 2019 5) Morrell MJ, *Neurology*, 2011

6) Ruoff et al., *Epilepsia Open*, 2020 (n=5) *N values are small and caution must be taken when interpreting results. 7) Meador et al., *Epilepsy Behav*, 2015 8) Loring et al., *Epilepsia*, 2015

STIMULATION TIME

Average time stimulation is delivered in 24 hrs:

- RNS – 6 mins (in patients with frequent detections, stimulation time may approach DBS)
- VNS – 130 mins
- DBS – 240 mins

RECORDING CHRONIC INTRACRANIAL DATA

Value of these data is evident in at least 6 domains:

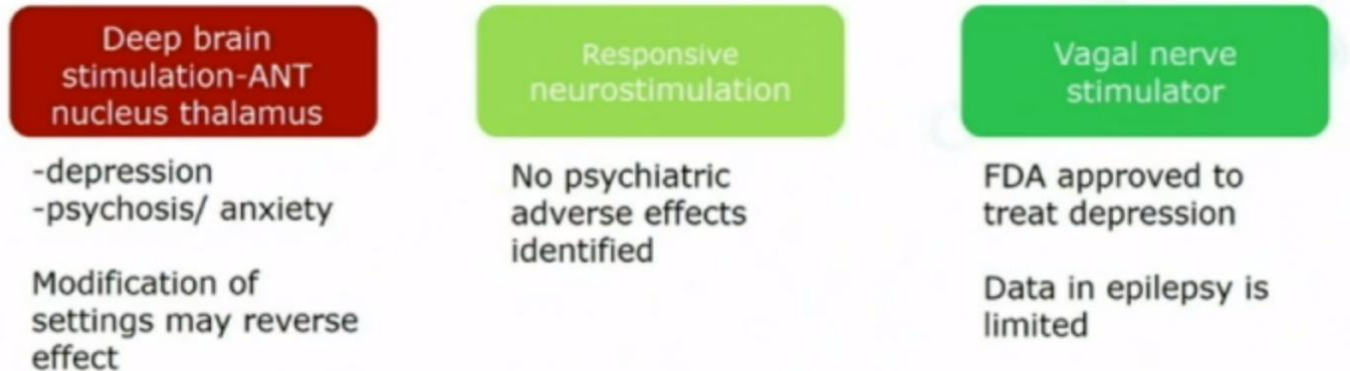
- 1) **informing additional surgical procedures**
 - among 157 RNS patients with presumed bilateral epilepsy, 25 patients (16%) had a mesial temporal lobe resection informed by chronic ambulatory iEEG.
Hirsch LJ, Mirro EA, Salanova V, et al. Mesial temporal resection following longterm ambulatory intracranial EEG monitoring with a direct brain-responsive neurostimulation system. Epilepsia 2020;61(3):408–20
- 2) **seizure forecasting:** interictal epileptiform activity oscillates with subject-specific **multidien (multiday) periodicity** (60% patients), in addition to the well known **circadian rhythms** (89% patients)
 - seizures occur preferentially during the rising phase of these multidien rhythms of interictal activity – information to adjust medication timing to increase efficacy and reduce side effects.
- 3) **medication management**

- in RNS patients with stable detection settings, rates of interictal epileptiform and ictal detections predicted whether a new ASM would be efficacious, within the first 1 to 2 weeks.

Quraishi IH, Mercier MR, Skarpaas TL, et al. Early detection rate changes from a brain-responsive neurostimulation system predict efficacy of newly added antiseizure drugs. Epilepsia 2020;61(1):138–48

- 4) lifestyle management
- 5) spell characterization
- 6) biomarker detection.

PSYCHIATRIC SIDE EFFECTS



SO HOW TO CHOOSE NEUROMODULATION TYPE?

- think about end user:

RNS – long term ECoG in compliant patient.

DBS – 10-15 yrs battery life (if using RC battery) but limited biomarker collection.

DBS and VNS may seem similar but DBS gets physically into epilepsy network.

- regardless of neuromodulation modality chosen, the optimal time interval to adjust parameters (if no efficacy) – 3 months.

DBS vs RNS if either is placed in seizure focus – different opinions (FDA approved only RNS).

COMBINATION OF NEUROMODULATIONS

N.B. neuromodulations can be combined - like two-three medications with different mechanism of action.

Focal epilepsy

- DBS may be working similarly to VNS, other studies show additive effects.
- RNS (added to VNS) doubles responder rate; maybe the same with ANT+RNS.
- RNS leads, longer DBS leads can capture two thalamic nuclei.

Generalized epilepsy

- DBS-CM (added to VNS) doubles responder rate in LGS.
- DBS effects on Parkinson's disease and Epilepsy are opposite - if one gets better, the other gets worse.

Deep Brain Stimulation (DBS)

see p. E27 >>

Vagus Nerve Stimulation (VNS)

see p. E23 >>

Responsive Neurostimulation (RNS)

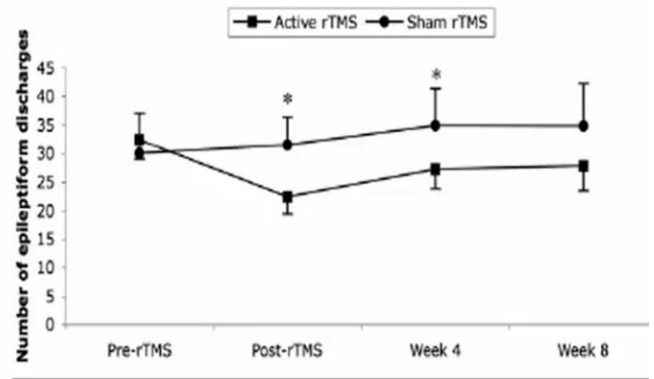
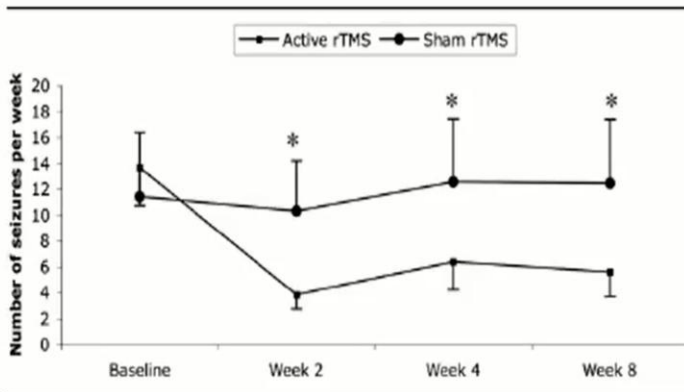
see p. E25 >>

Transcranial Magnetic Stimulation (nTMS)

- cortical stimulation with **low-frequency trains** decrease cortical excitability.
- some studies show ne efficacy.
- Randomized Double Blind Sham Controlled Trials

Study	Theodore (NIH) 2002 Neurology	Fregni (Sao Paulo) 2006 Ann Neurol	Cantello (Multicenter Italy) 2007 Epilepsia
N	24	21	43
Sz Type /Location	Mesial TLE 10 Lateral TLE 10 Frontal 3 Parietal 1	neocortical	mTLE 7 Neocortical 34
MRI	all lesions except MS were excluded	all malformation of cortical development (MCD)	Posttraumatic 6 MTS 5 Hemiatrophy 4 Cortical dysplasia 2 Tuberous Sclerosis 1
Stim Target	EEG identified <i>seizure focus</i>	MRI abnormality in 17, vertex in 4 with diffuse MCD	Vertex
Stim Parameters	1 Hz, 120% MT, 900 stim bid x 7 days	1 Hz, 70% max coil output, 1,200 stim/day x 5 days	0.3Hz, 120% MT, 1,000 stim/day x5 days
Outcome	No sig diff in stim vs sham, trend for better results in neocortical	Stimulated 2 weeks 72% p=0.003 4 weeks 53% p= 0.002 8 weeks 58% p=0.001 Sham – no change	No effect

- Fregni et al. 2006:



- rTMS may be effective in patients with epilepsy
 - In 21 patients with seizures due to malformations of cortical development, rTMS resulted in significant decrease in seizure count and the frequency of epileptiform discharges
- Chinese trial:

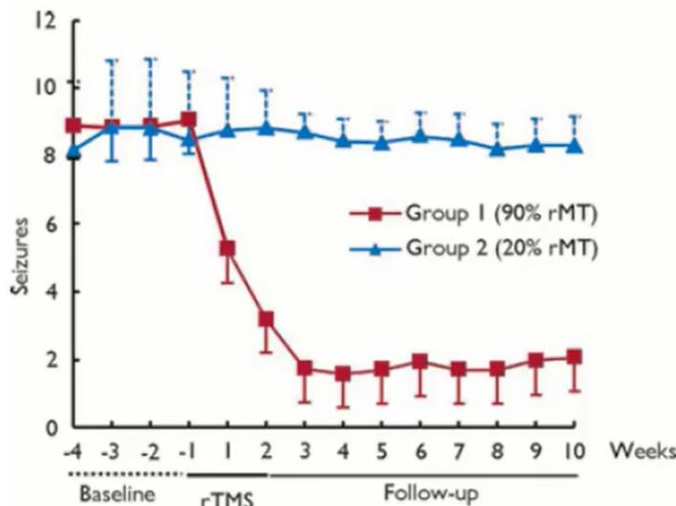


Table 2. Effect of rTMS on seizures (mean ± SD)

	Baseline	Follow-up	SRR (%)
Group 1 (n = 31)	8.9 ± 1.1	1.8 ± 3.7 ^a	79.8
Group 2 (n = 29)	8.6 ± 1.0	8.4 ± 10.1 ^b	2.3

^aSignificantly different from baseline ($p < 0.05$).
^bSignificantly different from group 1 ($p < 0.05$).
 SRR, Seizure Reduction Rate = [(Baseline seizures - Follow-up seizures) / Baseline seizures] × 100 (%).

Sun et al, 2012:

60 patients with refractory focal epilepsy randomized to receive 2 weeks of low-frequency rTMS at a moderate or very low intensity

4. CELL THERAPY

Transplantation of human stem cell-derived inhibitory neurons suppress recurrent seizures - Merab Kokaia (Sweden) at IEC 2023 (Dublin)

- TLE
- in-vitro human study + in-vivo animal study.
- injected into bilateral hippocampus.
- form synapses with adult human neurons.

- suppress seizures in chronic epilepsy animals.

POSTOPERATIVELY

- patient is returned to seizure monitoring room in epilepsy unit (vs. overnight ICU).

ASM

Patients generally need to remain on ASM therapy (for $\approx$ 2 years); some continue to require ASM therapy to remain seizure free (e.g. if pathology showed heterotopia, then other foci of heterotopia likely exist).

N.B. surgery is not indicated if patient expects to be ASM-free!

- even if almost seizure free can try to reduce medication.
- protocol: ASM withdrawal is initiated at 3 months in patients on ≥ 2 drugs and at 1 year for patients on a single drug.
- **seizure recurrence** occurs in 28.2%* on attempted withdrawal (regardless, 86% become seizure-free and 18% become drug-free after initial recurrence).
 *risk is only 17% if hx of **febrile seizures**, **normal postoperative EEG at 1 year**, and **duration of epilepsy of < 20 years**

60% of patients (53% in temporal lobe epilepsy) are able to stop all ASMs (**22% remain seizure-free and medication-free after > 5 years**)

- if patient is taking 2 ASMs, least effective drug is tapered after 1 year.
- if patient is seizure free at 2 years, remaining drug can be tapered.
- one seizure during or after withdrawal $\rightarrow$ resume single medication therapy.
- negative risk factors: multiple lesions, EEG abnormalities, post operative seizures, longer preoperative seizure duration - but even they deserve a chance to be free of medication if seizure free!
- Can weaning medication make things worse? Probably not (Kerling 2009, Jehi 2012)
 - **70% regain control** (Menon et al 2012).
 - those who recur are controlled with less aggressive regimen.
- **postoperative seizures**:
 - seizures **within first 24 hours ("honeymoon seizures")** do not correlate with poor long-term seizure outcome - may be due to irritation and edema of tissues adjacent to resection (neighborhood seizures).

Convulsions in immediate postop phase are common in first 3-5 days, if they are stereotypic of seizures prior to surgery, they may indicate lower chances of success and seizure freedom, but otherwise, they are not of huge concern unless they do not stop.

Acute postoperative seizures (up to 1 week after surgical resection) are not counted as evidence of recurrent epilepsy

- seizures **after 48 hours** (with adequate serum ASM levels) do not bode well for eventual outcome.
- N.B. many patients may continue to have **components of preoperative auras**;
- some may have **occasional seizures for few years** which then cease (**wind down**);
 - others may be **seizure-free for few years**, then have recurrence.

Alternative view – early withdrawal of ASMs - safe and does not affect long-term seizure outcome or cure, might unmask incomplete surgical success sooner (identifying patients who need continuous drug treatment and preventing unnecessary continuation of ASMs in others).

- if seizures recur postop, only minority do recur during ASM withdrawal.
- if seizures recur, restarting ASM helps regain control.

BEFORE DISCHARGE

- 1) neuropsychiatric evaluation? (usually delayed for 6-12 months)
- 2) serum anticonvulsant levels
- 3) EEG

SURGERY OUTCOMES - SEIZURES

Virtual resection tool - to predict sz success.

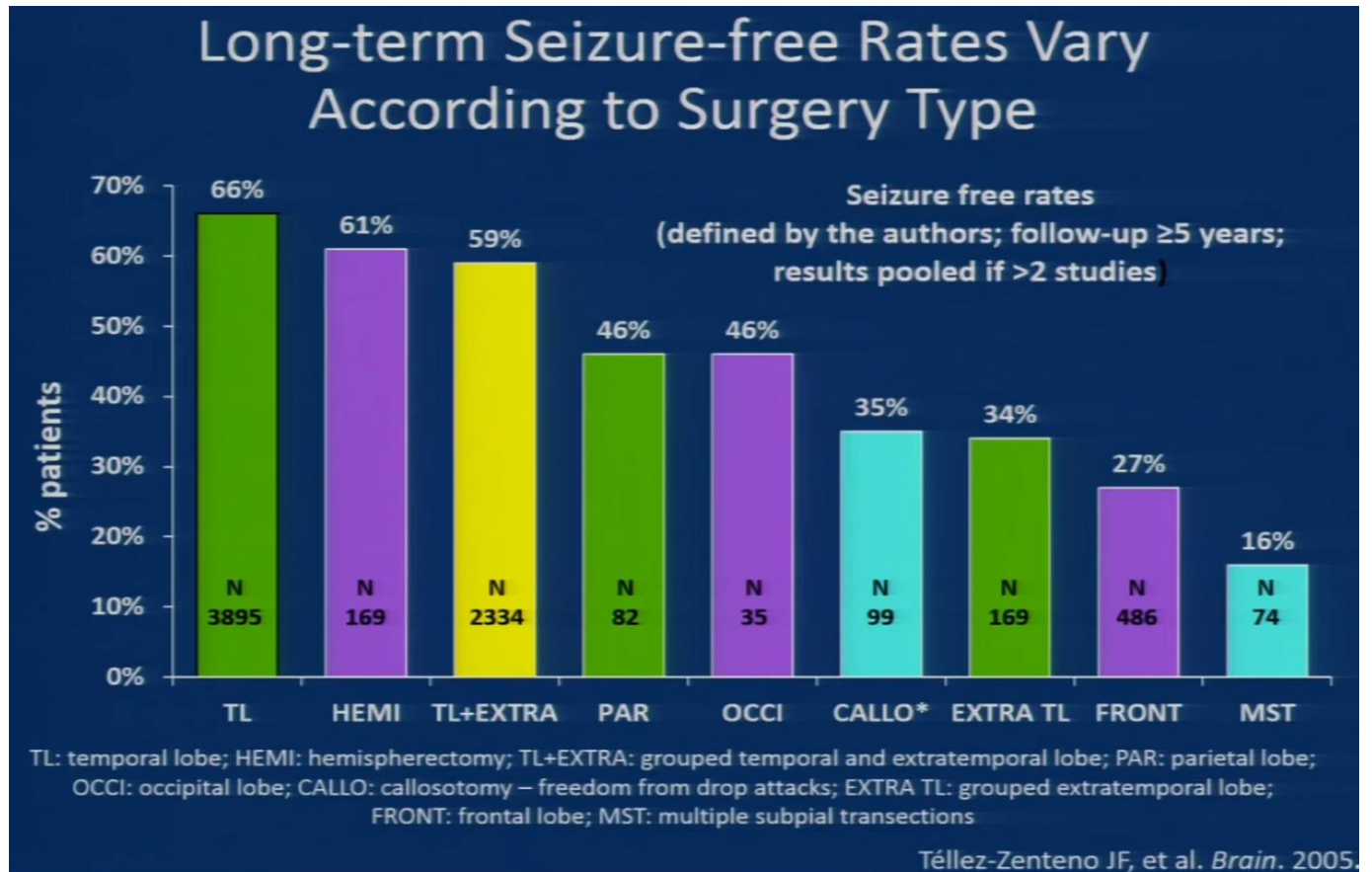
The only reliable predictor of seizure-free outcome – **resection of lesional focus!** (unclear if laser ablation qualifies for that).

- goal of epilepsy surgery:
 - palliative** epilepsy surgery – *improving quality-of-life* by seizure reduction (still can be very desirable for some patients, e.g. getting rid of convulsive seizures)
 - radical** epilepsy surgery – *seizure freedom*
- in medical therapy, > 50% reduction in seizures = success
- seizure control is assessed at 1, 3 & 6 most post op, and then annually.
 - 90% of seizures that recur do so within 2 years; per Dr. R. Gross – if seizures recur, they do it within 6 months (therefore, **outcomes are measured earliest at 12 months**).
 - neuromodulation antiseizure effect keeps improving over time.
 - “It takes time for epilepsy to develop, it takes time to undevelop”

Procedure	Seizure Free (%)	Improved* (%)
Anterior temporal lobectomy	76	24.0-27.2
Amygdala-Hippocampectomy	68.8	22.3
Extratemporal cortical resection	25.0-60	18.2-35.2
Lesionectomy	66.6	21.5
Hemispherectomy	67.4-77.3	18.2-21.1
Corpus callosotomy	7.6-8.0	60.9-80.0

*50% reduction of seizure frequency

Long-term (> 5 years) outcome combining all kinds of epilepsy surgeries – 22% seizure free and no ASMs



- ablative surgery is associated with IMPROVEMENT in *intelligence*, in *psychiatric* and *behavioral* disorders, in *social* and *vocational* function (abnormal brain tissue resection may remove undesirable functional effects which are interfering with function of other cortical areas).
N.B. cognitive function starts to improve after successful surgery but after the lag period.
- parents often wonder whether their child's personality will change or if he / she will become more depressed or anxious after undergoing resection of temporal or frontal lobe surgery – *surgery doesn't appear to have major effect on mood or anxiety in children*, and some kids even do better.
- RUNNING DOWN phenomenon** - gradual decline of seizures over several months or years until seizure freedom is achieved after surgery.

CLASSIFICATION OF OUTCOMES

These scales were created to assess **outcomes relative to seizure freedom**.

N.B. scales do not capture extensive quality-of-life changes that can occur in patients with intracranial neuromodulation without seizure freedom!

Jerome ENGEL'S classification of postoperative outcome:

Class I: Free of disabling seizures

- Completely seizure free since surgery
 - Nondisabling simple partial seizures only since surgery
 - Some disabling seizures after surgery, but free of disabling seizures for at least 2 years
 - Generalized convulsions with ASM discontinuation only
- class 1 includes patients with *residual auras*.
 - auras do not bother patient if they are infrequent.

2) depending on nature of auras (e.g. intense fear), they can affect quality of life.

Class II: Rare disabling seizures (“almost seizure free”)

- A. Initially free of disabling seizures but has rare seizures now
- B. Rare disabling seizures since surgery
- C. More than rare disabling seizures since surgery, but rare seizures for the last 2 years
- D. Nocturnal seizures only

Class III: Worthwhile improvement (> 90% reduction)

- A. Worthwhile seizure reduction
- B. Prolonged seizure-free intervals amounting to greater than half the followed-up period, but not < 2 years

Class IV: No worthwhile improvement (< 90% reduction)

- A. Significant seizure reduction
- B. No appreciable change
- C. Seizures worse

ILAE classification of postoperative outcome:

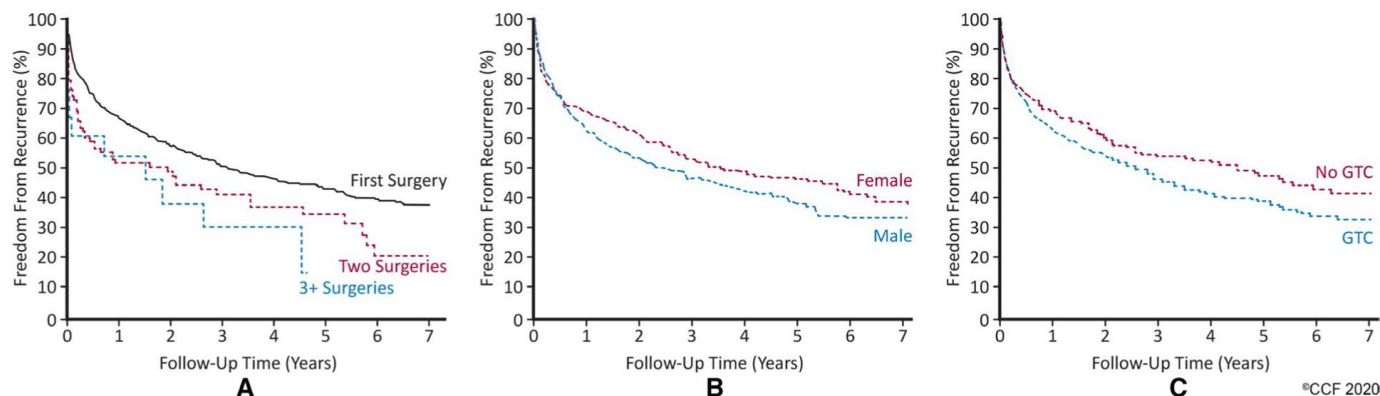
- 1. Completely seizure free; no auras
- 2. Only **auras**; no other seizures
- 3. **1-3 seizure days** per year; ± auras
- 4. **4 seizure days** per year to 50% reduction of baseline seizure days; ± auras
- 5. **Less than 50% reduction** of baseline seizure days; ± auras
- 6. **More than 100% increase** of baseline seizure days; ± auras

SEIZURE FREEDOM

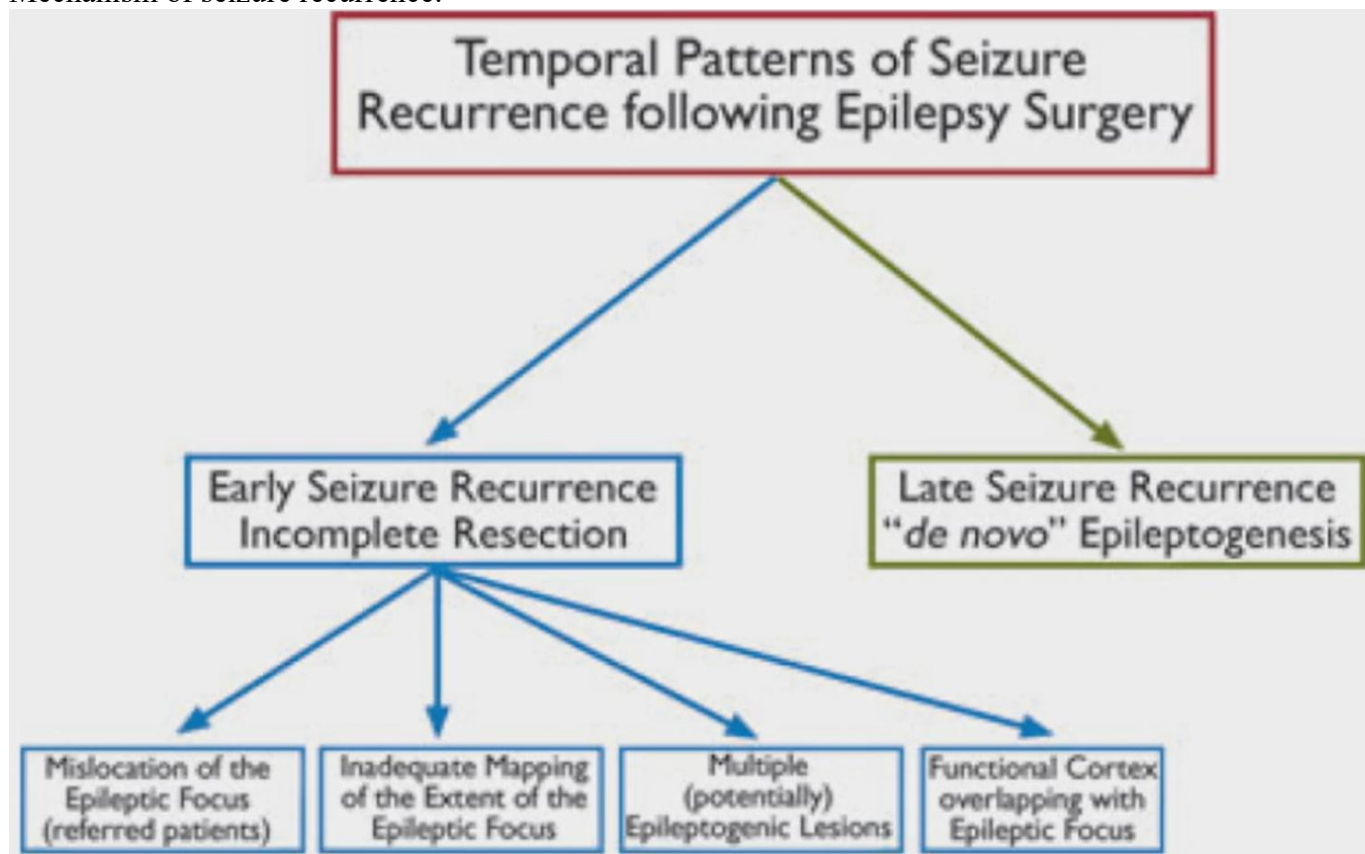
Factors predictive of freedom from seizures:

- 1) MRI-detectable **lesion** (unless functional constraints limit extent of resection); better temporal than extratemporal.
Lesional epilepsy has odds of seizure freedom **2.5 times higher** than nonlesional cases!
- 2) concordant interictal **epileptiform discharges**.
- 3) more extensive **resections**.

Kaplan-Meier curves showing seizure freedom in relation to: A, number of surgeries; B, sex of the patient; and C, history of generalization. GTC, generalized tonic-clonic seizures:



Mechanism of seizure recurrence:



TEMPORAL EPILEPSY

Seizure freedom (at 1 year)

ATL – 75% (< 50% in long term)

SAH – 67%

LITT – 58% Engel I (77% Engel I or II)*

RF – 27% (orthogonal trajectories), 78% (along hippo axis)

*60-89% seizure freedom in patients with radiographic evidence of hippocampal sclerosis

N.B. LITT has slightly lower (than open resections, including SAH) seizure outcomes but wins in minimal invasiveness with better neuropsychological outcomes

Predictors of surgery success

- 1) low **seizure frequency**
- 2) lack of **secondary generalization**
- 3) **unilateral** preoperative EEG and MRI findings
- 4) lack of need for **invasive EEG** recordings
- 5) only **specific pathological diagnosis** predicts **long-term seizure freedom**: 40% of patients with gliosis or nonspecific pathology were seizure free 10 years after TLE surgery, as opposed to 80% of those with a specific pathological abnormality.

RESECTIONS

- **Dr. Holloway**: with a typical left temporal lobe epilepsy coming from mesial sclerosis - 90% chance of a 90% reduction in the seizures and a 60-75% chance of a cure.

CANADIAN trial: surgery vs. meds for refractory temporal lobe epilepsy – 58%* vs 8% seizure freedom

Wiebe S et al. Effectiveness and Efficiency of Surgery for Temporal Lobe Epilepsy Study Group. A randomized, controlled trial of surgery for temporal lobe epilepsy. *N Engl J Med* 2001; 345 : 311 – 318.

Surgery for temporal lobe epilepsy is not only safe but also superior to prolonged medical treatment.

*64% if excluding those who did not have surgery

- **RCT with class 1 evidence!**
- 80 patients agreed to participate with 40 randomized to each arm of the trial.
- patients randomized to medical treatment were put on a 1-year waiting list for surgery (the standard practice in Canada) – avoids ethical conflict of not offering surgery but only one year of follow-up.
- patients randomized to surgery were admitted for pre-operative evaluation within 48 h of randomization.

Outcome	Surgery group	Medical group	Statistical significance
Freedom from disabling seizures	58% free	8% free	$p < 0.001$
Freedom from all seizures	38% free	3% free	$p < 0.001$
Change in frequency of disabling seizures	100%	34%	$p < 0.001$
Mean severity of residual seizures (Scale = 10–48)	21.4	26.5	No significant difference
Mean quality of life from 3 to 12 months (Scale 0–100)	72	59	$p < 0.001$
Percentage employed or attending school at 1 year	56.4	38.5	No significant difference

- **depression** occurred in 18 % of patients in the surgical group and in 20 % of patients in the medical group.
- subsequently, **AAN, AES, and the AANS** published a Practice Parameter in 2003 recommending **surgery as the treatment of choice for drug-resistant TLE** with the highest, **Level A**, rating.

ERSET (Early Randomized Surgical Epilepsy Trial) – surgery vs meds within 2 years of failure of 2 meds – 73% vs 0% seizure freedom

Engel J Jr, McDermott MP, Wiebe S, et al. Early surgical therapy for drug-resistant temporal lobe epilepsy: a randomized trial. *JAMA*. 2012;307(9):922-930.

- mesial TLE diagnosis by rigid criteria.
- 2 year follow up postop.
- 0/23 (0%) in the medical arm, and 11/15 (73%) in the surgical arm were completely seizure free (no auras) for 2 years in the intention to treat analysis.

COMPARISON OF DIFFERENT RESECTION TECHNIQUES

Subtemporal versus transsylvian SAH

Viola Lara Vogt et al. Neuropsychological outcome after subtemporal versus transsylvian approach for selective amygdalohippocampectomy in patients with mesial temporal lobe epilepsy: a randomised prospective clinical trial. *Journal of Neurology, Neurosurgery, and Psychiatry* 2017 December 22

- 47 patients randomised to subtemporal versus transsylvian approaches.
- cognitive functions were assessed before and 1 year after surgery.
- ILAE 1a was achieved in 62% of all patients without group difference with no significant effects of approach on cognition (incl. verbal recognition memory declined irrespective of approach).
- post hoc tests: the **subtemporal approach was associated with a greater memory losses** (worse outcome for verbal learning and delayed free recall as well as for semantic fluency); **left side of surgery** was associated with decline in naming regardless of approach.

Transcortical versus transsylvian SAH

Lutz MT, Clusmann H, Elger CE, et al. Neuropsychological outcome after selective amygdalohippocampectomy with transsylvian versus transcortical approach: a randomized prospective clinical trial of surgery for temporal lobe epilepsy. *Epilepsia* 2004; 45:809–816.

- **no difference** in **seizure freedom** (76.9% vs. 73.2%, respectively).
- **fluency improvements and gains** were significantly higher in the **transcortical group**.

Sparing of superior temporal gyrus

Hermann B, Davies K, Foley K, et al. Visual confrontation naming outcome after standard left anterior temporal lobectomy with sparing versus resection of the superior temporal gyrus: a randomized prospective clinical trial. *Epilepsia* 1999; 40:1070–1076.

- **no difference** between the groups with respect to **seizure outcomes** (60% resected vs. 55% preserved) or **confrontation naming**.

2.5-cm versus 3.5-cm mesial temporal resection (hippocampus, parahippocampal gyrus)

Schramm J, Lehmann TN, Zentner J, et al. Randomized controlled trial of 2.5-cm versus 3.5-cm mesial temporal resection in temporal lobe epilepsy—part 1: intent-to-treat analysis. *Acta Neurochir (Wien)* 2011; 153:209–219.

- **no difference** in regard to **Engel Class I seizure outcome** (74% for 2.5 cm vs. 72.8% for 3.5 cm group), **neurological complications**, **visual field defects**, or **surgical complications**.

Partial (to anterior edge of cerebral peduncle) vs. complete (to level of superior colliculus) hippocampectomy

Wyler AR, Hermann BP, Somes G. 1995. Extent of medial temporal resection on outcome from anterior temporal lobectomy: a randomized prospective study. *Neurosurgery* 37:982–990; discussion 990–991

- significantly higher **seizure freedom** in **complete** resection versus partial resection (69% vs. 38%).
- **no differences** in **cognitive outcomes** or **complications** between the two groups.

LITT

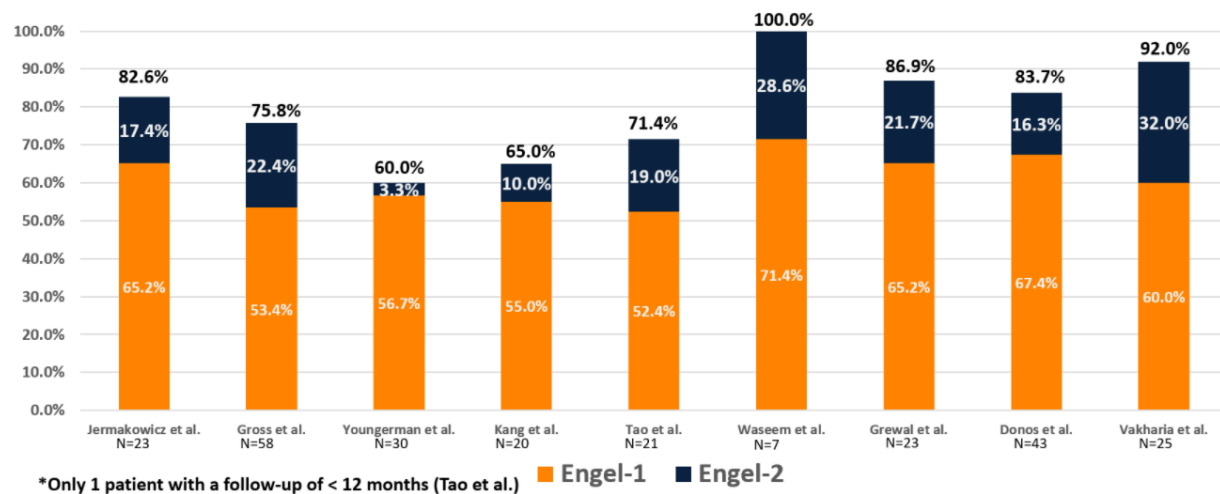
If LITT has > 43% seizure freedom and < 40% morbidity risk, then it is equivalent to open resection.

M Attiah et al. Epilepsy Res 115:1, 2015

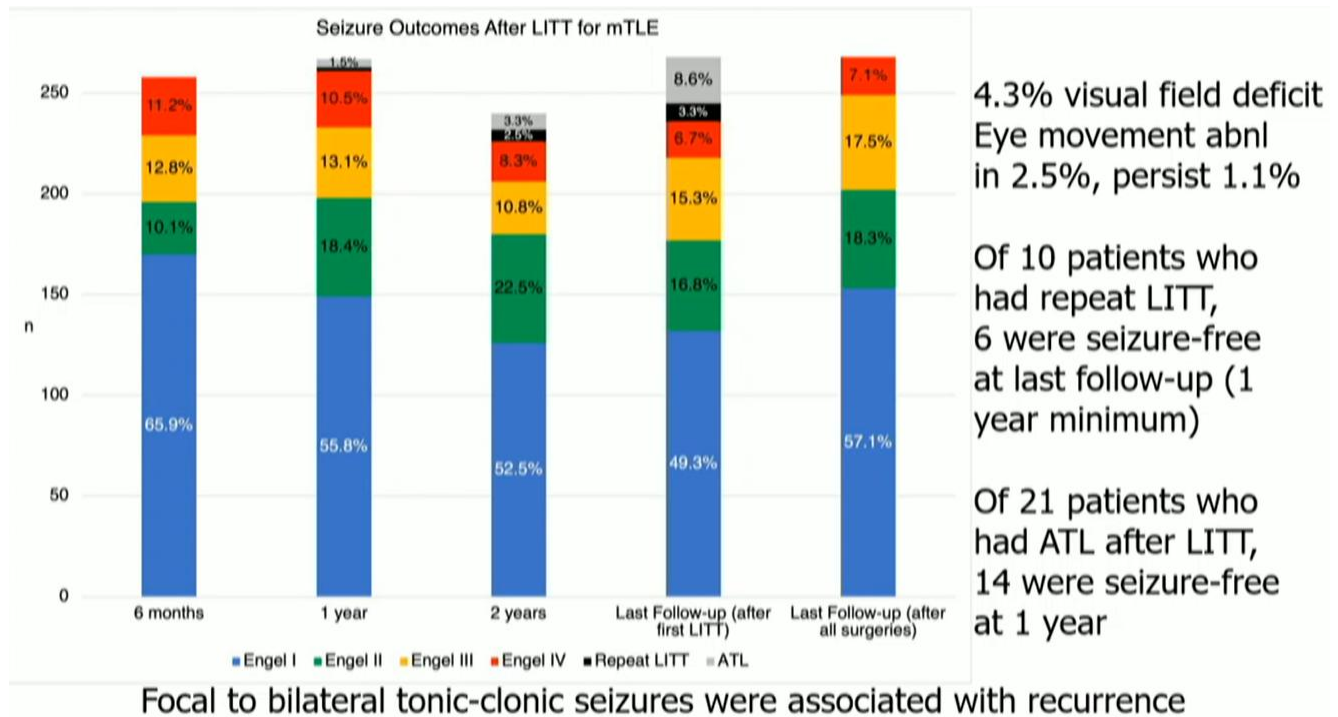
LITT vs Open – patient survey: sz freedom outcome difference > 23% is defining factor (however, LITT is only 10% inferior than open TLE)

ENGEL CLASSIFICATION:

Nine Studies / 250 patients with Laser Ablation (7-70 months*)



Grewal et al., World Neurosurg. 2019;122:e32-e47.



BE Youngerman et al. JNNP 2023 DOI: 10.1136/jnnp-2022-330979

Seizure recurrence after LITT:

- same semiology – “LITT missed target” → repeat LITT / open resection
- different semiology (plus, MRI shows good ablation volume) → SEEG

SRS

- Cleveland Clinic: results unsatisfactory, CCF trial terminated due to serious adverse events.

ROSE trial - SRS vs. surgery

Barbaro NM, Quigg M, Ward MM, Chang EF, Broshek DK, Langfitt JT, Yan G, Laxer KD, Cole AJ, Sneed PK, Hess CP, Yu W, Tripathi M, Heck CN, Miller JW, Garcia PA, McEvoy A, Fountain NB, Salanova V, Knowlton RC, Bagić A, Henry T, Kapoor S, McKhann G, Palade AE, Reuber M, Tecoma E. Radiosurgery versus open surgery for mesial temporal lobe epilepsy: The randomized, controlled ROSE trial. *Epilepsia*. 2018 Jun;59(6):1198-1207.

- stereotactic radiosurgery** (24 Gy to the 50% isodose targeting mesial structures) versus standardized **anterior temporal lobectomy** for pharmacoresistant unilateral mesial temporal lobe epilepsy (MTLE).
- randomized, single-blinded, controlled trial.
- 14 centers in the USA, UK, and India.
- 58 patients (31 in SRS, 27 in ATL).
- outcomes at 36-month follow-up:
 - seizure remission** (absence of disabling seizures between 25 and 36 months): **52% SRS and 78% ATL** patients achieved seizure remission (difference between ATL and SRS = 26%, upper 1-sided 95% confidence interval = 46%, P value at the 15% non-inferiority margin = 0.82).
 - verbal memory (VM)**: mean VM changes from baseline for 21 English-speaking, dominant-hemisphere patients did not differ between groups; consistent worsening occurred in **36% of SRS and 57% of ATL** patients.
 - quality of life (QOL)**: QOL improved with seizure remission.

- adverse events were anticipated cerebral edema for some SRS patients, and cerebritis, subdural hematoma, and others for ATL patients.
- conclusion:
 - ATL has an advantage over SRS in terms of proportion of seizure remission, and both SRS and ATL appear to have effectiveness and reasonable safety as treatments for MTLE.
 - SRS is an alternative to ATL for patients with contraindications for / reluctance to undergo open surgery.

Case series

- 21 patients with mesial temporal epilepsy/20 evaluable.
 - 5/20 symptomatic radiation-induced mass effect, 3 hospitalized - treated with steroids in most
 - 50% new visual field defect.
 - 65% seizure-free at two years.

Regis J, Rey M, Bartolomei F, et al. Epilepsia 45:5-4-515, 2004.

- 15 patients, > 5 yr follow-up, 24 Gy treatment.
 - 60% seizure free (on drugs).
 - 0% seizure free (off drugs).
 - mean time to effect, 12 months.
 - 60% temporary mass effect
 - increase in seizures early on

Bartolomei et al. Neurology (2008 May 6) 70(19):1658-63

- 30 patients, 17 @ 20 Gy, 13 @ 24 Gy
 - 67% seizure free @ 36 months HD.
 - 77% Seizure free @ 36 months LD.
 - 12% significant verbal memory deficit.
 - mean time to effect, 12 months.
 - increase in seizures early on.

Barbaro et al. Neurology (2008 May 6) 70(19):1658-63

COST-EFFECTIVENESS

TEMPORAL EPILEPSY

Cost-effectiveness of surgery for drug-resistant temporal lobe epilepsy in the US

Sheikh SR et al. Cost-effectiveness of surgery for drug-resistant temporal lobe epilepsy in the US. Neurology. September 08, 2020; 95 (10)

- surgery is cost-effective compared to medical management in surgically eligible patients by virtue of being cost-saving (\$328,000 vs \$423,000) and more effective (16.6 vs 13.6 QALY) than medical management in the long run.
- surgical evaluation is cost-effective in patients with DR-TLE even if the probability of being deemed a surgical candidate is only 5%.
- from a societal perspective, surgery becomes cost-effective within 3 years, and 89% of simulations favor surgery over the lifetime horizon.
- patients with DR-TLE should be referred for surgical evaluation without hesitation on cost-effectiveness grounds!

SEIZURE WORSENING / DE NOVO SEIZURES

- facilitation of seizure propagation due to alteration of complex epileptogenic networks as a result of surgery.
- studies have shown that there are mixed trajectories of seizure freedom after epilepsy surgery, with relapses after initial seizure freedom as well as improvement over time in some patients who have recurrent seizures.

Risk factors:

multiple seizure semiology

incomplete resection

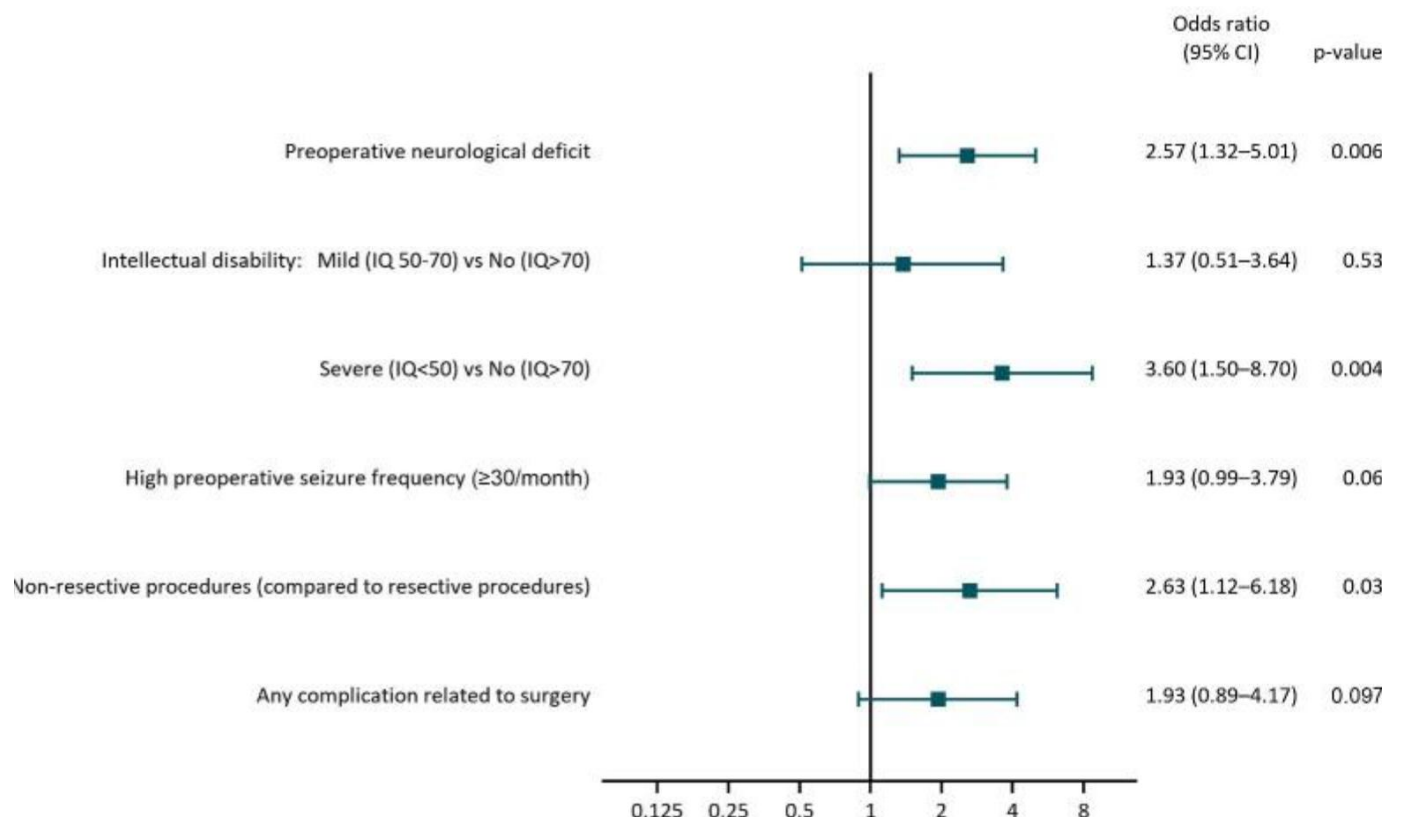
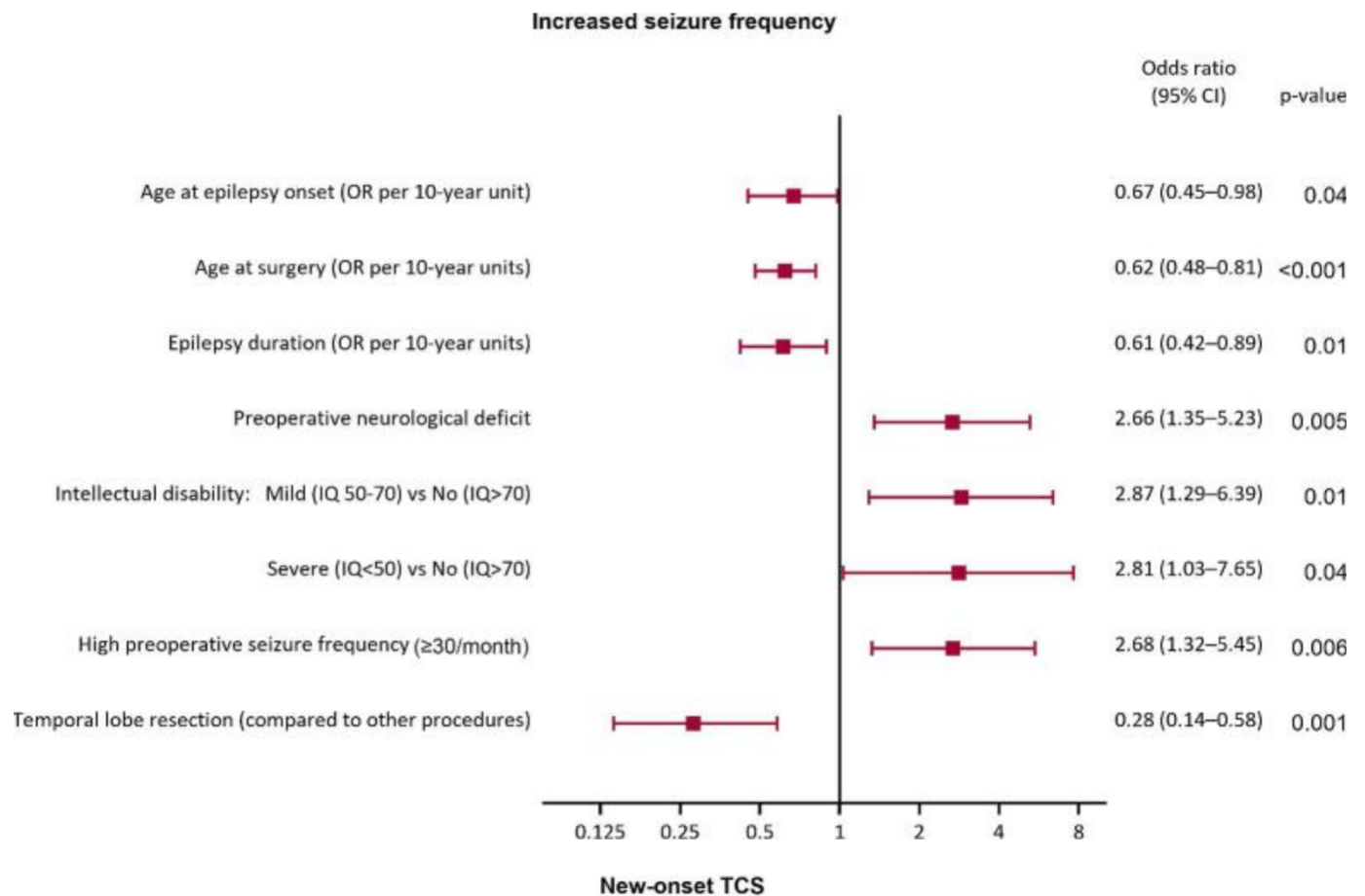
malformations of cortical development (e.g. postoperative status epilepticus has been reported in cases with incomplete resection of focal cortical dysplasia)

+ Swedish study below:

Population-Based Register Study (Sweden)

Johan Bjellvi et al. Risk Factors for Seizure Worsening After Epilepsy Surgery in Children and Adults: A Population-Based Register Study. Neurosurgery, Volume 87, Issue 4, 1 October 2020, Pages 704–711

- 2-year seizure outcome - 1407 procedures.
- 4.0% had increased seizure frequency compared to baseline (2.8% had >100 % increase in seizure frequency - ILAE class VI).
- new-onset TCS (grand mal) - 3.9% (6.6% in patients without preoperative TCS).
- risk factors:
 - 1) reoperations compared to first surgeries
 - 2) lower age of onset
 - 3) lower age at surgery
 - 4) shorter epilepsy duration
 - 5) preoperative neurological deficit
 - 6) intellectual disability
 - 7) high preoperative seizure frequency
 - 8) extratemporal procedures
 - 9) nonresective procedures (for new-onset TCS)



Risk of
worsening
seizures after
epilepsy
surgery:

- Overall: 10%
- New onset GTC: 1.4%
- New onset of Status: 2.2%
- SUDEP: 1.4%

Sarkis et al., 2012

SURGICAL REFRACTORINESS / RECURRENT SEIZURES / SURGERY FAILURE

“Widespread, ‘malignant’ epileptogenic network with a tendency to develop new epileptogenic zones upon resection of the prior zone, causing a progressive disease state”

Yardi Ruta et al. Long-term outcomes of reoperations in epilepsy surgery. *Epilepsia*. Volume 61, Issue 3 p. 465-478. 27 February 2020

- 898 patients (one-third were pediatric patients and 68% underwent temporal lobe resection), 110 had reoperations (92 had one resection prior to the index surgery and 18 patients had ≥ 2 prior resective surgeries).
- 2 years after index surgery - Engel I score:
 - 69% of patients with no prior surgeries
 - 42% of patients with 1 prior surgery
 - 33% of patients with ≥ 2 prior resections ($P < .001$).
- odds of seizure freedom diminish with each additional operation.
- risk factors for “surgical refractoriness”: male sex ($P = .011$), history of secondary generalization ($P = .016$), increasing number of prior surgeries ($P = .002$).

Epilepsy is a dynamic process both before and after surgery:

- drug resistance is a *time-dependent* process rather than an automatic consequence of a cortical lesion - focal epilepsy matures over years to decades up to the point of drug resistance; best illustrated in MTS, where a “honeymoon” period of seizure freedom is typically seen on or off ASMs before drug-resistant seizures develop, or in malformations of cortical development where, although the lesion is present at birth, seizures begin in teenage years or young adulthood.
- drug resistance is a *space-dependent* process - extent of the epileptic network spreads beyond the physical epileptic lesion and inversely correlates with seizure control in new-onset epilepsy - mechanical removing focus of DRE with epilepsy surgery interrupts but doesn't necessarily reverse molecular or structural changes that extend beyond the resection bed.
- after surgery, there is still a need in some cases for ongoing ASM use to maintain seizure control:
 - although 17–25% of postoperative seizure recurrences occur with ASM withdrawal, seizure control is regained after reinitiation of ASMs in 60–70% of these cases - group of “seizure free but only while on ASMs,” surgery converted DRE to a pharmacoresponsive disease.

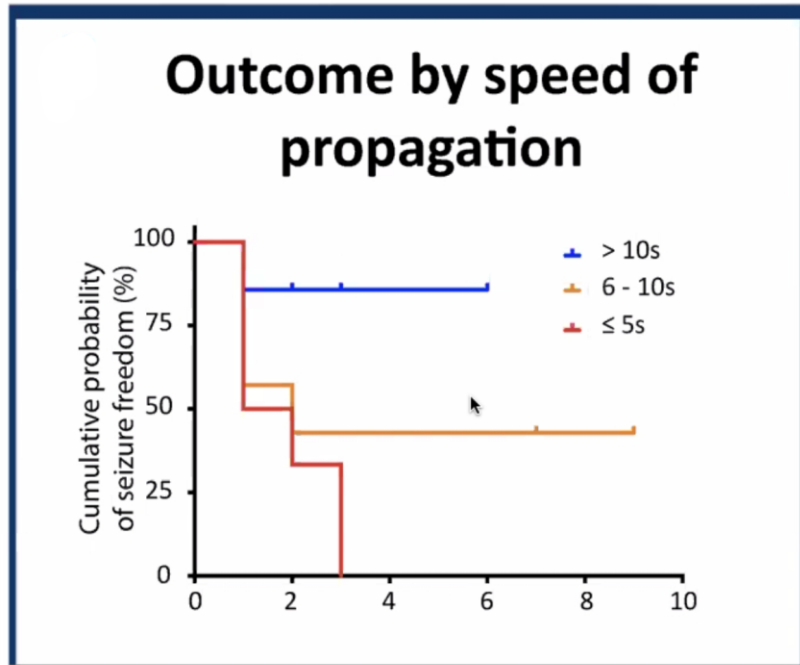
TEMPORAL EPILEPSY

Causes

1. **Incomplete resection** of epileptogenic zone
 2. **Incorrectly diagnosed** epileptogenic zone
 3. **Dual pathology** – discrete lesions in same temporal lobe (e.g. MTS and FCD)
 4. **Temporal +** epilepsy
 5. **Bitemporal** epilepsy
 6. **New** epileptogenesis
- patients with **surgically refractory** medial temporal lobe epilepsy (MTLE) exhibit distinct pattern of structural network organization involving temporal lobes and extratemporal regions as seen on MRI-DTI; compared with controls, not seizure-free patients exhibit higher connectivity between structures in 1) ipsilateral medial and **lateral temporal lobe**, 2) ipsilateral medial temporal and **parietal lobe**, and 3) **contralateral temporal pole** and parietal lobe

Bonilha L "Presurgical connectome and postsurgical seizure control in temporal lobe epilepsy." Neurology. 2013; 81(19):1704-10 (ISSN: 1526-632X)

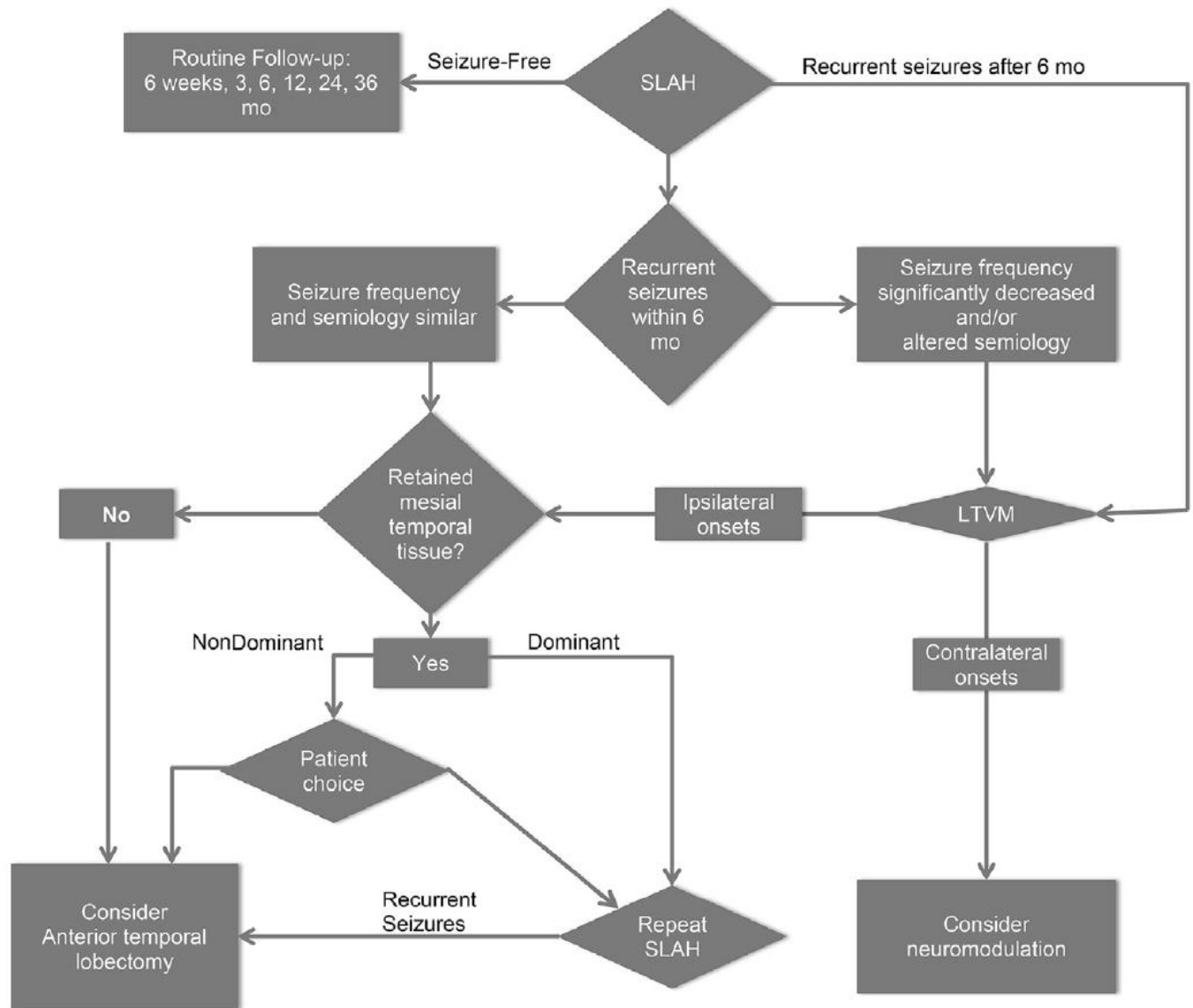
 - networks involving key components of medial temporal lobe and structures traditionally not removed during surgery may be associated with seizure control after surgical treatment of MTLE
 - **rapid seizure propagation (< 5 sec)** out of temporal lobe is a good predictor of **ATL failure**:



Patients with rapid seizure propagation were significantly more likely to fail ATML (* p < 0.05)

- Over 80% of patients with slow propagation are seizure free over 5 years later.
- No patients with rapid seizure propagation are seizure free at 5 years.
- Rapid seizure propagation may be an important marker for a more widespread and/or robust seizure network in MTL epilepsy patients

Decision tree for further surgical treatment in non–seizure-free patients after SLAH:



Robert E. Gross, Jon T. Willie, Daniel L. Drane. The Role of Stereotactic Laser Amygdalohippocampotomy in Mesial Temporal Lobe Epilepsy. *Neurosurg Clin N Am* 27 (2016) 37–50

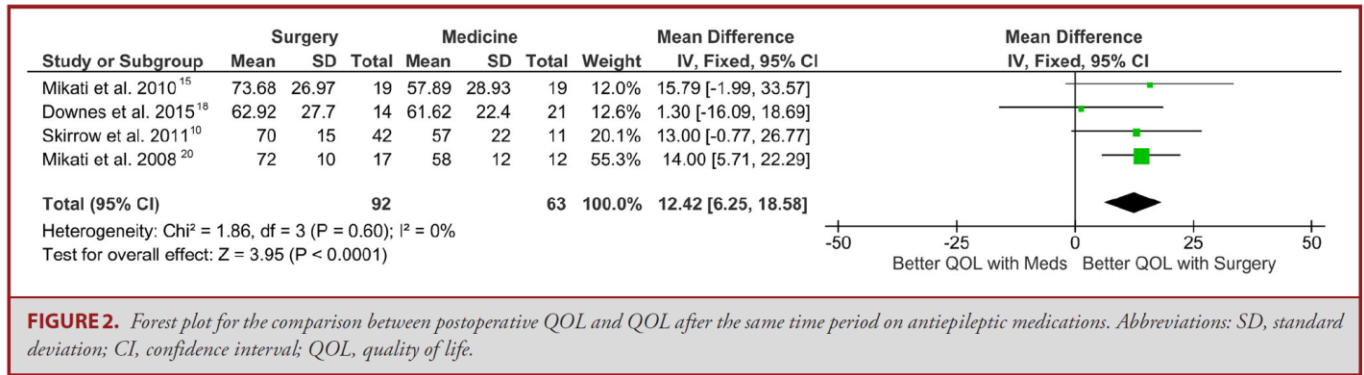
Temporal lobectomy after LITT – tissues sclerotic, increased risk of **ant choroidal injury** (H: CUSA)

SUDEP

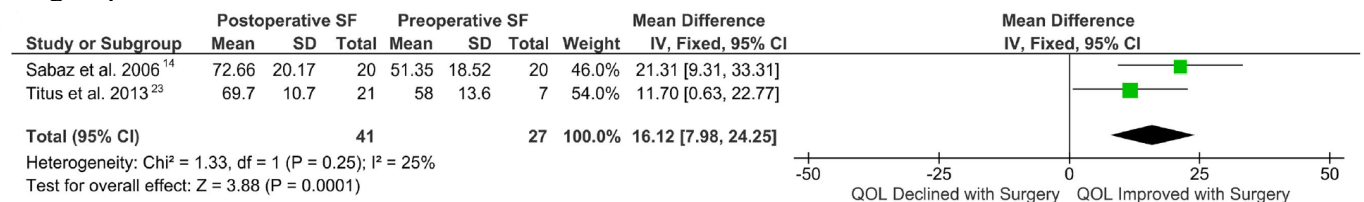
DBS and RNS are more effective than VNS in reducing SUDEP.

SURGERY OUTCOMES - QUALITY OF LIFE for patients, **fixed** postop **deficits** are easier to manage than **unpredictability** of epilepsy.

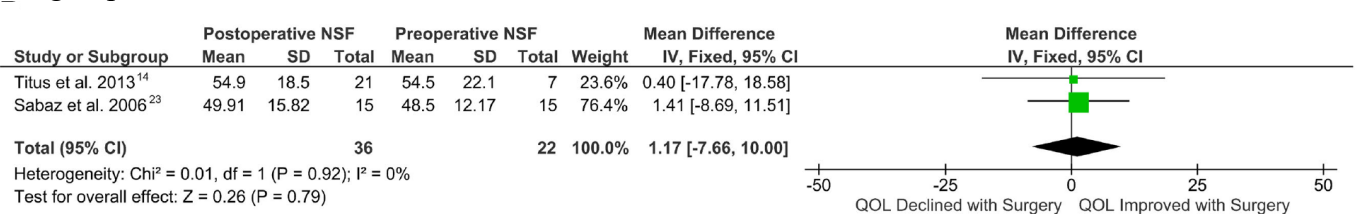
Quality of life in pediatric patients (surgery vs. medications):



Subgroup with seizure freedom:



Subgroup with residual seizures:



Maragkos et al. 2020

- when looking at the secondary outcome of IQ, there was a **significant improvement of about 10 IQ points in patients treated surgically** than matched controls treated medically.

SURGERY OUTCOMES - PSYCHIATRIC

- patients with drug-resistant epilepsy have an up to **5-fold increased rate of death by suicidality** compared to the general population, with highest rates in people with **TLE*** and **history of surgery**.
 *impairment of limbic regions leading to mood disturbance, aggression, and impulsivity.

N.B. **suicides tend to occur once good seizure control is obtained** - so-called “burden of normalization”.

De Novo Psychopathology

Novais F et al. Predicting De Novo Psychopathology After Epilepsy Surgery: A 3-Year Cohort Study. *Epilepsy Behav.* 2019;90:204-208. doi:10.1016/j.yebeh.2018.11.037. PMID: 30573340.

- 106 patients
- psychiatric evaluations were made before surgery and every year, during a 3-year follow-up period.
- resective surgery (93%) and ANT-DBS (7%).

- survival analysis model was used to determine pre- and postsurgical predictors of de novo psychiatric events after surgery.
- 15% developed de novo psychiatric disorders that were never identified before surgery.
 - significant predictors - multilobar epileptogenic zone ($P = .001$) and ANT-DBS ($P = .003$).
 - hypotheses:
 - c) failure to establish adequate baseline.
 - d) surgery eliminates seizures, which could be an endogenous psychiatric treatment (ECT – a medically monitored generalized seizure - is a particularly effective treatment for depression).

SURGERY OUTCOMES - COGNITIVE

MEMORY

- new postop memory deficit does not adversely impact quality of life when patients become seizure free.

Langfitt JT, Westerveld M, Hamberger MJ, et al. Worsening of quality of life after epilepsy surgery: effect of seizures and memory decline. Neurology. 2007;68(23):1988-1994

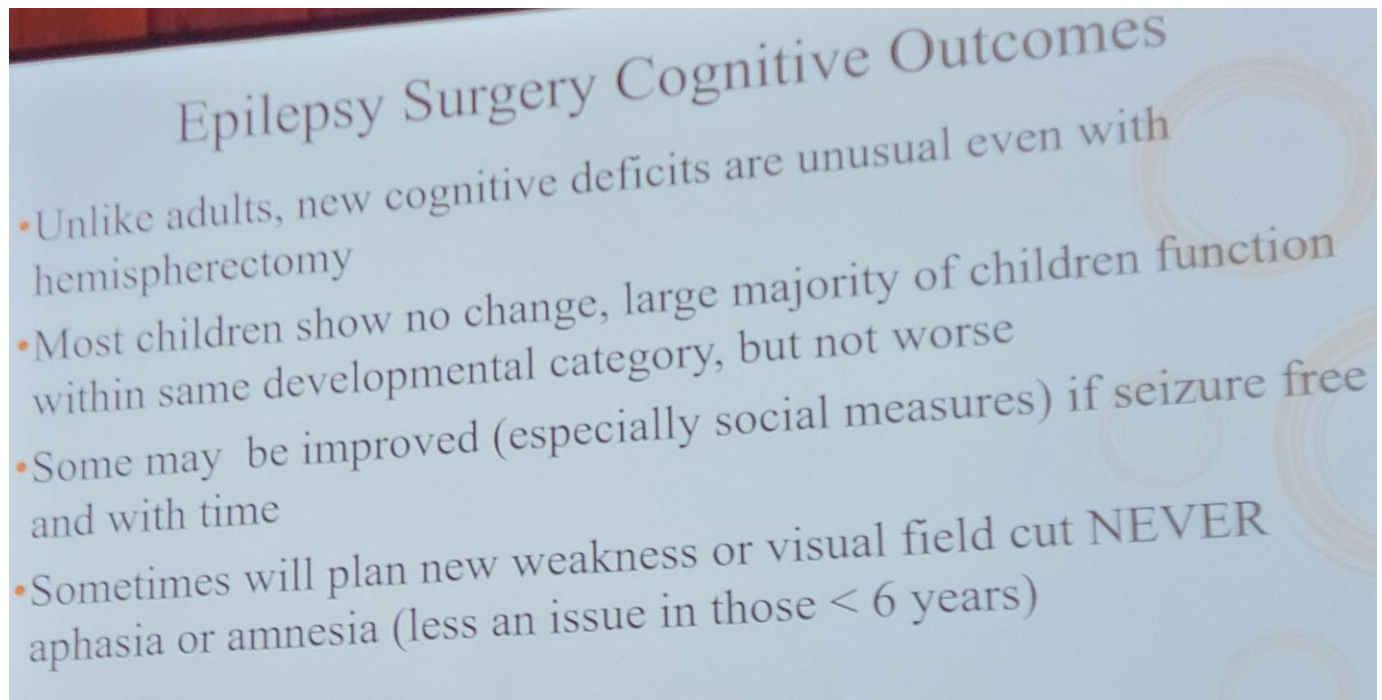
PEDIATRIC

- early surgery is especially important in **children**, when epileptogenic discharges interfere with normal development! (language can shift to opposite hemisphere if surgery is performed while patient is < 6 years).

N.B. even patients only few months of age are treated surgically if surgery is treatment of choice!

Consolidation of cognitive systems:

- Motor helping hand < 2 years
- Language 6 (temporal) -10 years (frontal)
- Visual spatial 10 years
- Memory material specificity in teens
- Unilateral systems can “transfer” function; if does not transfer then more impaired
- Bilateral systems paradoxically less compensation (e.g. executive function)
- Implications for timing of epilepsy surgery can be more aggressive with resections in the young



TEMPORAL EPILEPSY

Lower preoperative neuropsychological scores mitigate the risk of postoperative decline!

ROLE OF STRUCTURES

Hippocampus-sparing ATL vs. LITT amygdalohippocampectomy in language-dominant temporal lobe

George Z. et al. Neuropsychological outcomes of hippocampus-sparing anterior temporal lobectomy versus stereotactic laser amygdalohippocampectomy in language-dominant temporal lobe epilepsy. Epilepsia. 2025;00:1–15.

- **background:** hope that hippocampal preservation will lead to better verbal memory outcomes.
- **hs-ATL** (no MTS) → significant declines in **logical memory** ($p = .04$) despite sparing hippocampus, which were not seen following LITT in patients regardless of MTS status.
- **LITT** (no or with MTS):
 - improved **semantic fluency, immediate verbal memory, and executive function** despite hippocampal ablation;
 - significant decline in **letter fluency** (with subgroup analyses demonstrating significantly greater declines in MTS(+) patients (.02).

Verbal memory may depend on a broader network incorporating extramesial structures.

Approach	LITT	hs-ATL
uncus and amygdala	resected	resected
hippocampus and parahippocampal gyrus	ablated	spared
temporal pole and ventral and lateral temporal gyri	spared	resected

Conclusions:

Cognitive advantages of LITT targeting mesial temporal structures while maximally preserving cortical structures and white matter tracts critical for language networks.

LITT has better preservation of immediate verbal memory and executive function despite hippocampal ablation vs. hs-ATL was associated with declines in immediate and delayed verbal memory despite partial hippocampal sparing.

Data failed to indicate any relationship between hippocampal preservation and improved verbal memory.

- hippocampal preservation was associated with declines in immediate and delayed verbal memory for stories, which were not seen with targeted hippocampal ablation - this suggests the lateral temporal regions and related pathways (uncinate fasciculus, ILF, inferior fronto-occipital fasciculus, MLF, and superior longitudinal fasciculus) may be more vital for verbal memory encoding than mesial structures.

Temporal pole resection may impair association zones vital to object recognition and naming functions. Therapies (like LITT) that maximally preserve lateral, ventral, and anterior temporal areas may offer advantages for cognitive function, despite disrupting critical nodes like hippocampus in patients with relatively preserved language function.

OPEN RESECTIONS

- children decline less (vs. adults) after open resections; actually, children show improvement of IQ in long-term (maybe result of ASM withdrawal) – do surgery in kids ASAP.

Dr. Daniel Drane: best neuropsychological outcomes – from subtemporal selective amygdalo-hippampectomies

Modifications to the Subtemporal Selective Amygdalohippampectomy Using a Minimal Access Technique: Seizure and Neuropsychological Outcomes

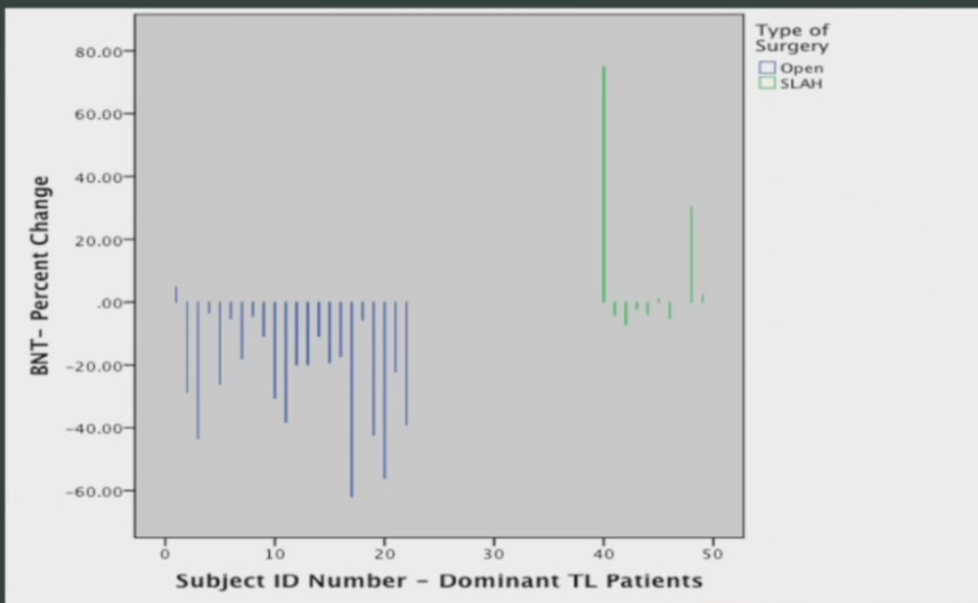
Andrew S. Little, MD, Kris A. Smith, MD, Kristin Kirlin, PhD*, Leslie C. Baxter, PhD*, Steve Chung, MD†
Rama Maganti, MD†, David M. Treiman, MD†

LITT

Dr. Daniel Drane: mesial temporal LITT can be done without language or object recognition deficits!!! (unless ablating temporal pole, fusiform gyrus [famous face naming decline], ventral language stream)

vs. $\geq 40\%$ ATL patients have naming deficits at 1 year follow up:

Post-Surgical Change in Man-Made Object Naming (Boston Naming Test) Performance by Surgery Type



Drane et al. (2015)
Epilepsia.

Note. BNT = Boston Naming Test; SLAH = stereotactic laser amygdalohippocampotomy; open = open resection; TL = Temporal Lobe.

- 1) **memory decline** - same as with open SAH.
LITT preserves **naming** and **contextual verbal memory** - all white matter fibers (vs. verbal memory is in hippo)
- 2) neuropsychiatric symptoms, worsening of a preexisting affective disorder (4.3%).
- 3) **language decline** – but better than with ATL.
LITT is better for **language preservation** vs. ATL or SAH
 - through investigation of connectomes involved in TLE, naming and object recognition deficits are postulated to occur through injury to white matter in temporal stem which tends to be damaged when approaching the mesial temporal structures during ATL and SAH.
D.L. Drane, J.G. Ojemann, V. Phatak, D.W. Loring, R.E. Gross, A.O. Hebb, et al. Famous face identification in temporal lobe epilepsy: support for multimodal integration model of semantic memory. Cortex, 49 (2013), pp. 1648–1667
 - theoretically, LITT should circumvent this complication by limiting similar damage to white matter of temporal stem.
 - trend toward better preservation of naming with SAH versus ATL in dominant hemisphere.
A. Mansouri et al. Neurocognitive and seizure outcomes of selective amygdalohippocampotomy versus anterior temporal lobectomy for mesial temporal lobe epilepsy. Epilepsy Res Treat, 2014 (2014), pp. 1–8
 - LITT avoids neurocognitive adverse effects of open resection on naming (dominant side) and object recognition (nondominant side).
Cognitive declines related to the approach (collateral damage) impair naming and verbal learning (dominant hemisphere) or object recognition and figural learning (nondominant hemisphere).
Robert E. Gross, Jon T. Willie, Daniel L. Drane. The Role of Stereotactic Laser Amygdalohippocampotomy in Mesial Temporal Lobe Epilepsy. Neurosurg Clin N Am 27 (2016) 37–50
 - in several reports, LITT largely preserved **naming and object recognition** following language dominant ablations, functions that commonly decline following ATL or SAH.
 - verbal memory may decline following dominant side LITT; however, the risk appears to be lower than with open surgery.

- Kang parsed out verbal memory changes and found a decline in **noncontextual (word list) verbal memory**, which is localized to the mesial structures, but preservation of **contextual (narrative) verbal memory**, which is supported by the temporal neocortex.

Kang JY, Wu C, Tracy J et al. . Laser interstitial thermal therapy for medically intractable mesial temporal lobe epilepsy. Epilepsia. 2016;57(2):325-334

PENSAR study

BIBLIOGRAPHY for ch. “Epilepsy and Seizures” → follow this [LINK](#)